

LAMINAR-TURBULENT TRANSITION IN PIPES WITH BODY FORCES : CONTINUOUS, DISCONTINUOUS OR BOTH?

Guru Kalyan Jayasingh

Nigel Goldenfeld

Department of Physics, University of California San Diego, USA

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Ref: GKJ & NG, Tricritical directed percolation controls the laminar-turbulent transition in pipes with body forces, (in press at *Physical Review Letters*)

Acknowledgments: Björn Hof, Bowen Yang and Mukund Vasudevan

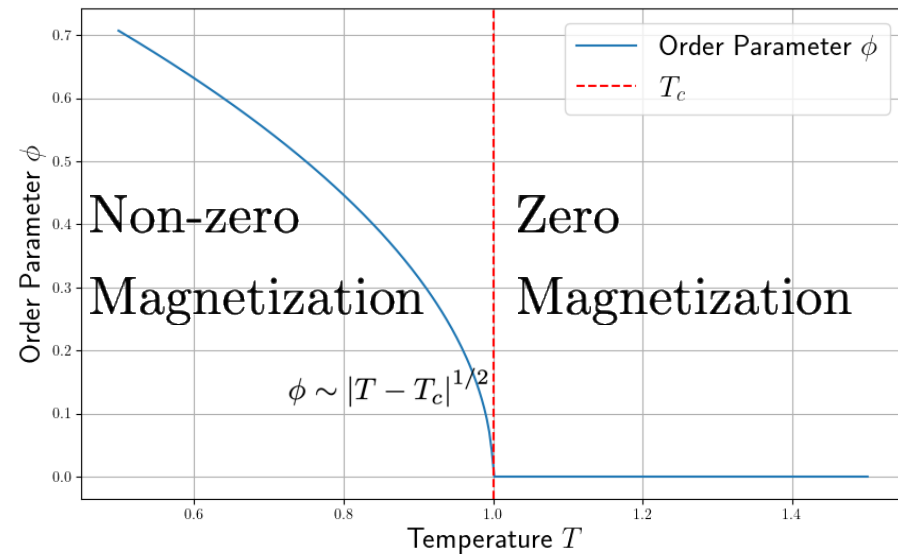
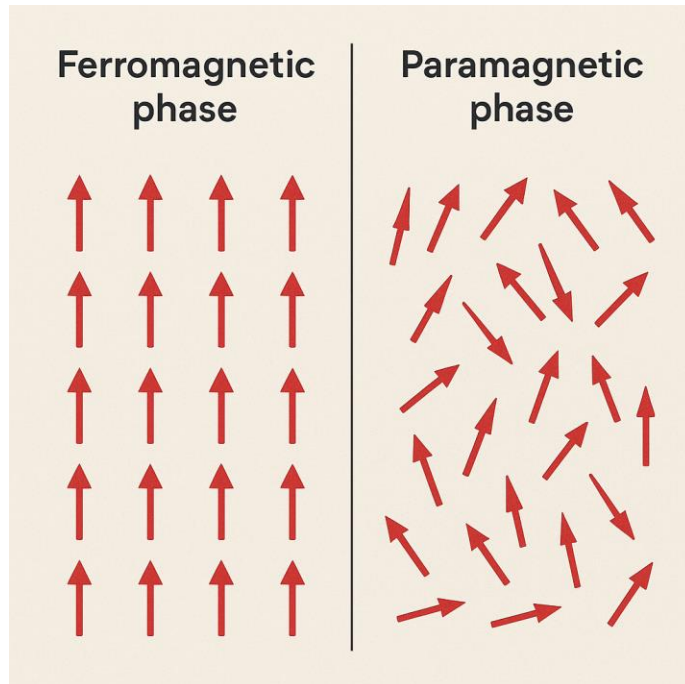
Outline

- Emerging consensus of transition in pipe flows
 - Flow phenomenology, criticality near transition
 - Emergent Directed Percolation at transition
- Transition in body force influenced flows
 - Critical point under perturbation?
Experimental data
 - Universality? Relation to straight pipes?
 - Can we use methods of statistical mechanics
viz. minimal models, landau theory etc. to describe
the transition?

Review of Critical Points

- Critical point:
 - Requires tuning one parameter (e.g. $T - T_c$)

$$\mathcal{F}_{GL} = a_2\phi^2 + a_4\phi^4 \quad a_2 = \alpha(T - T_c) \quad a_4 > 0 \quad \frac{\partial \mathcal{F}_{GL}}{\partial \phi} = 0$$

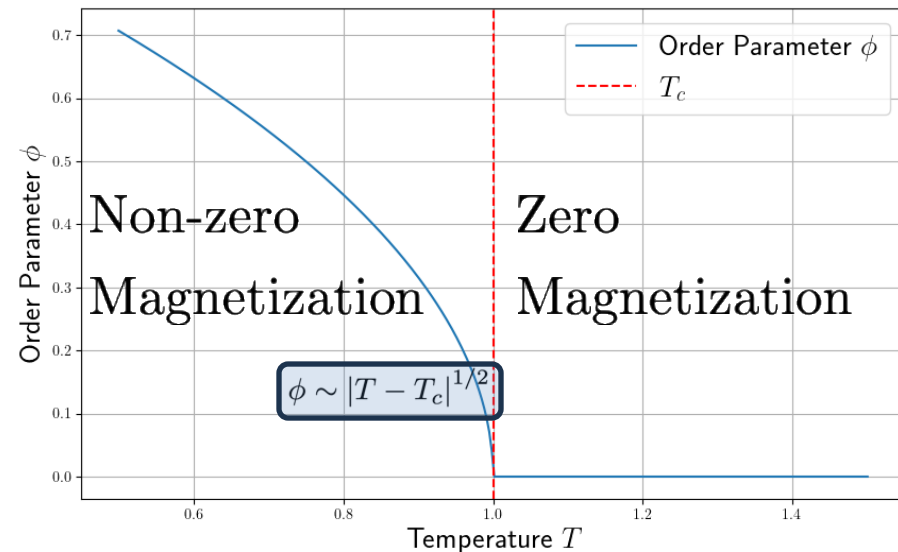
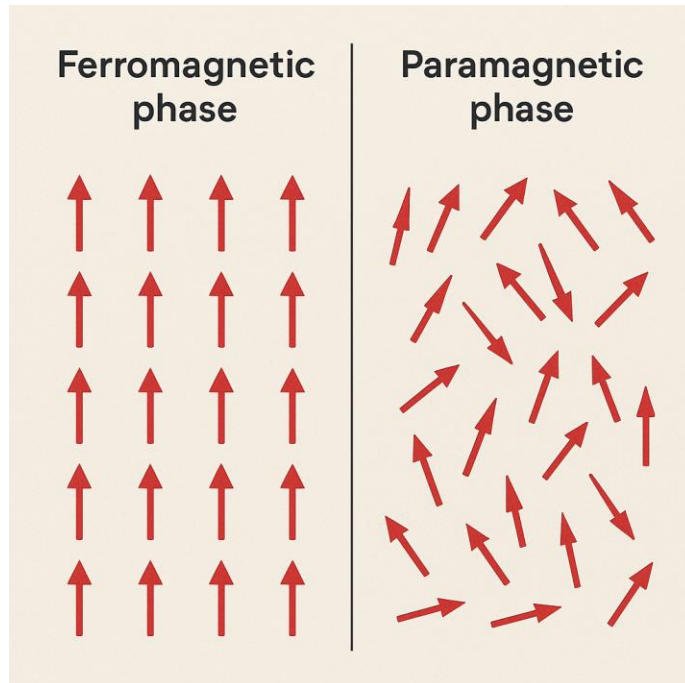


$\phi \neq 0 \implies$ Ordered $\phi = 0 \implies$ Disordered

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Tricritical point in phase transition

- Tricritical point
 - Requires tuning two parameters

$$\mathcal{F}_{GL} = a_2\phi^2 + a_4\phi^4 + a_6\phi^6 \quad a_2 = \alpha(T - T_c) \quad a_4 \gtrless 0 \quad a_6 > 0$$

Temperature, Pressure (e.g.) $He^3 - He^4$

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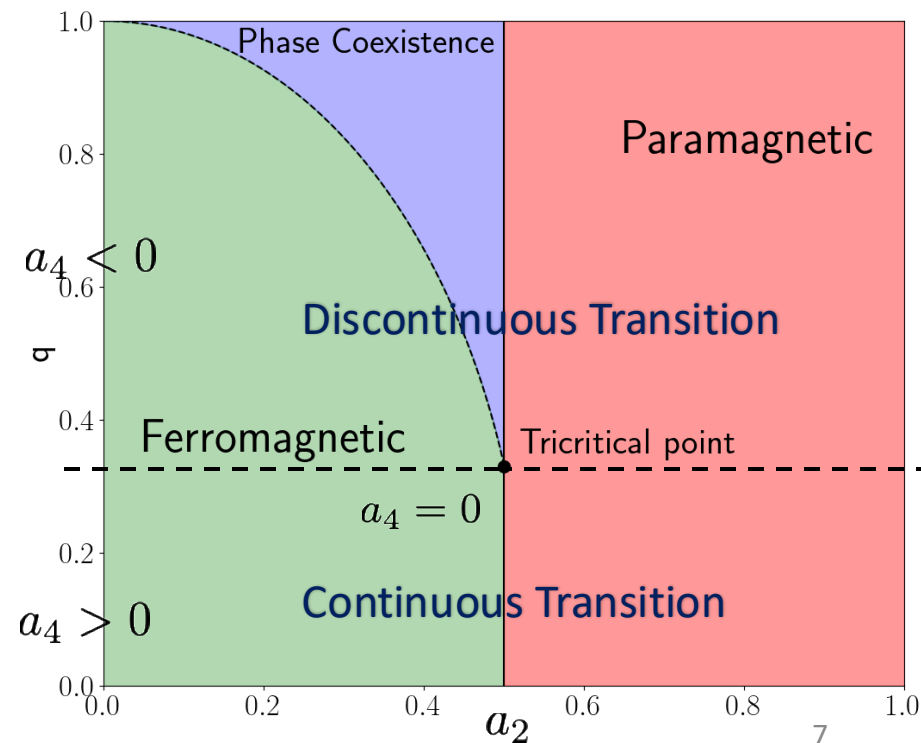


Two tunable
parameters

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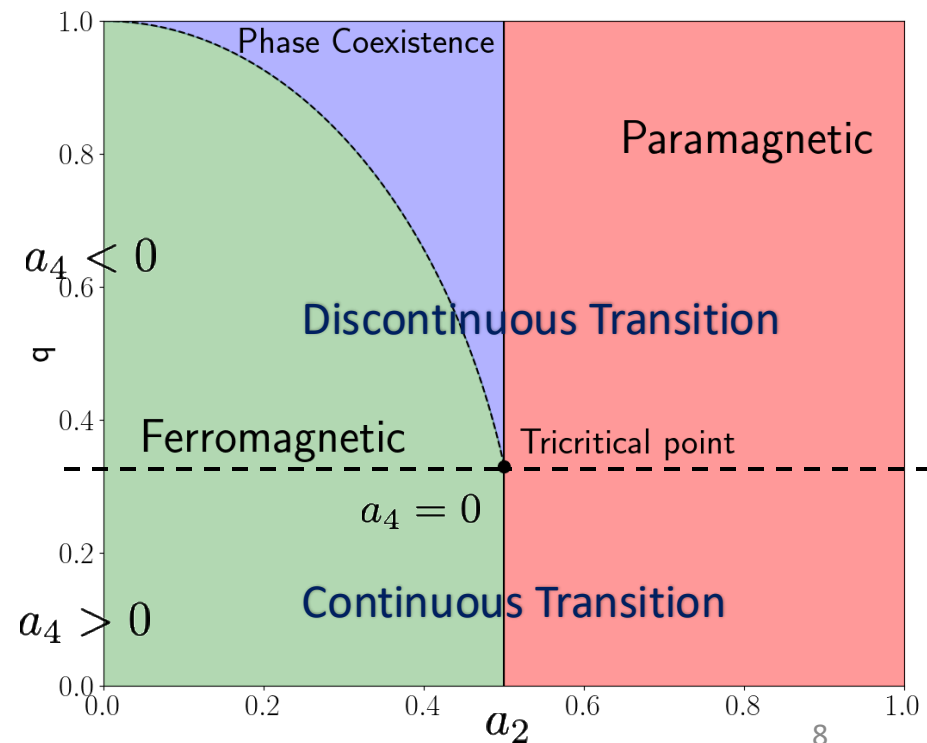


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- $a_4 < 0$ discontinuous
- $a_4 > 0$ continuous

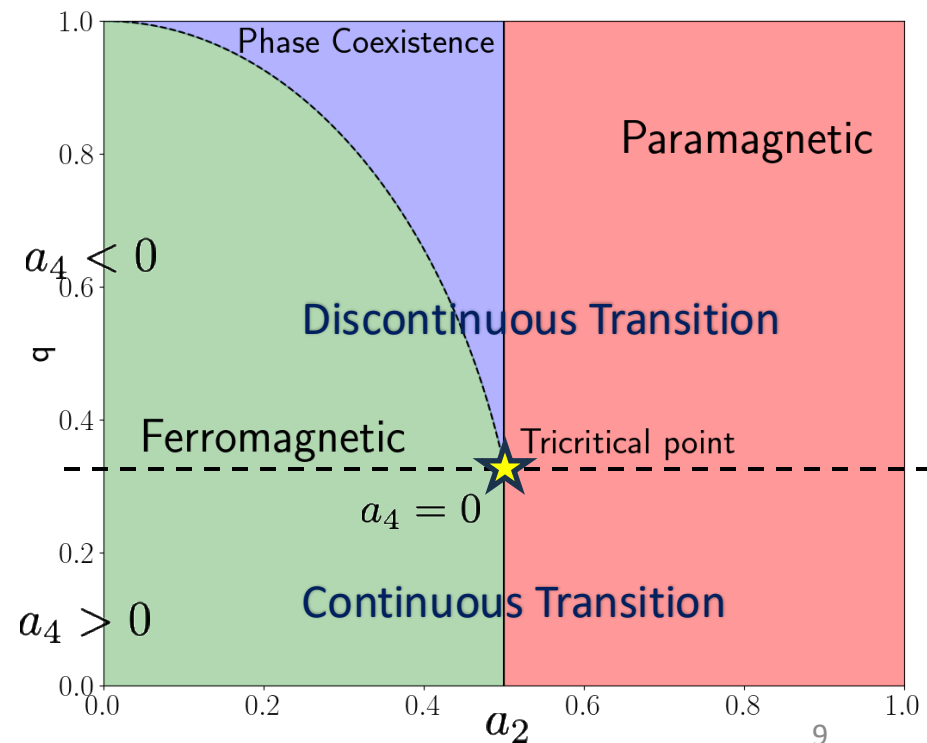


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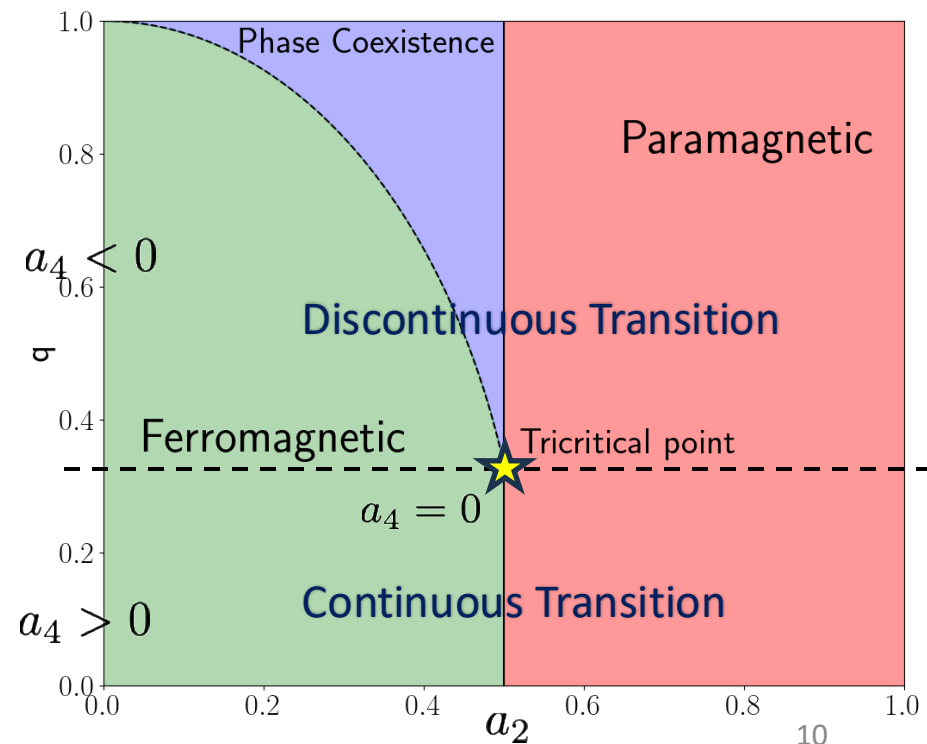


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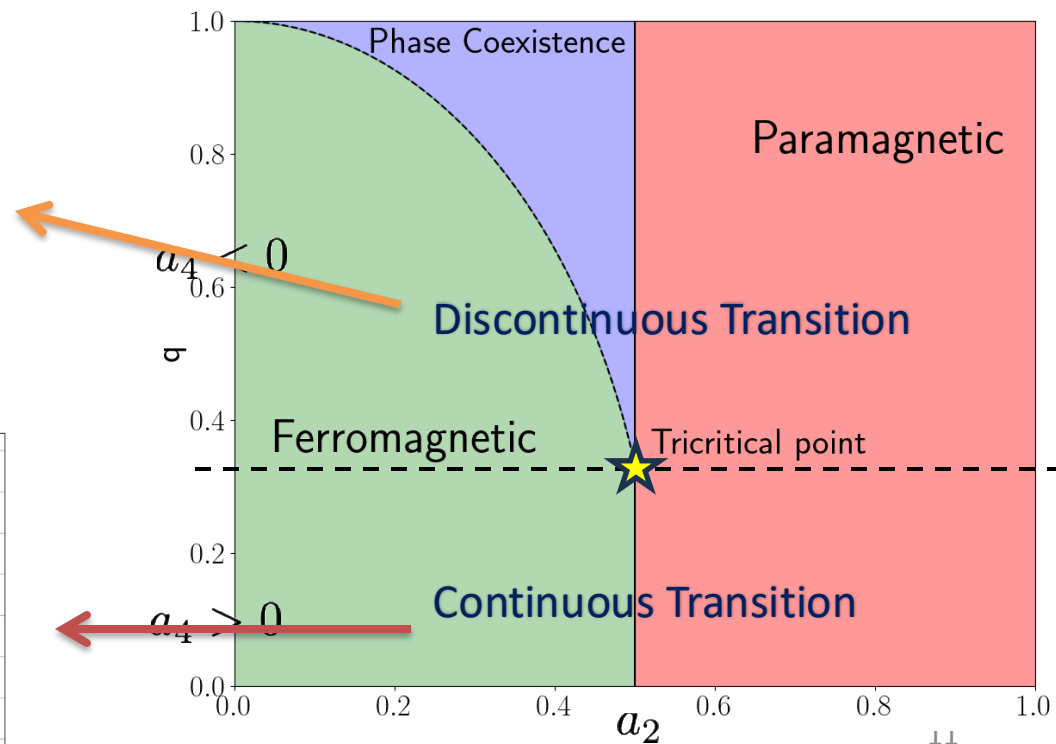
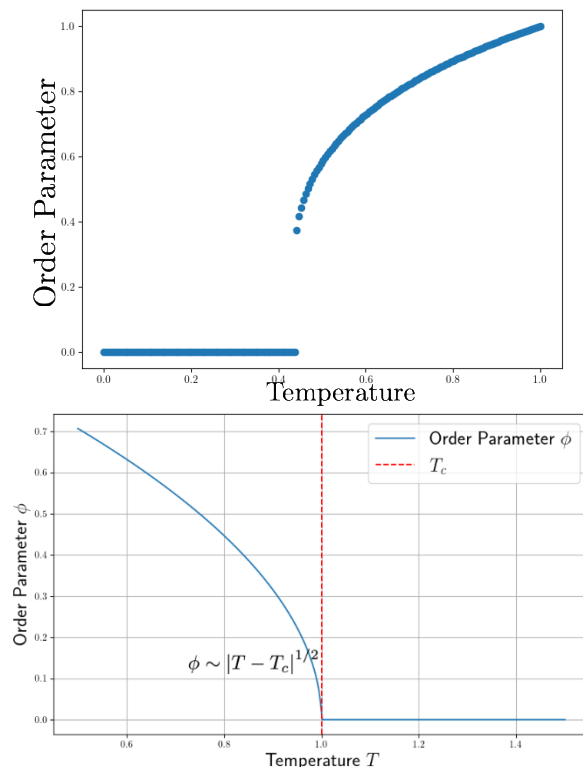
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- Upper critical dim = 3



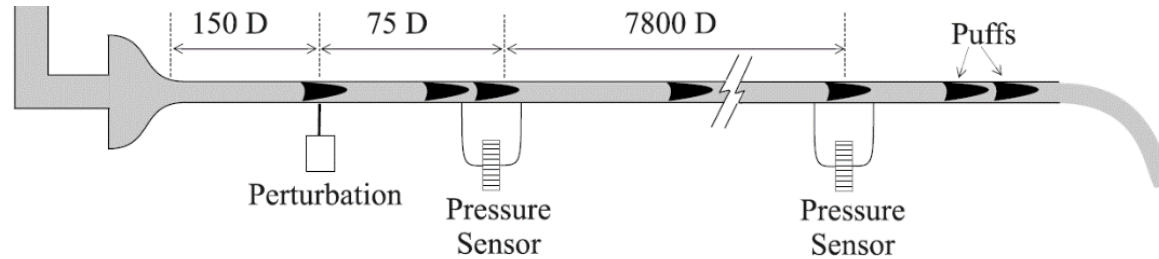
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Phase diagram of pipe turbulence



Vasudevan & Hof (2018)

laminar flow

steady
predictable

laminar-turbulent transition

fully-developed turbulence

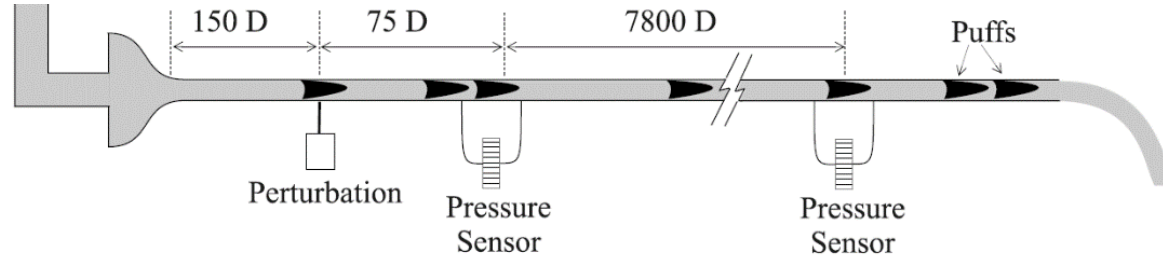
fluctuating
unpredictable

$Re < 1600$

$Re \sim 2000$

$Re > 10^5$

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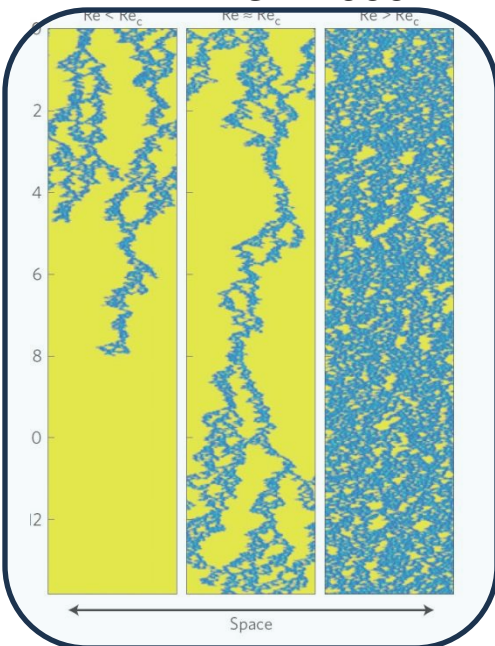
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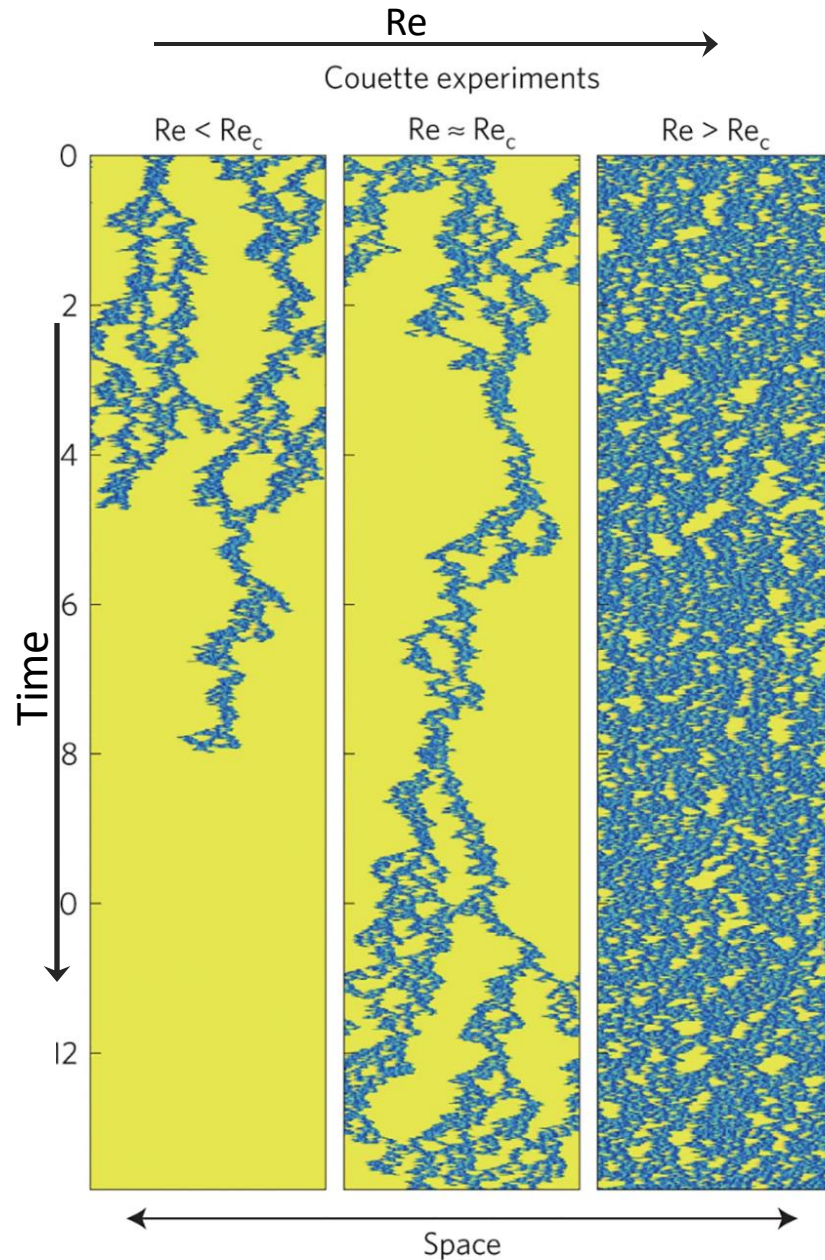
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Spatiotemporal
dynamics

Experiments in Taylor Couette flow



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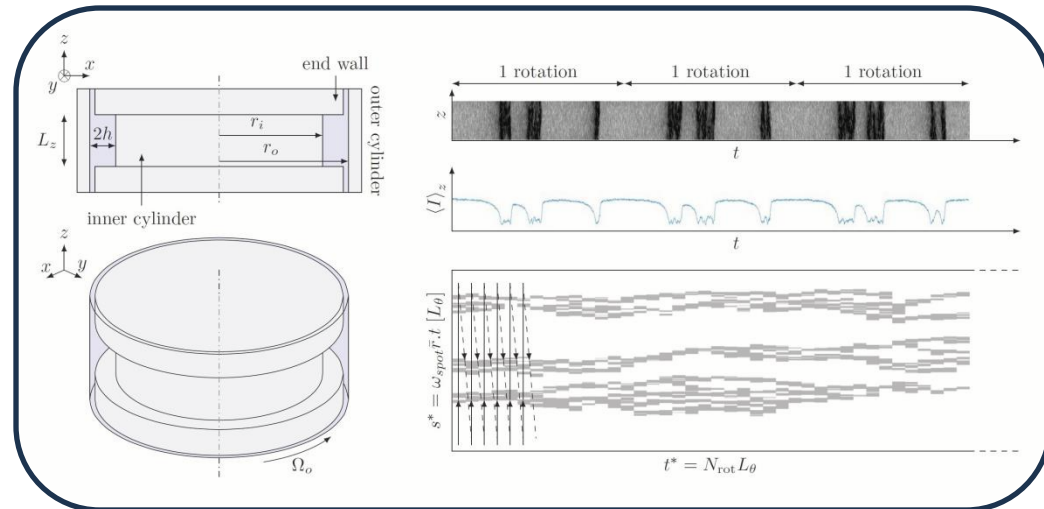
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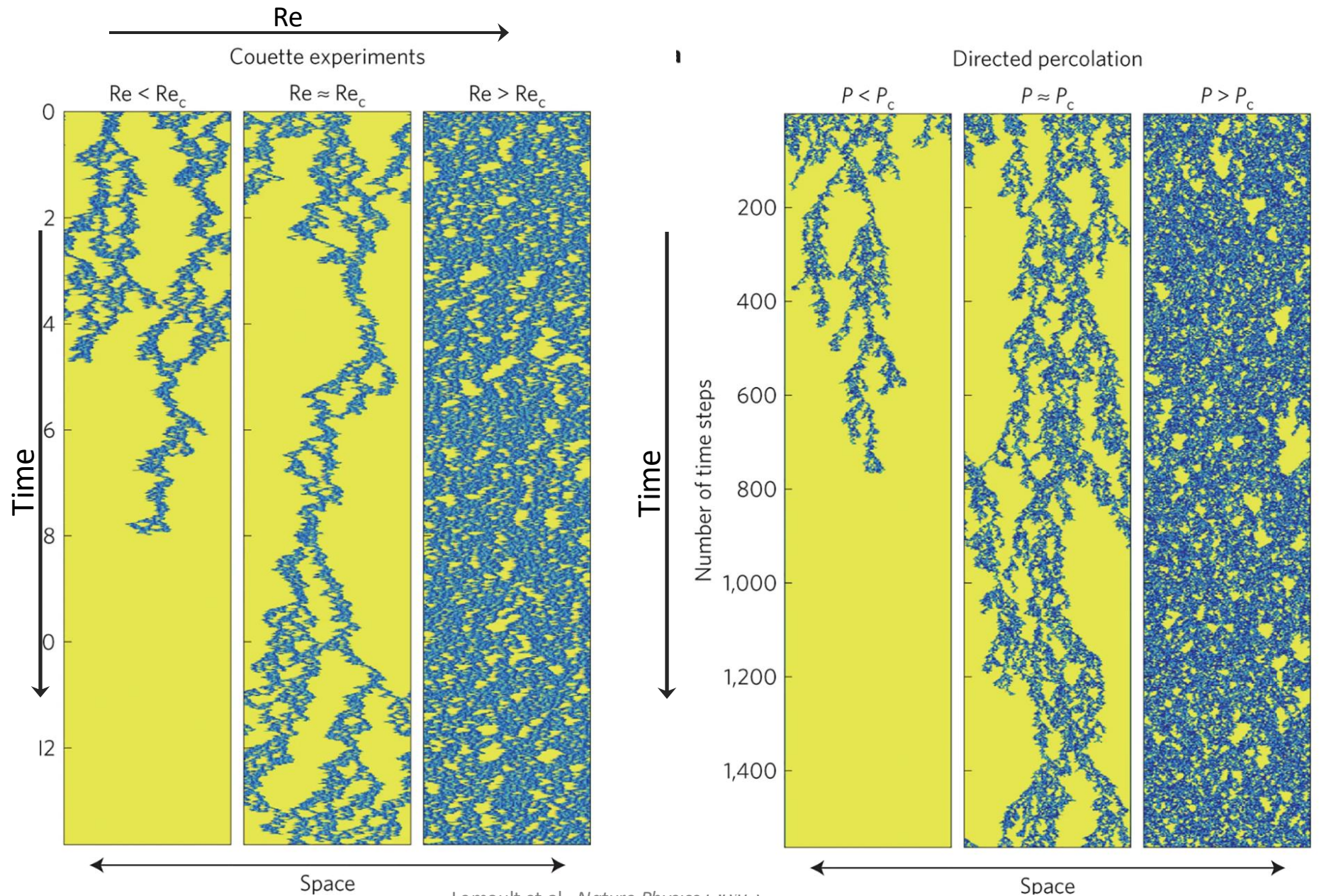
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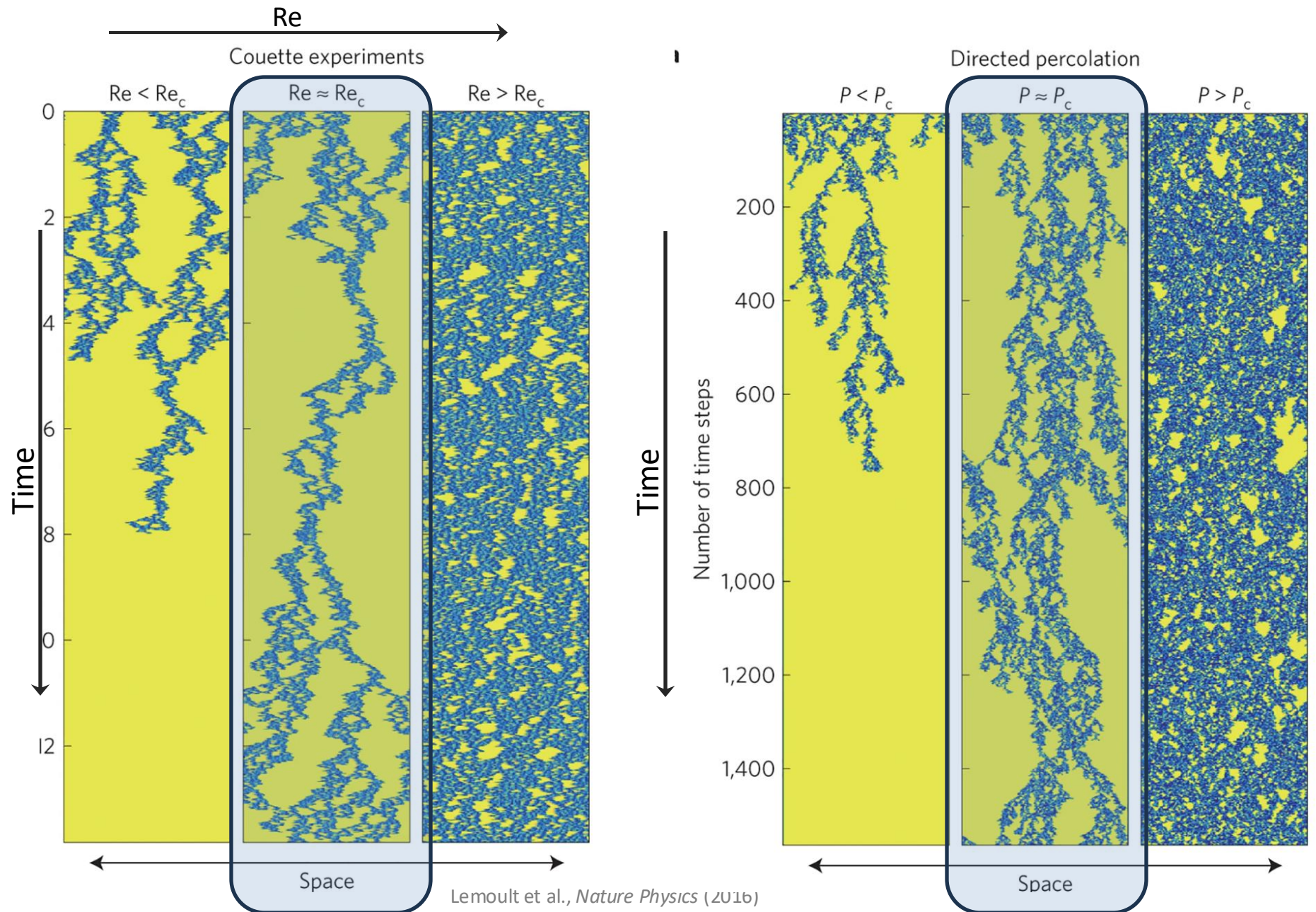
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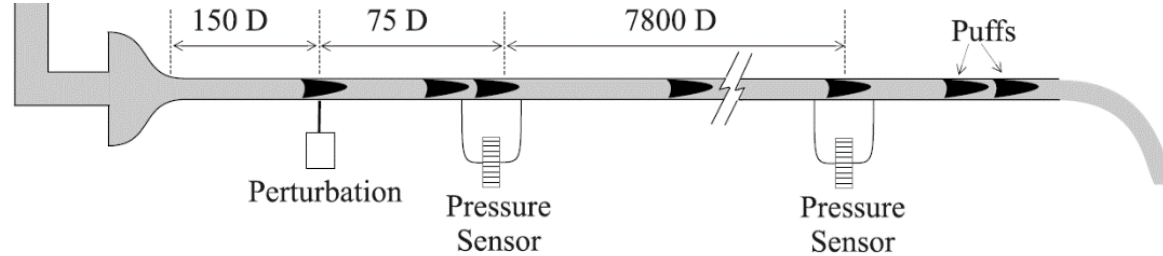
Turbulent strands show Directed Percolation dynamics



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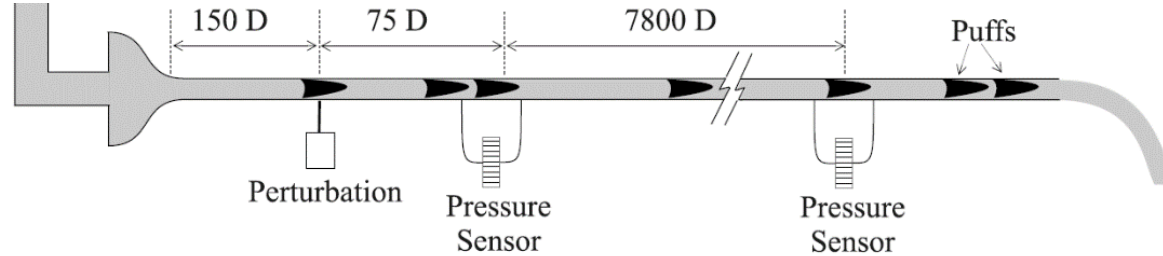
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Phase diagram of pipe turbulence



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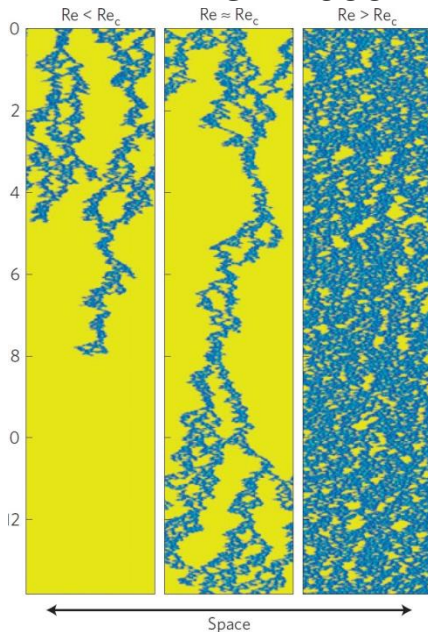
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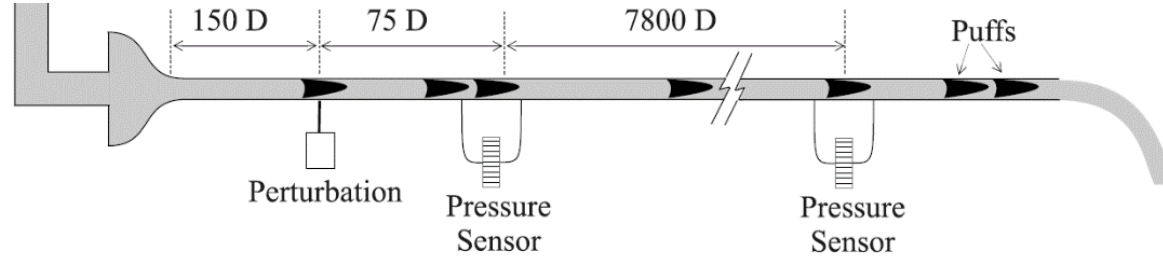
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**Directed percolation
(DP)**



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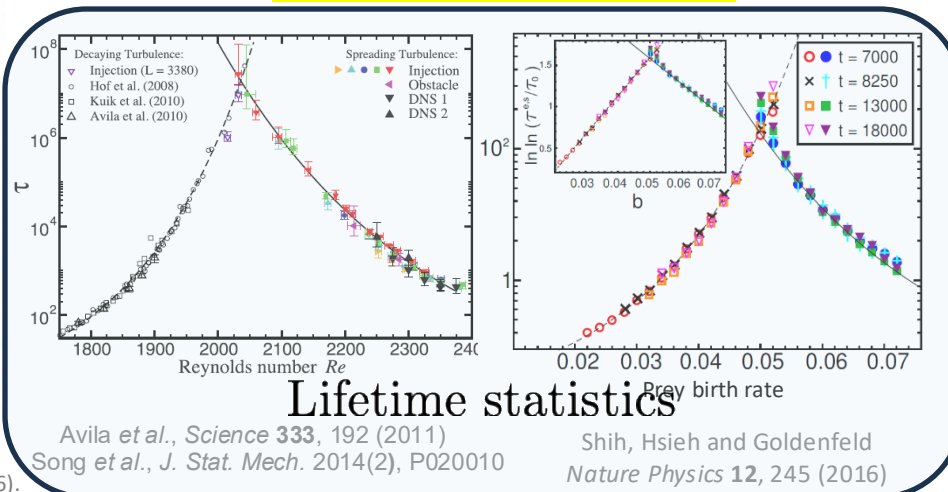
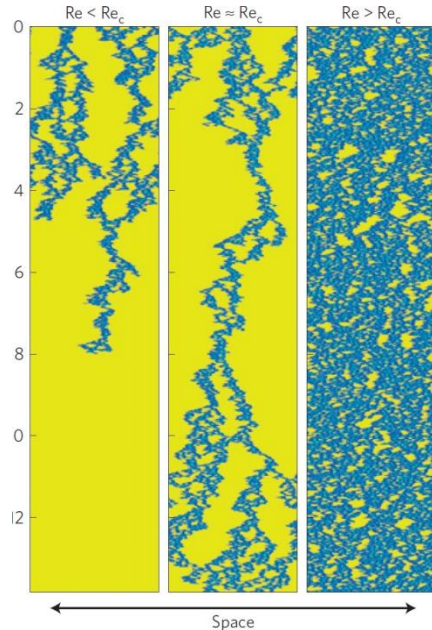
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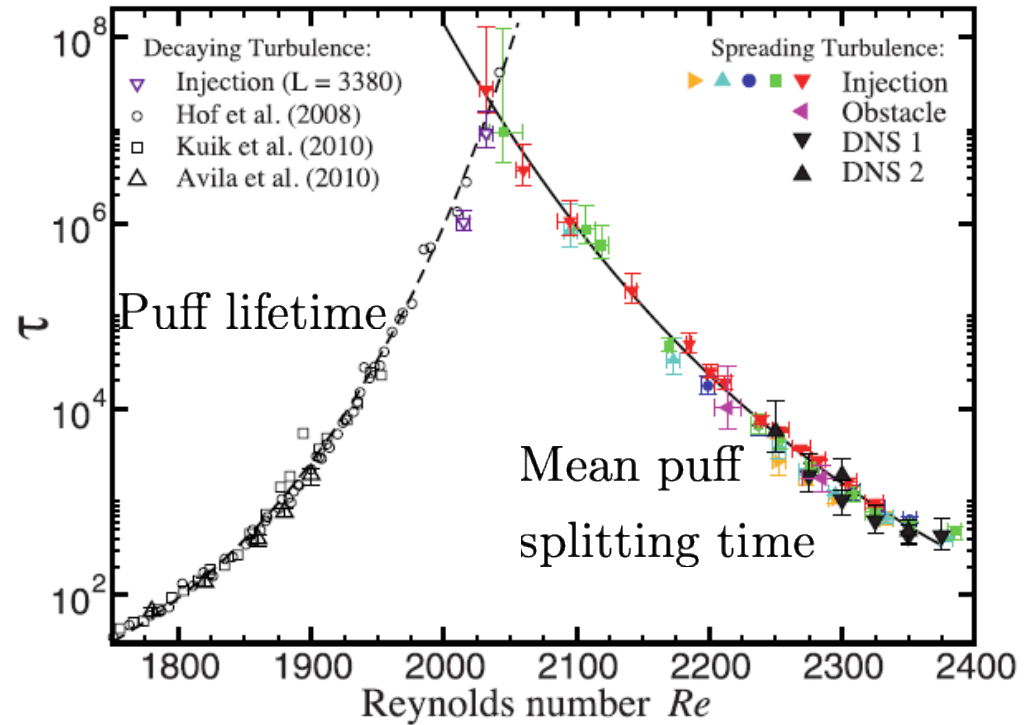
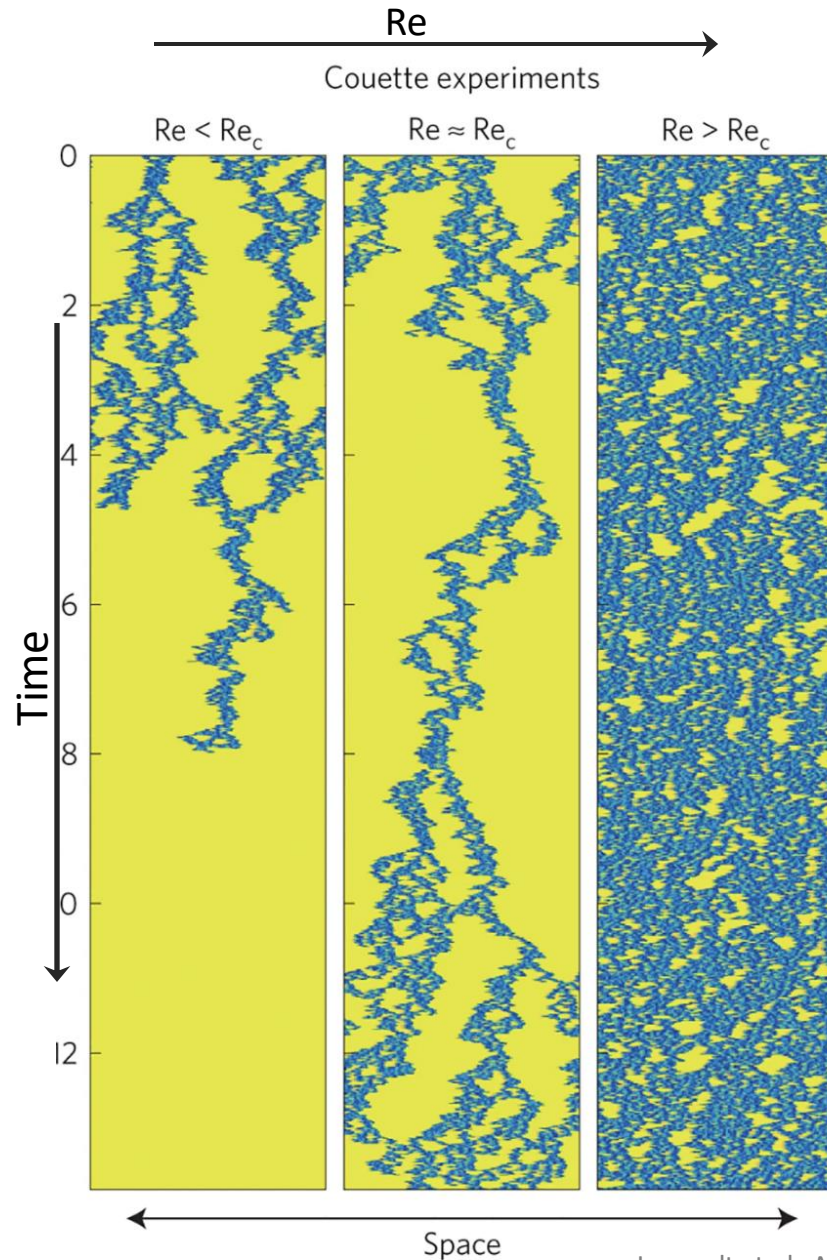
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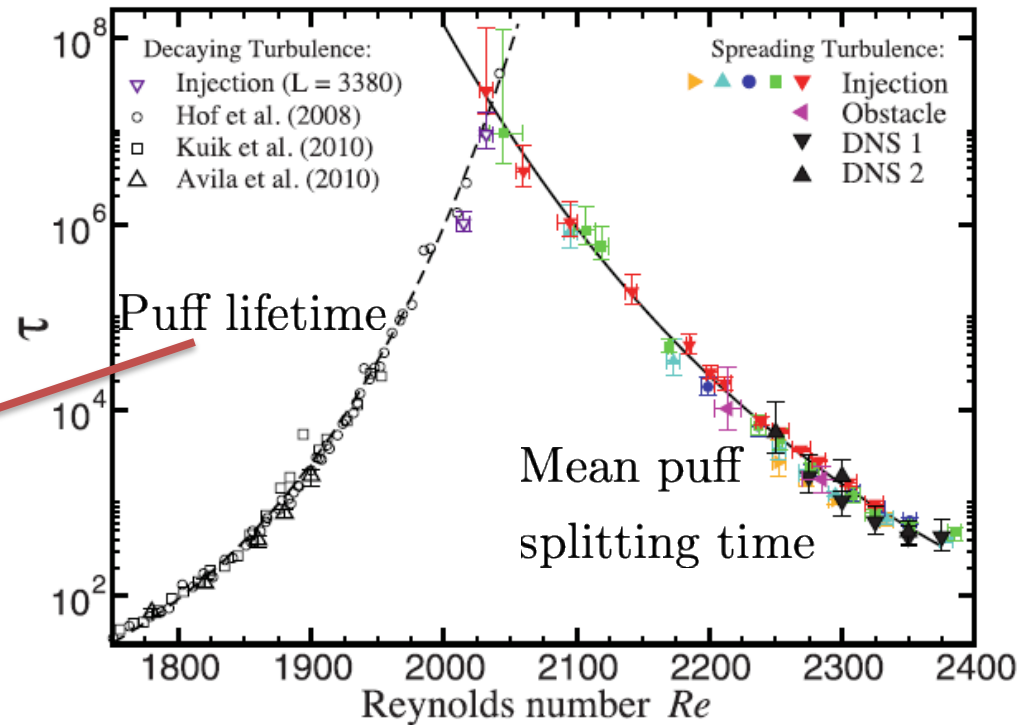
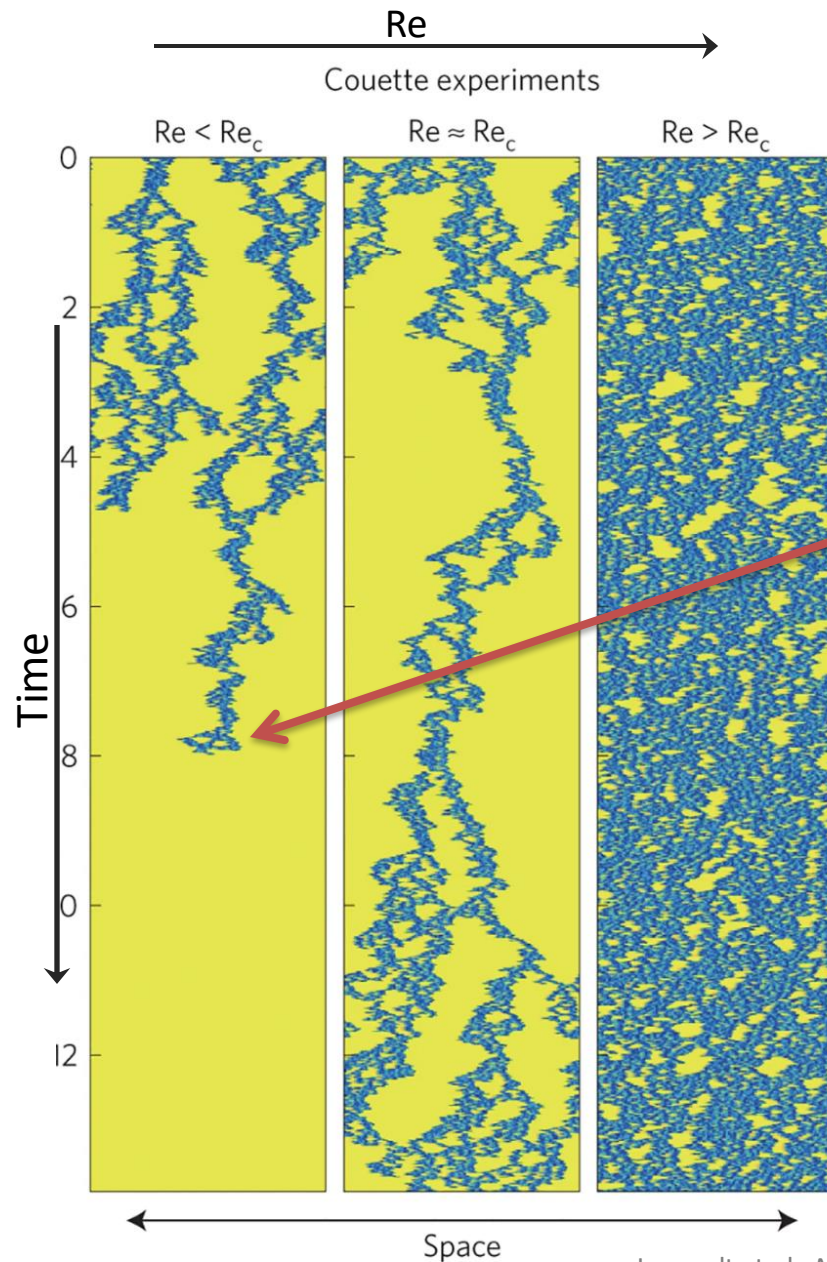
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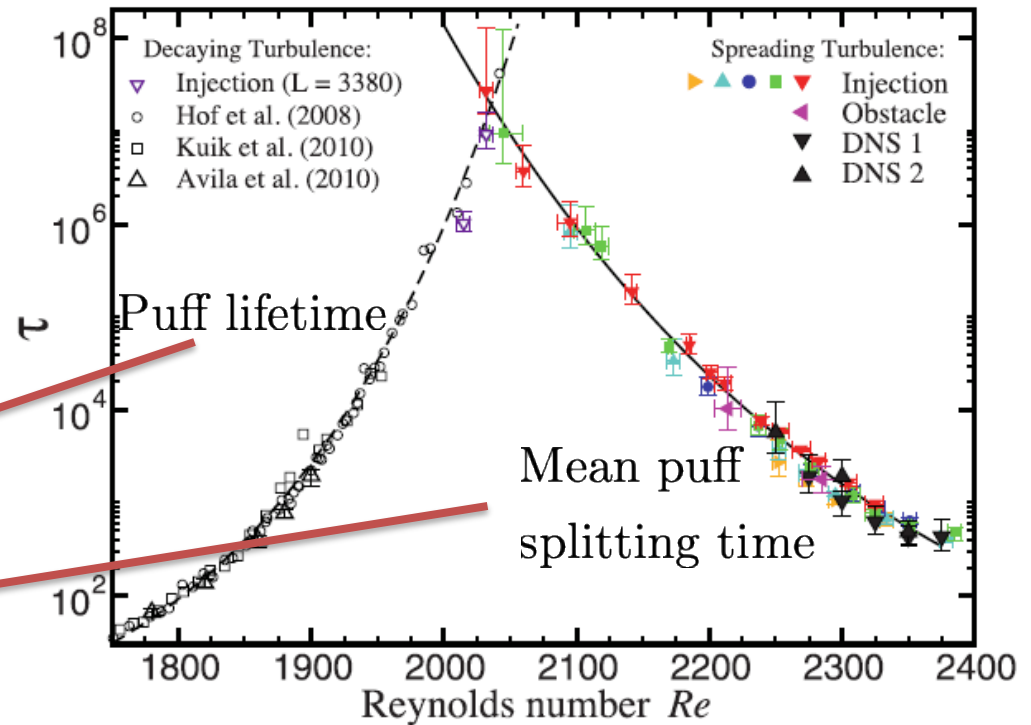
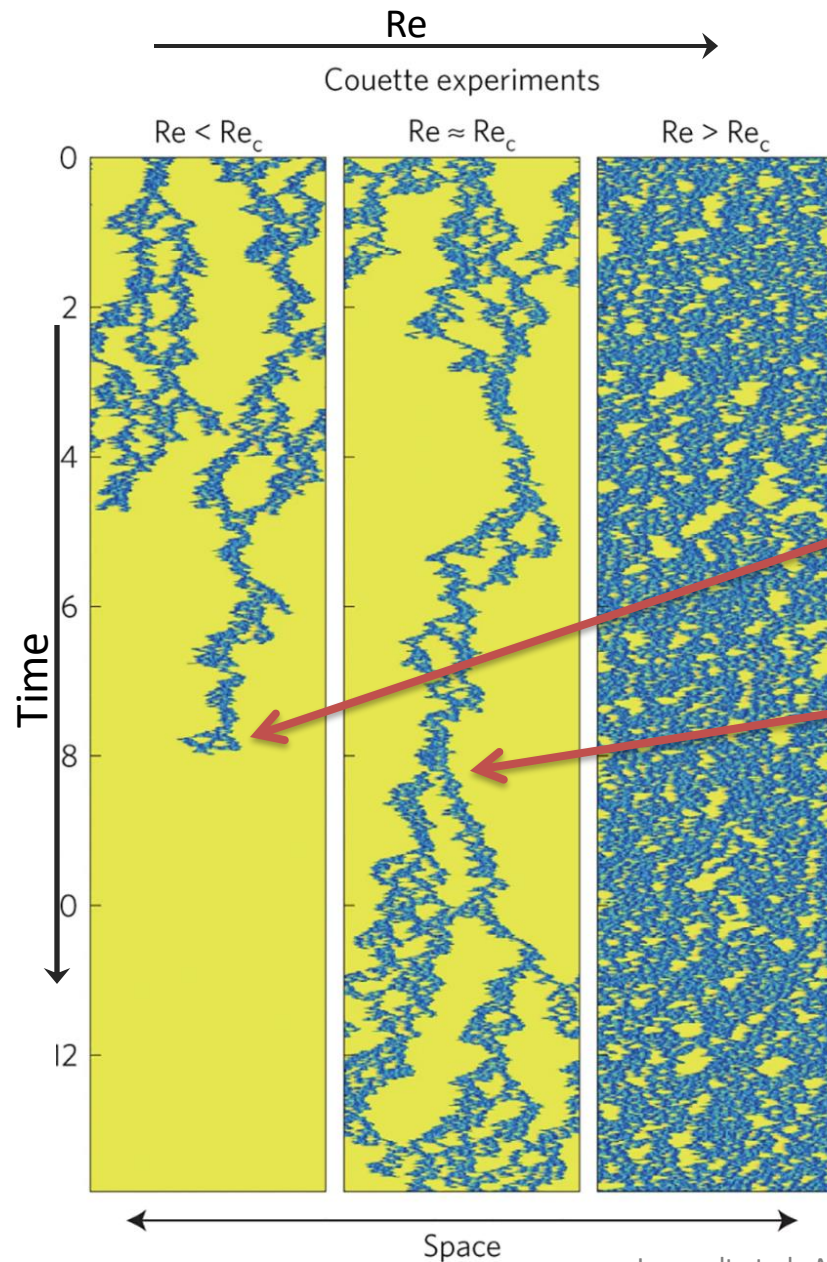
Lifetime statistics



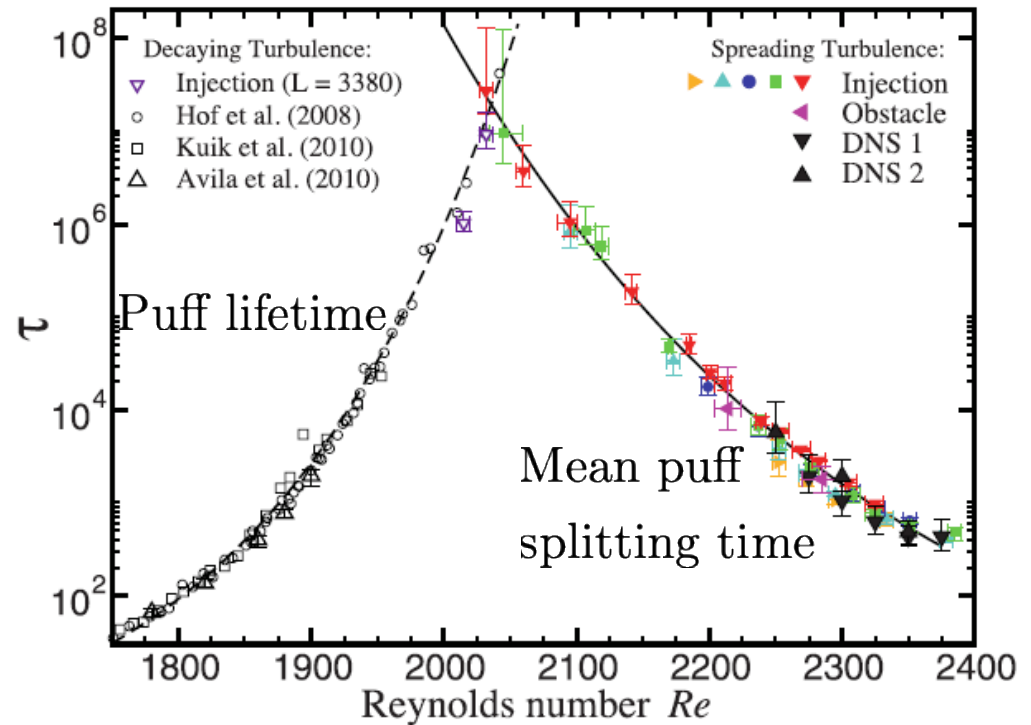
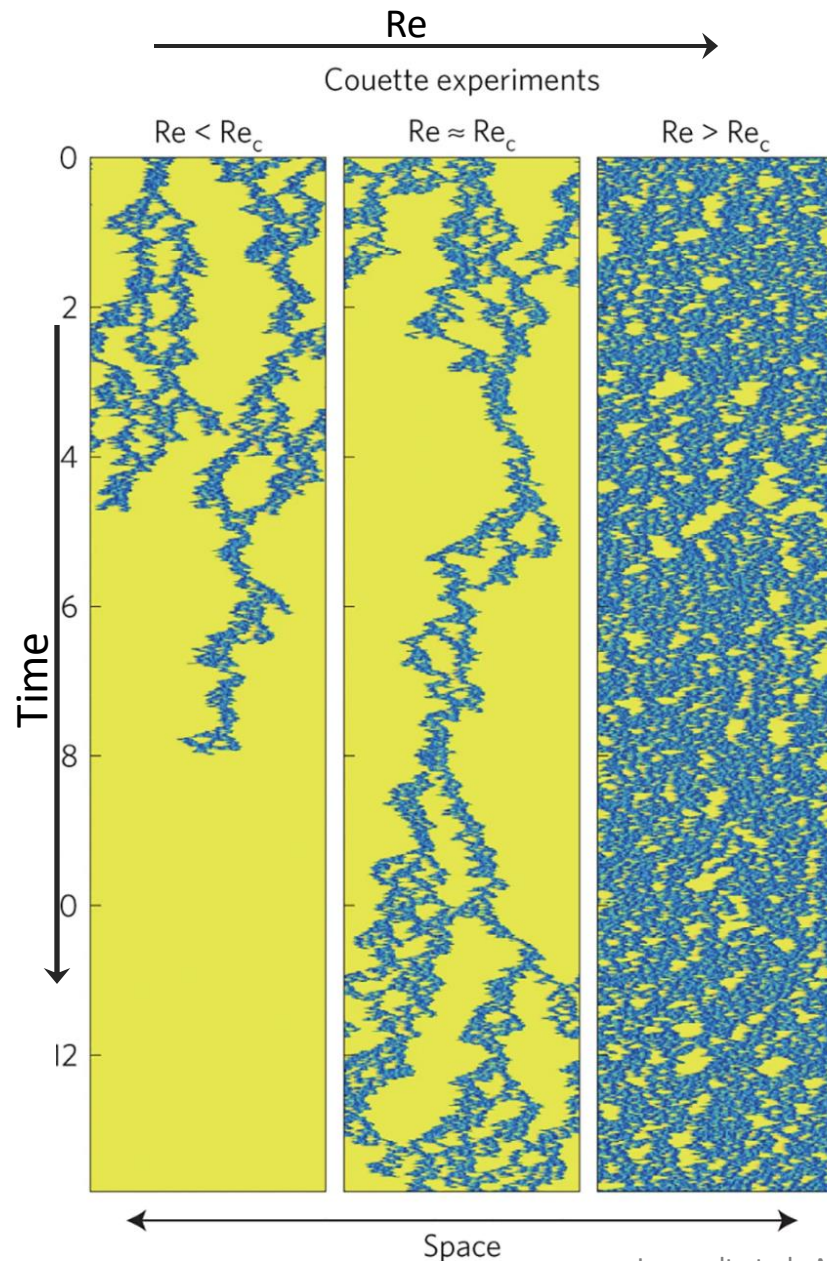
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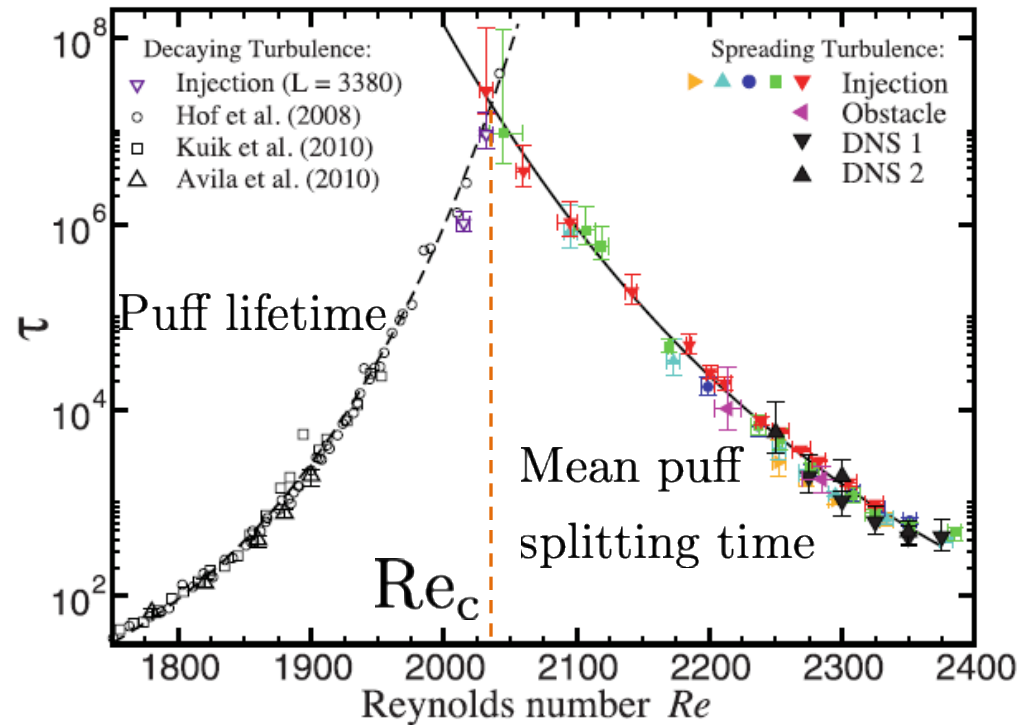
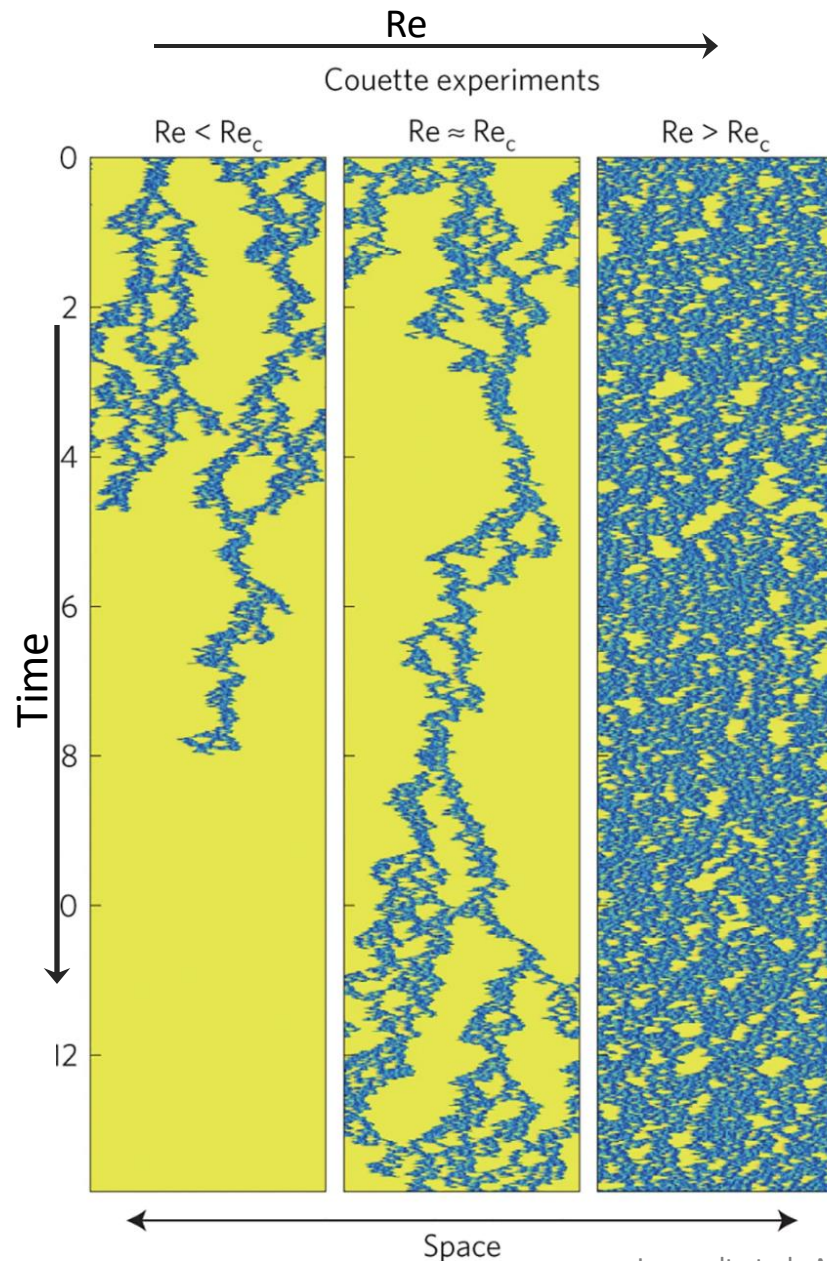
Lifetime statistics



$$\frac{\tau}{\tau_0} \sim \exp(\exp Re)$$

Super-exponential scaling
of lifetimes

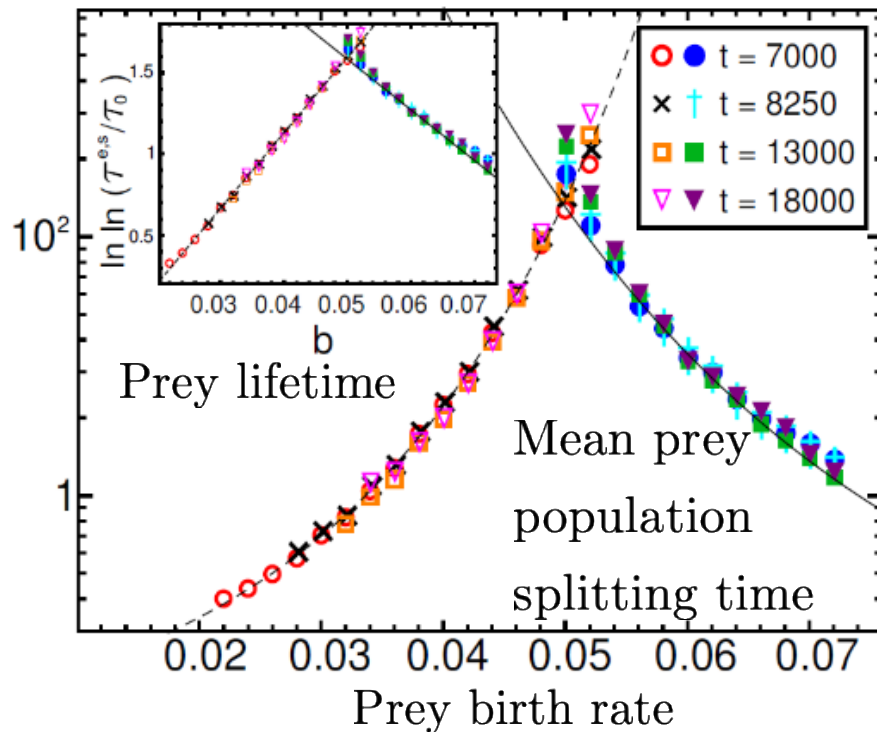
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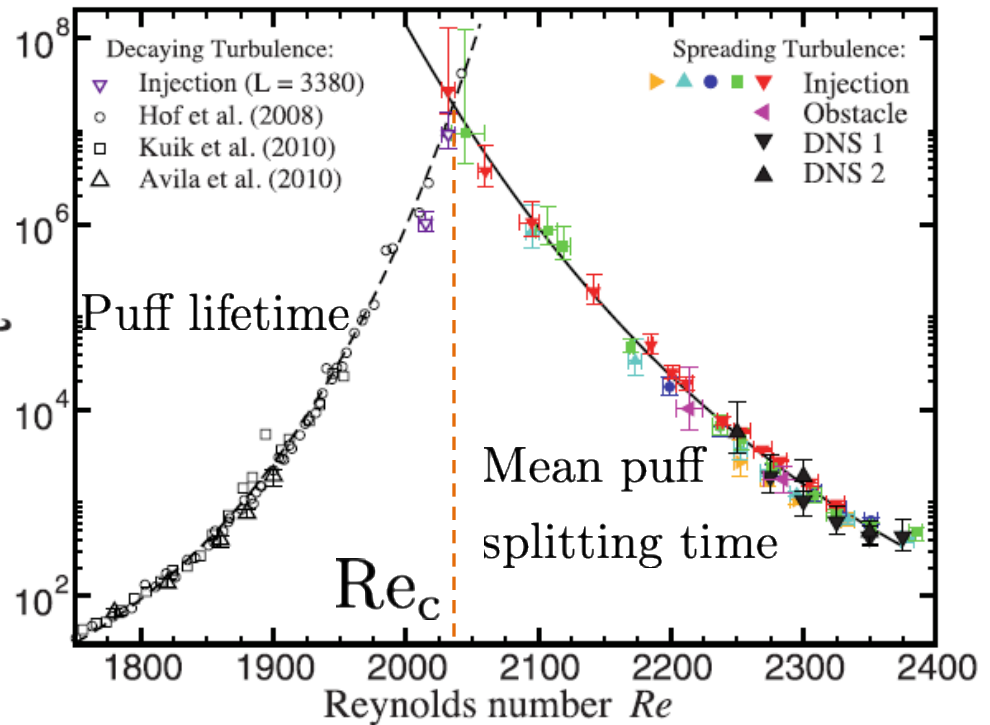
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Super-exponential scaling
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Predator-prey model recapitulates lifetime statistics

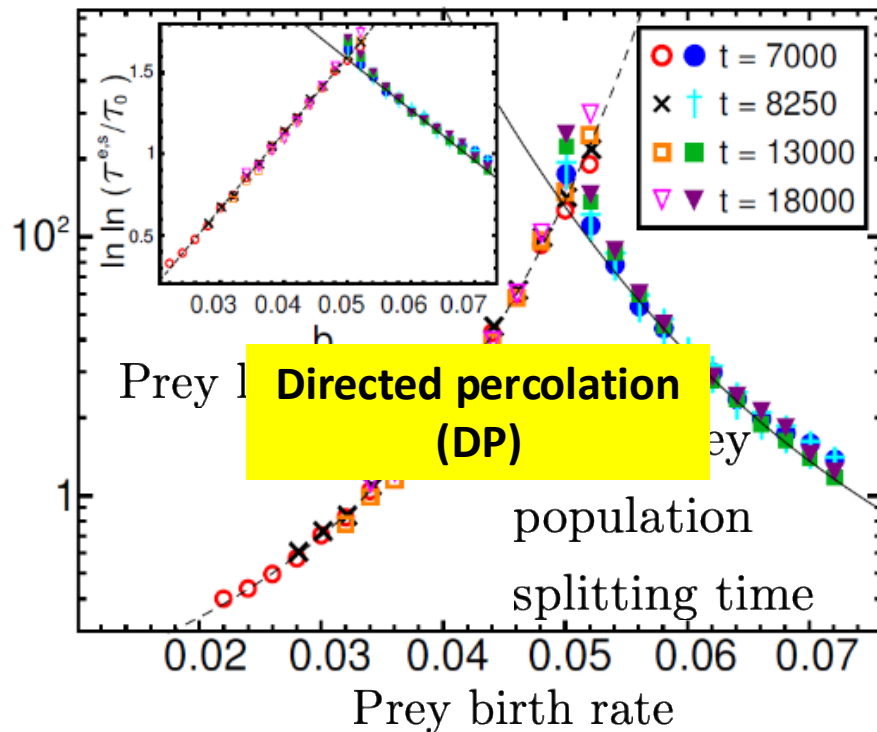


Stochastic predator-prey dynamics

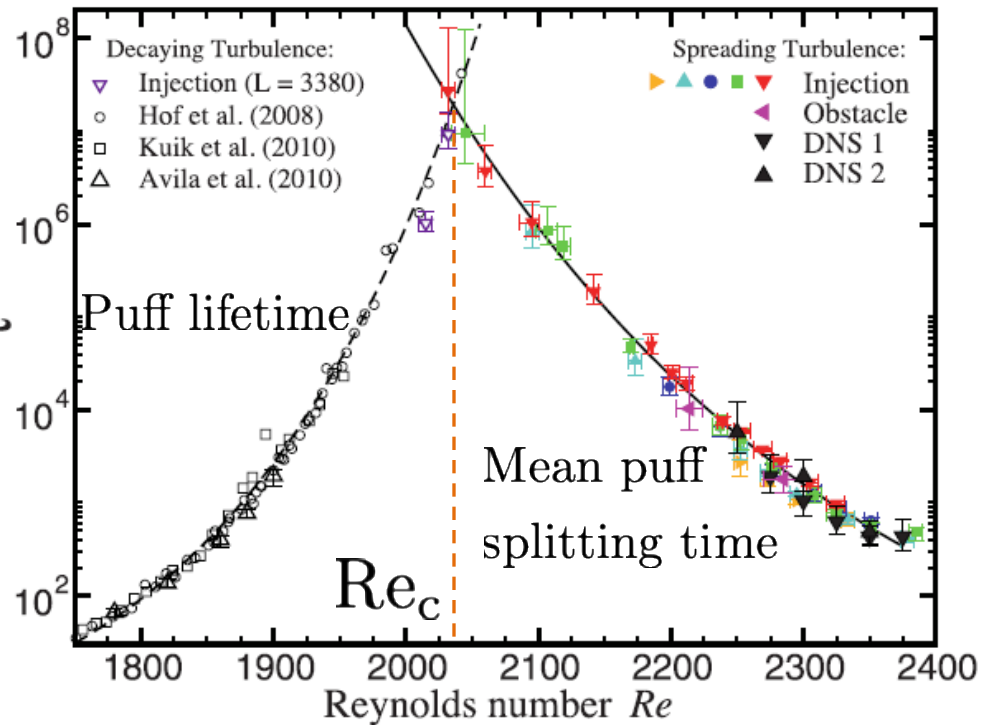


Incompressible Navier-Stokes

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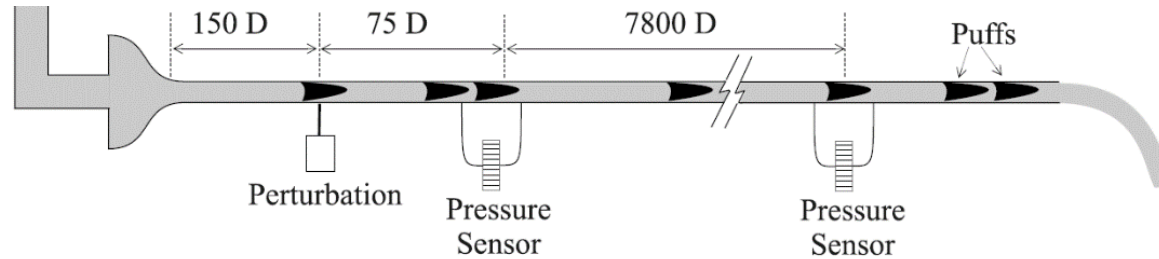


Stochastic predator-prey dynamics



Incompressible Navier-Stokes

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Vasudevan & Hof (2018)

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**laminar-turbulent
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critical behavior

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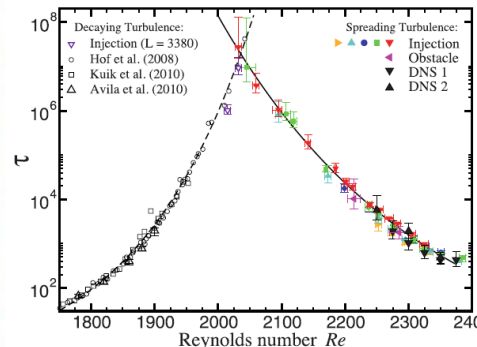
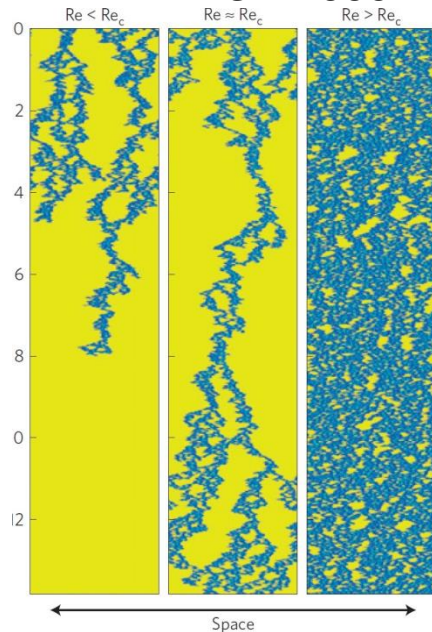
fluctuating
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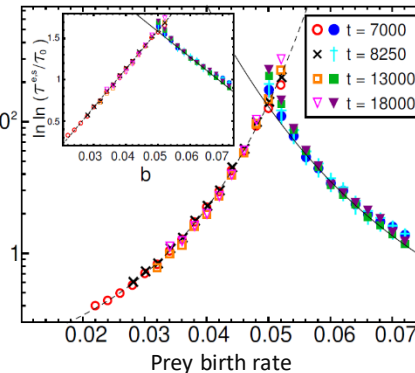
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**Directed percolation
(DP)**

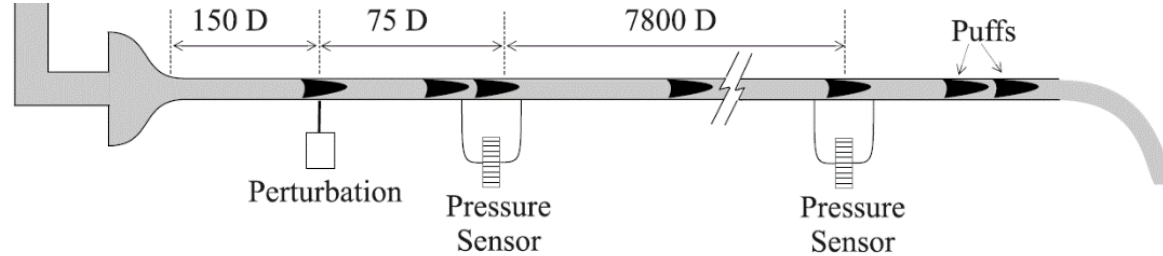


Avila et al., *Science* **333**, 192 (2011)
Song et al., *J. Stat. Mech.* 2014(2), P020010



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Nature Physics **12**, 245 (2016)

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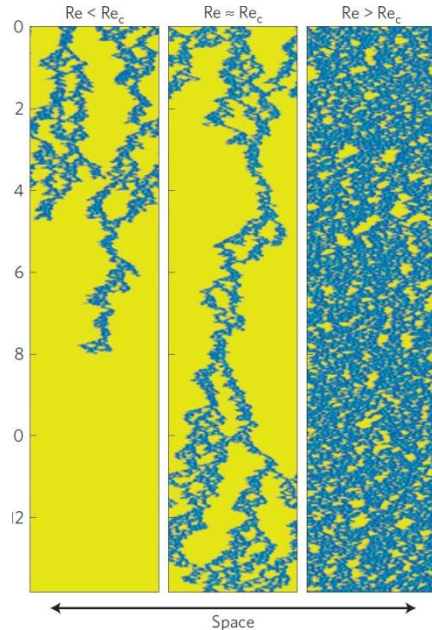
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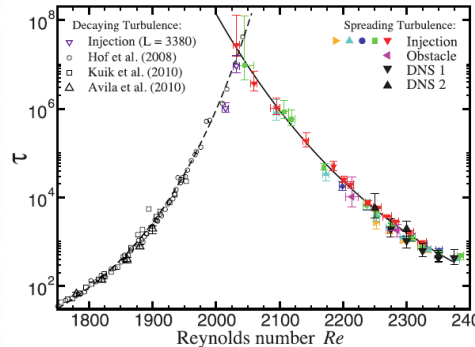
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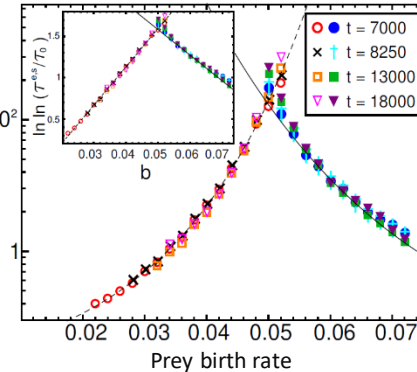
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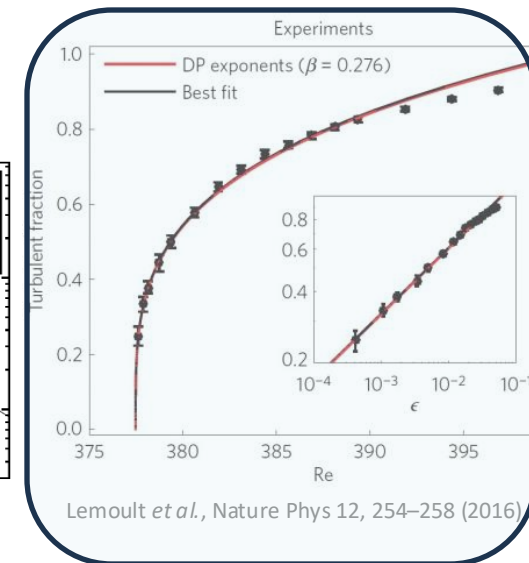


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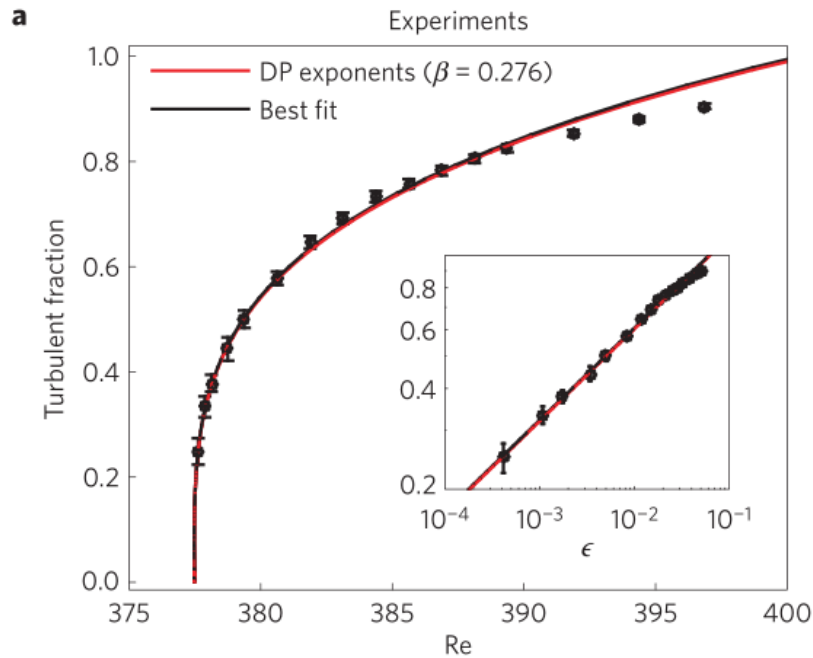
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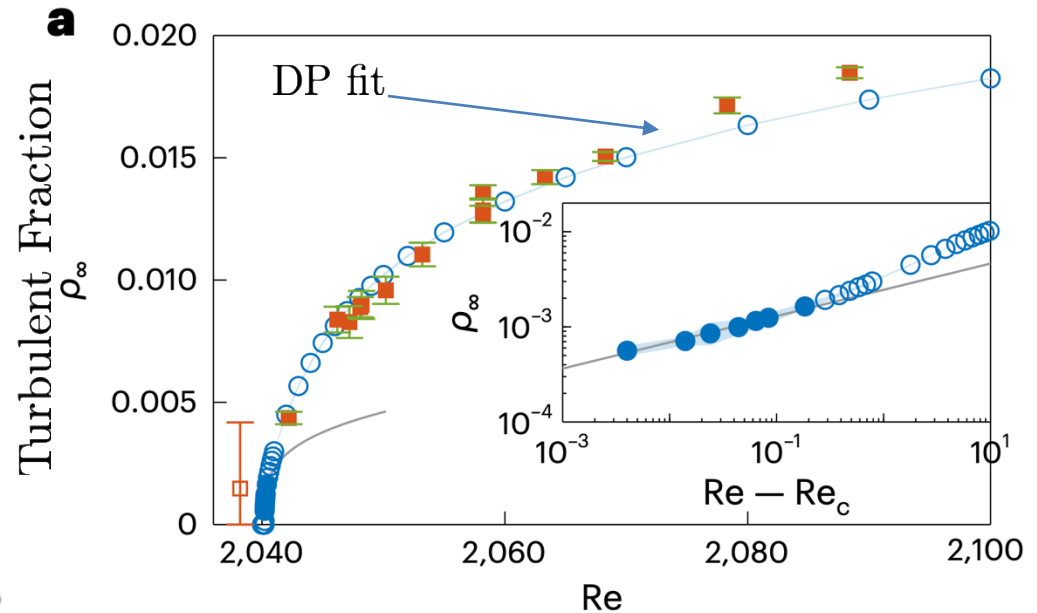
Scaling laws 28

Scaling laws match directed percolation



Lemoult et al., *Nature Physics* (2016)

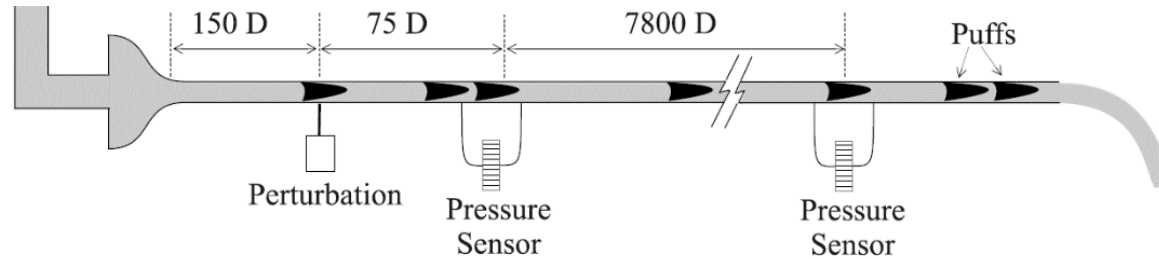
Couette flow



Lemoult et al., *Nature Physics* (2024)

Pipe flow

Phase diagram of pipe turbulence



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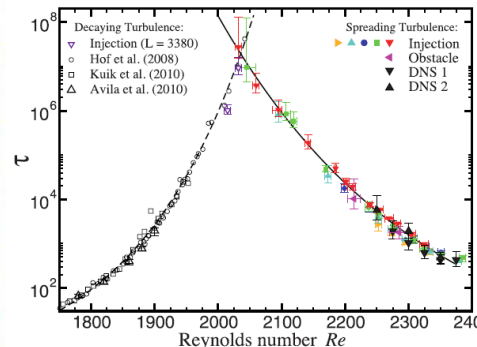
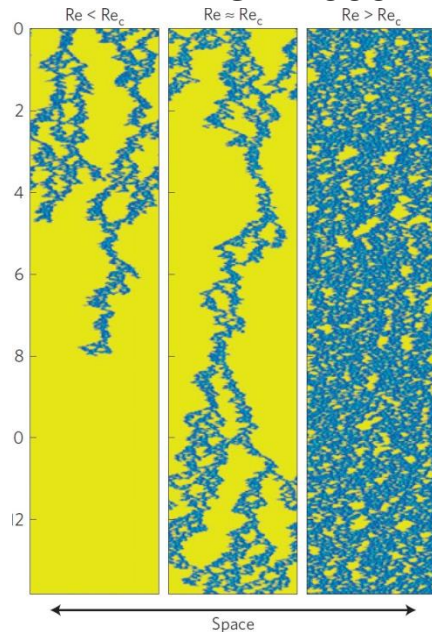


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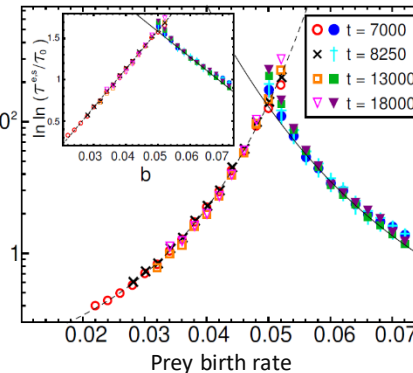
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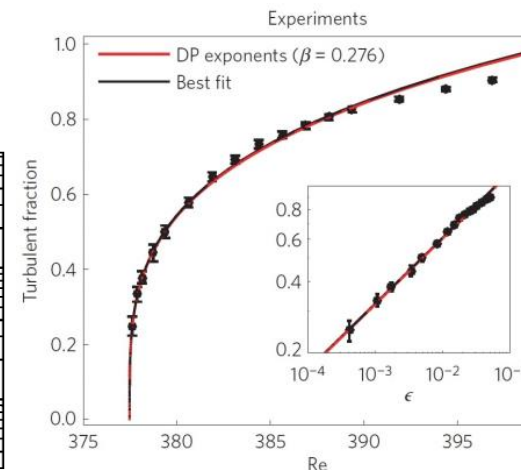
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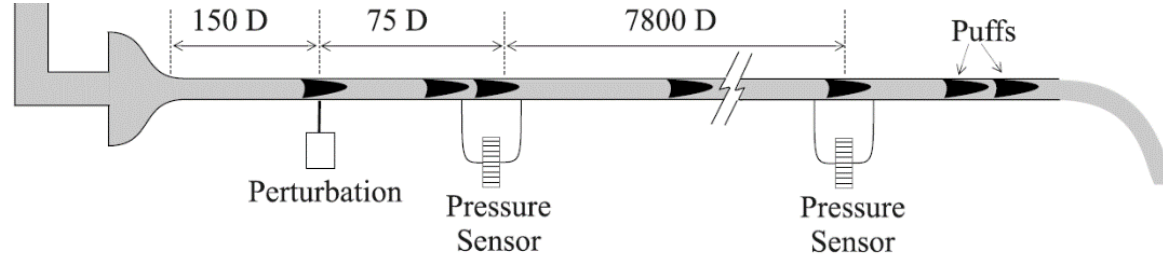


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Phase diagram of pipe turbulence



Vasudevan & Hof (2018)

laminar flow

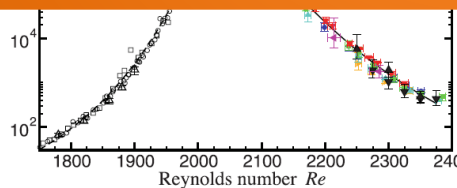
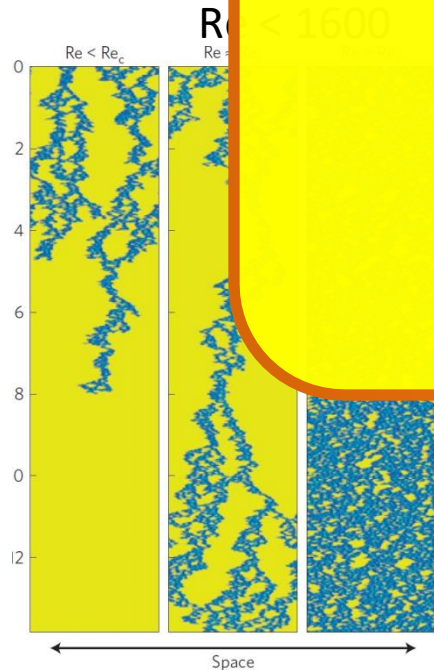
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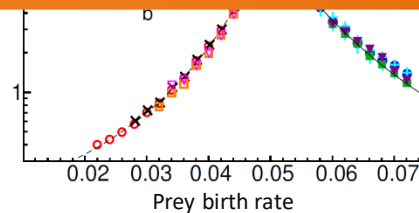
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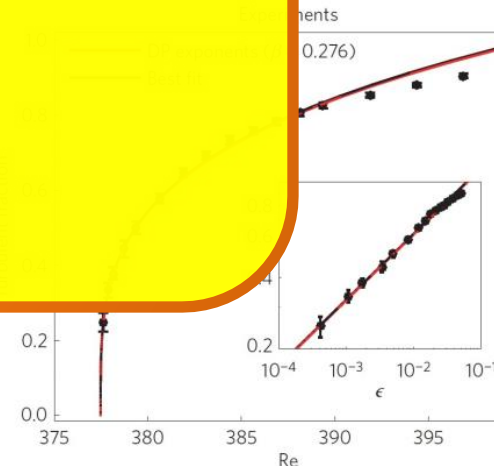
Physical mechanism?



Avila *et al.*, *Science* **333**, 192 (2011)
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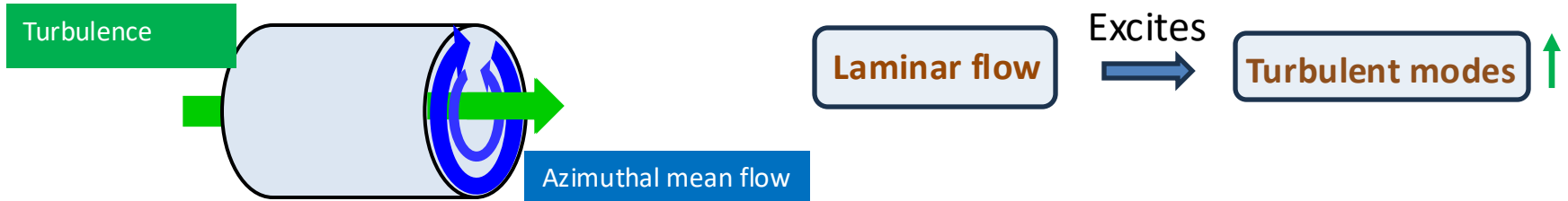
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Transitional flow has an emergent predator-prey dynamics

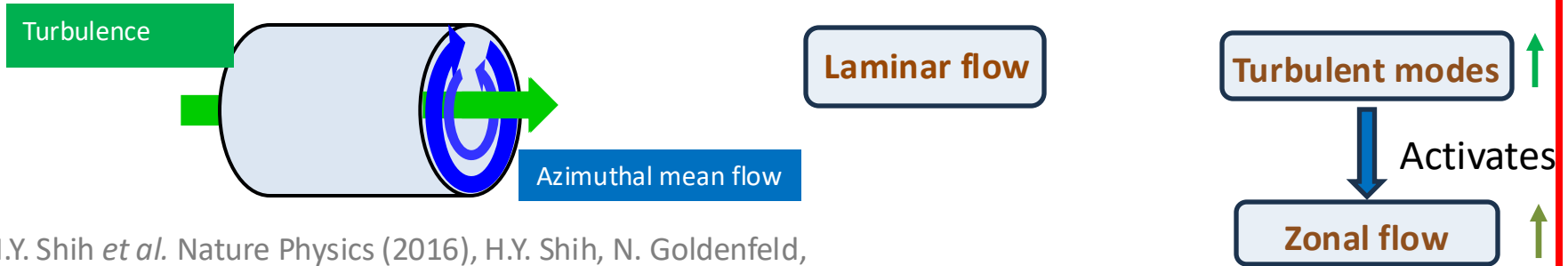
Activation-inhibition relation between turbulence and azimuthal mean flow



H.Y. Shih *et al.* Nature Physics (2016), H.Y. Shih, N. Goldenfeld, Journal of Statistical Physics (2017); Diamond *et al.* (1994).

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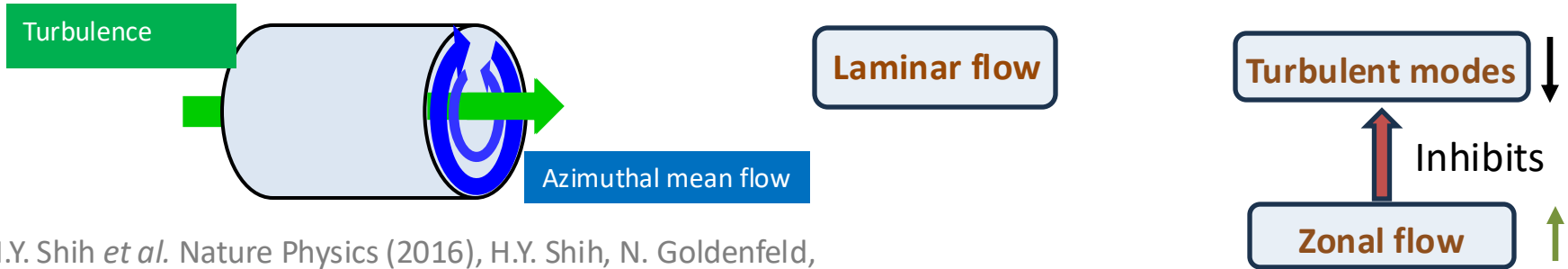
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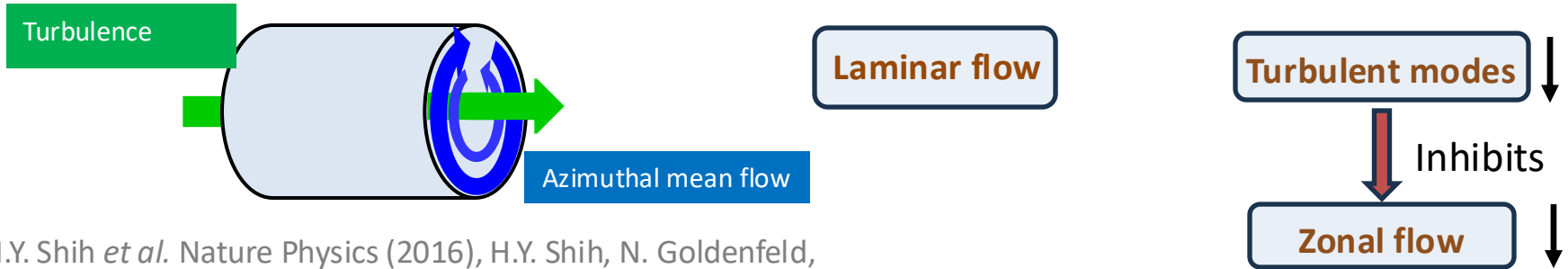
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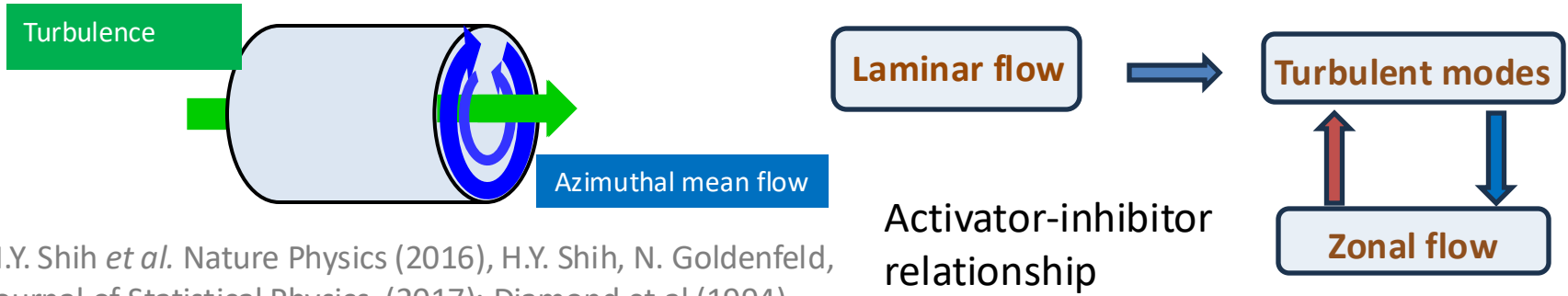
Activation-inhibition relation between turbulence and azimuthal mean flow



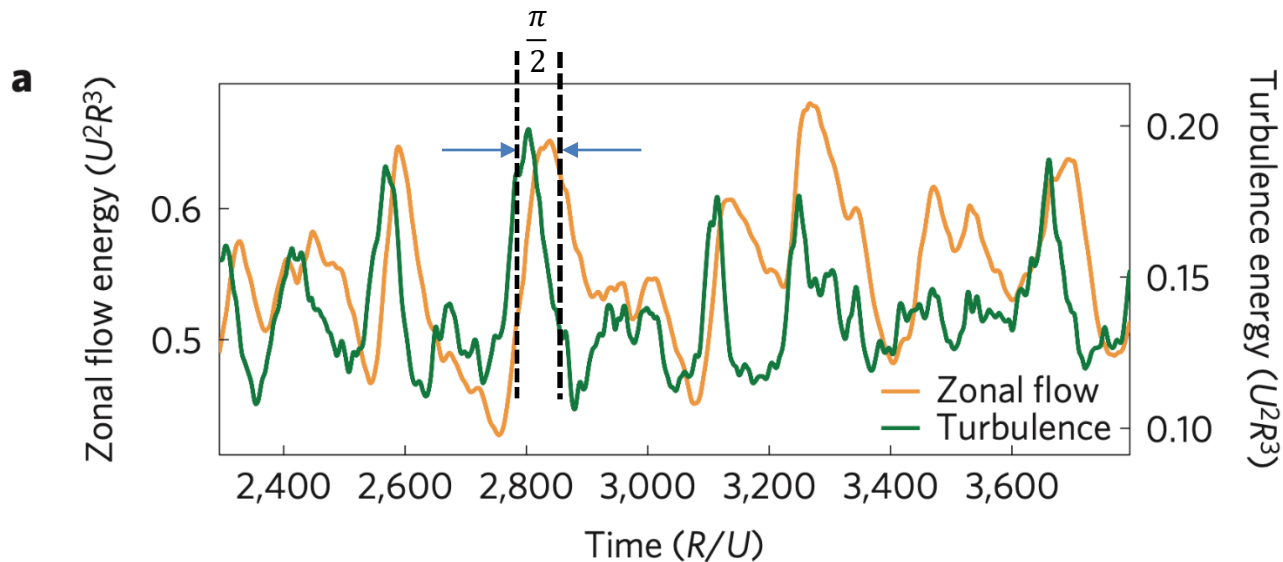
H.Y. Shih *et al.* Nature Physics (2016), H.Y. Shih, N. Goldenfeld, Journal of Statistical Physics (2017); Diamond *et al* (1994).

Transitional flow has an emergent predator-prey dynamics

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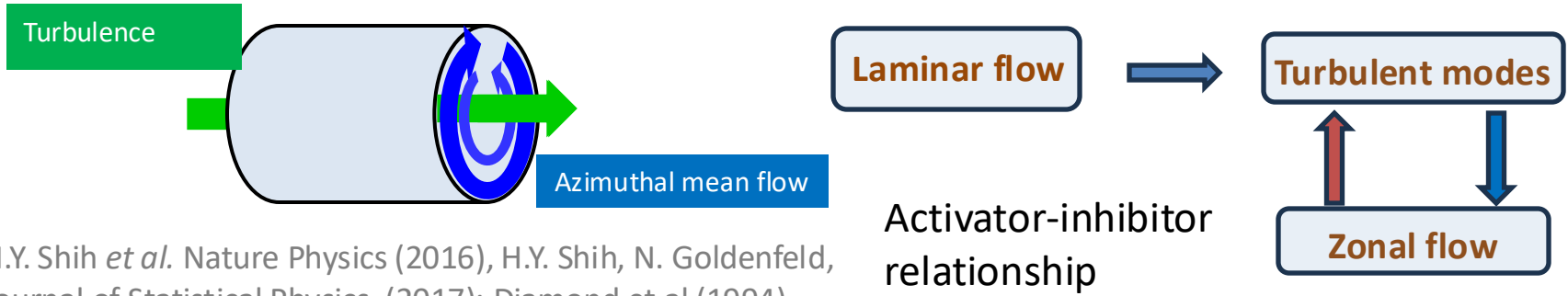


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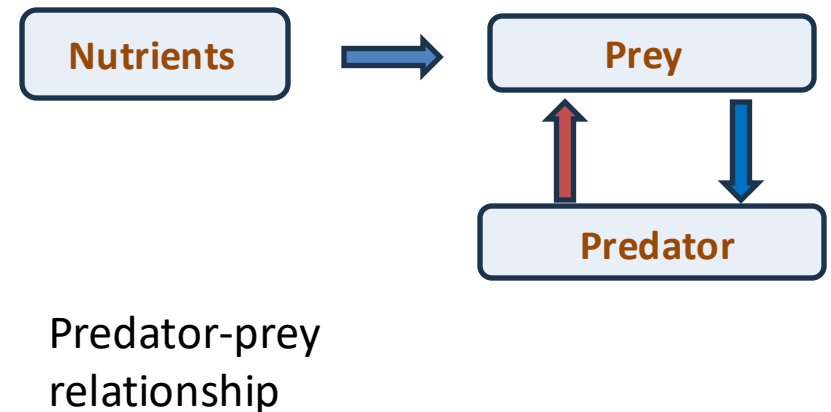
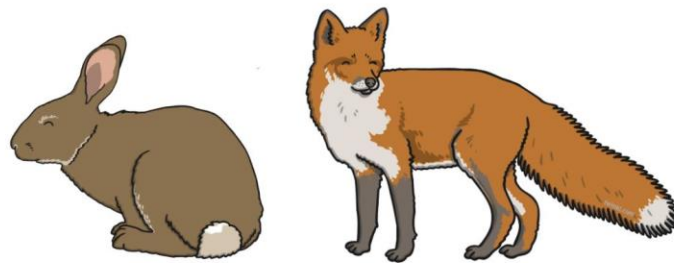


Transitional flow has an emergent predator-prey dynamics

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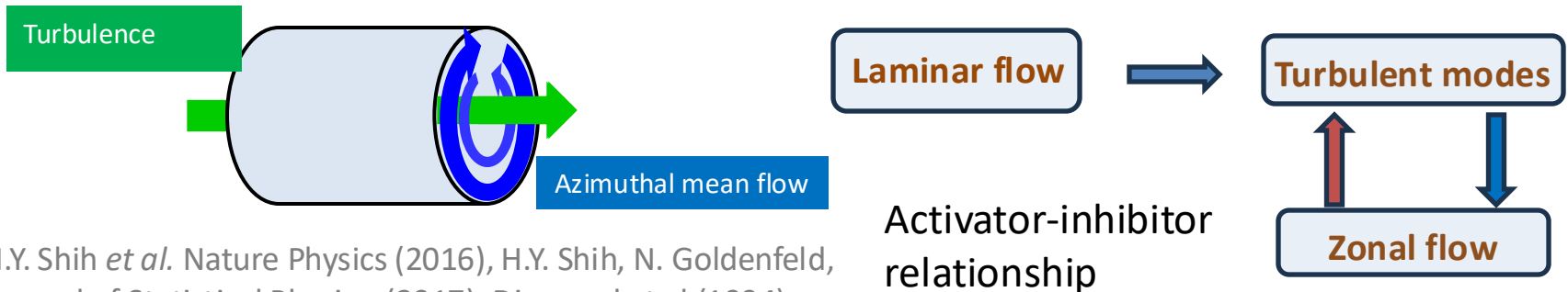


Stochastic dynamics of predator and prey in an ecosystem



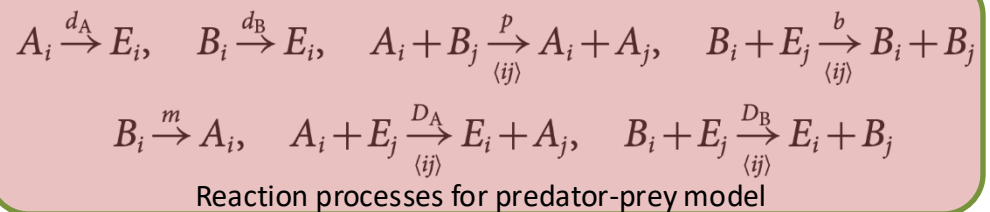
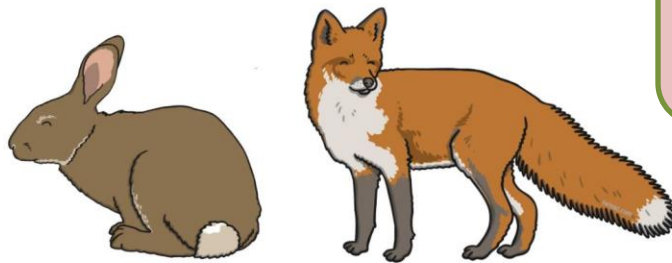
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Directed Percolation Universality Class

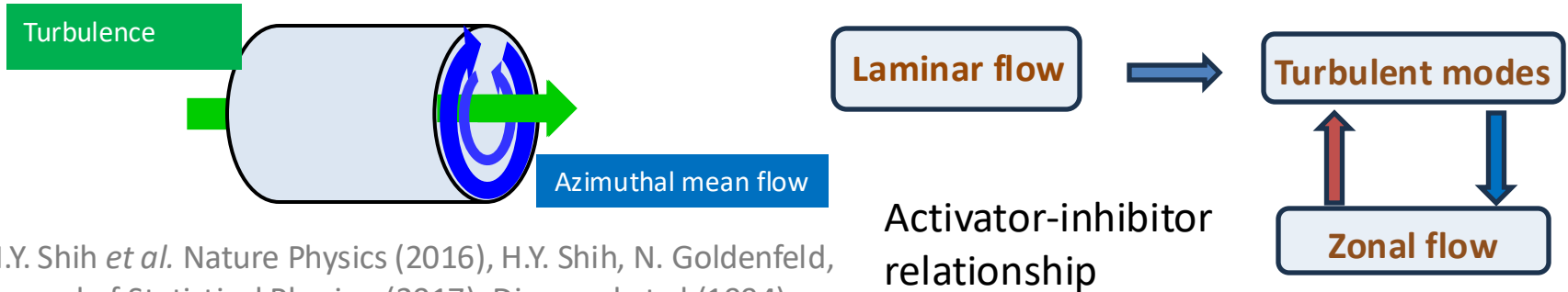
H.Y. Shih *et al.* Nature Physics (2016); Wang *et al.* PRL (2022);
G. Lemoult *et al.* Nature Physics (2024)

E. Kim and P. H. Diamond. Zonal flows and transient dynamics of the L-H transition. Phys. Rev. Lett. (2003).

Picture credit: <https://www.twinkl.com/teaching-wiki/prey>

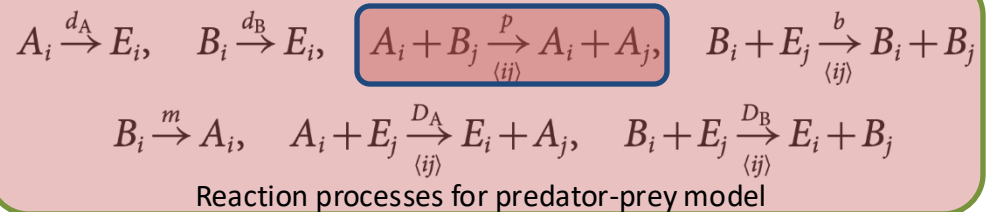
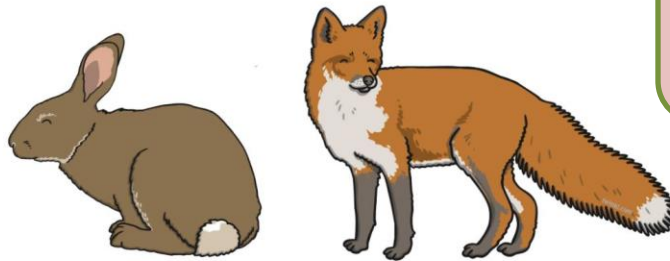
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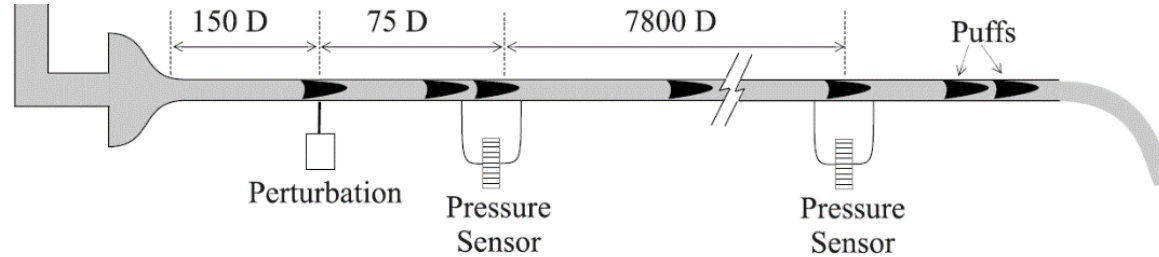
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Phase diagram of pipe turbulence



Vasudevan & Hof (2018)

laminar flow

steady
predictable

**laminar-turbulent
transition**

critical behavior

fully-developed turbulence

fluctuating
unpredictable

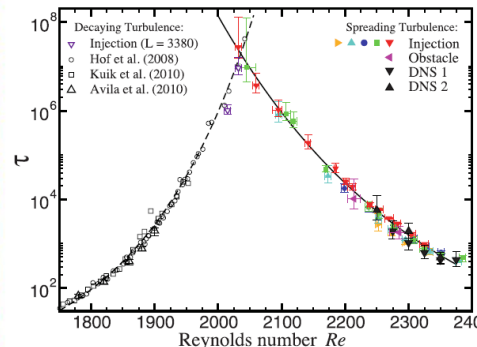
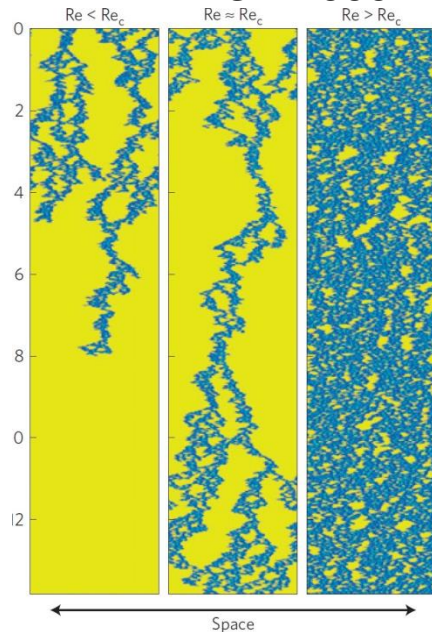


$Re \sim 2000$

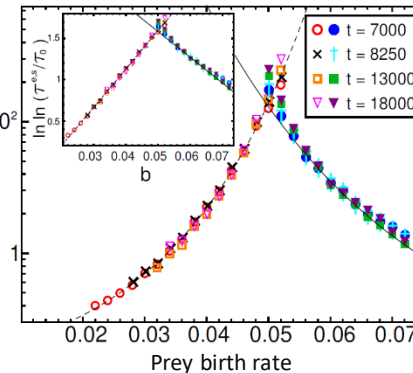
$Re < 1600$

$Re > 10^5$

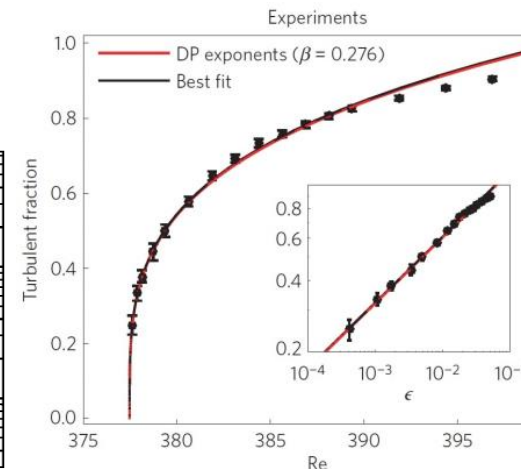
**Directed percolation
(DP)**



Avila *et al.*, *Science* **333**, 192 (2011)
Song *et al.*, *J. Stat. Mech.* 2014(2), P020010

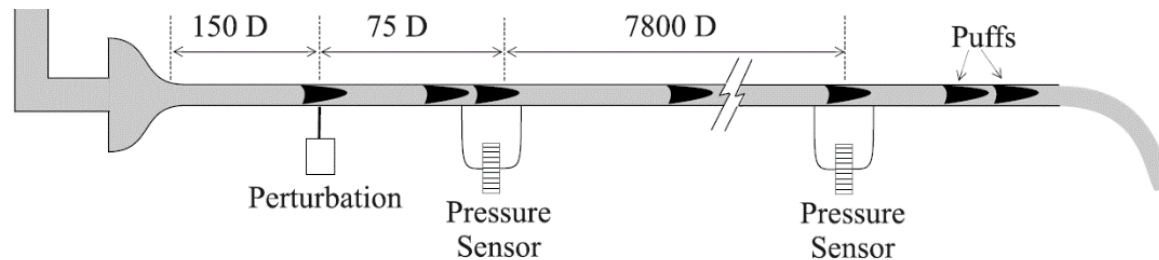


Shih, Hsieh and Goldenfeld
Nature Physics **12**, 245 (2016)



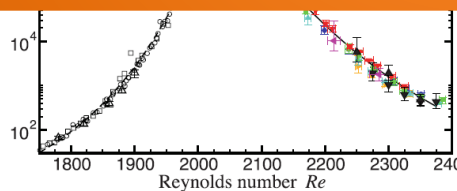
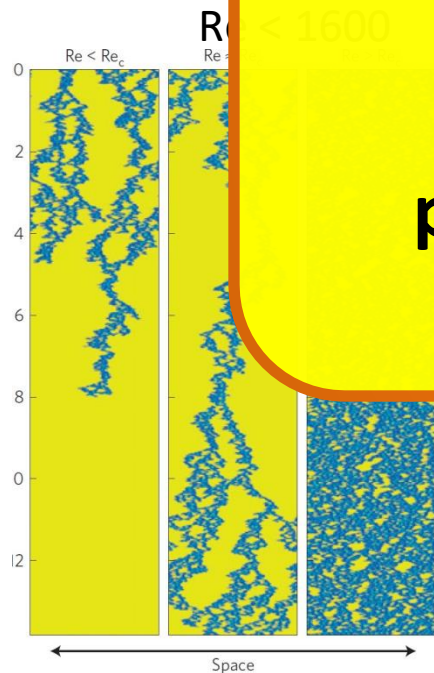
Lemoult *et al.*, *Nature Phys* **12**, 254–258 (2016).

Phase diagram of pipe turbulence

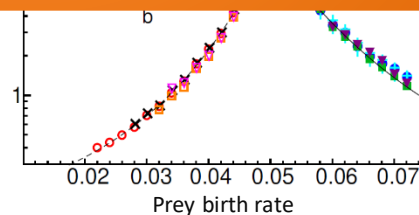


Wall bounded shear flows show critical scaling at transition.

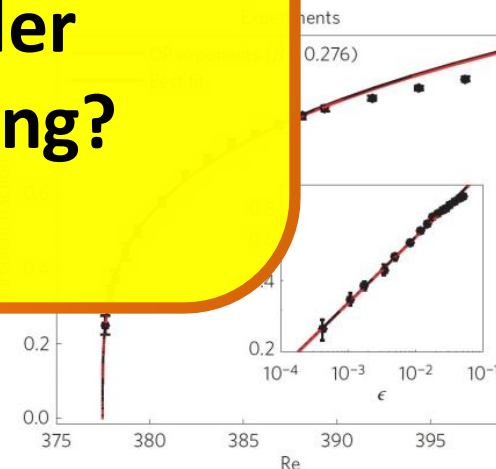
General? Valid for flows under perturbation or external forcing?



Avila *et al.*, *Science* **333**, 192 (2011)
Song *et al.*, *J. Stat. Mech.* 2014(2), P020010



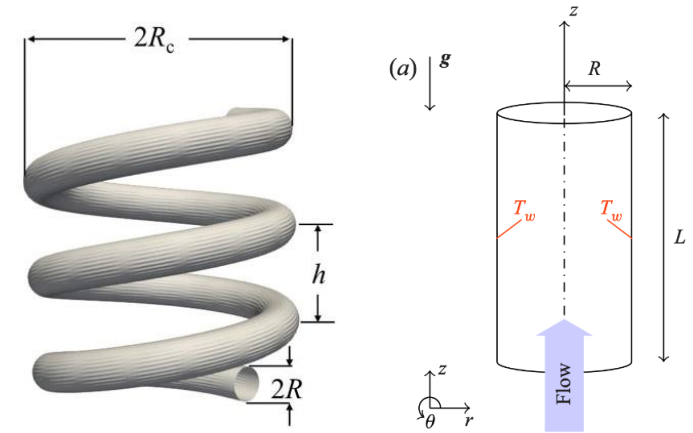
Shih, Hsieh and Goldenfeld
Nature Physics **12**, 245 (2016)



Lemoult *et al.*, *Nature Phys* **12**, 254–258 (2016).

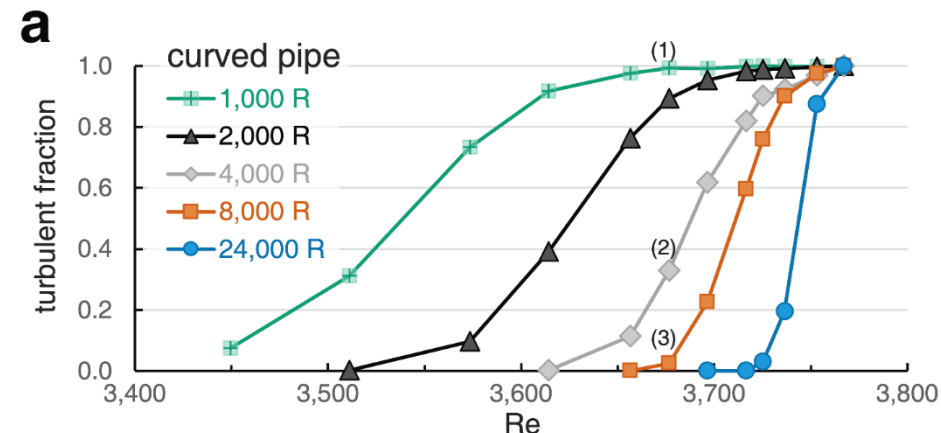
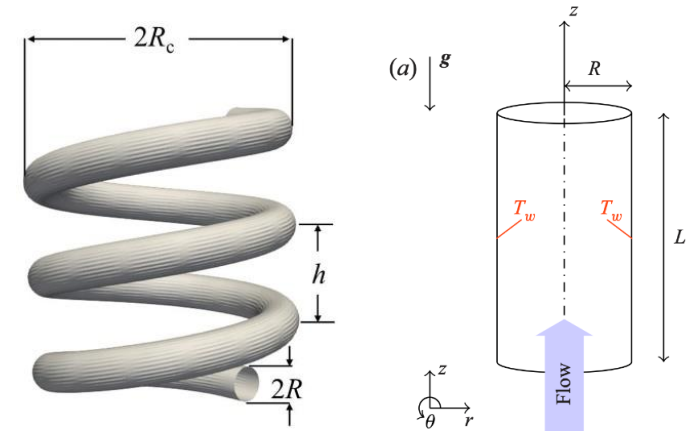
Body force perturbed flows

- Zhuang *et al.* investigated transition dependence under centrifugal force (curved pipes) and buoyancy (heated pipes).
- Setup:
 - Inlet turbulent fluid, measure turbulent fraction (T.F.) at different distances along the pipe.
 - Tune Re and repeat measurements.
 - Plot result by fixing x , and looking at T.F. vs Re



Body force perturbed flows

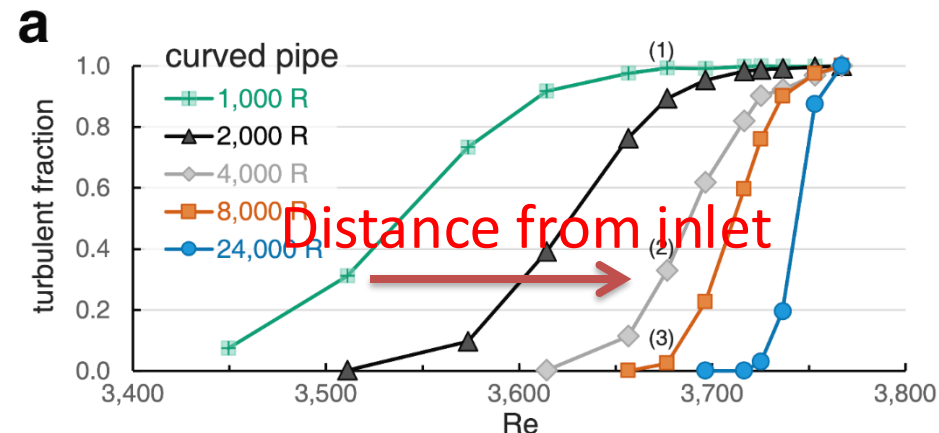
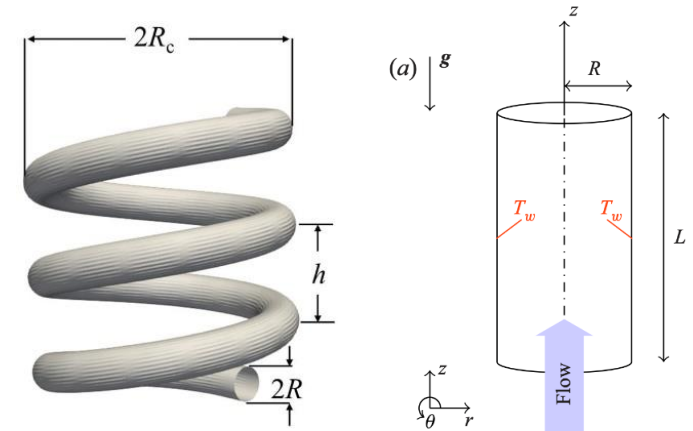
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Re dependence of the turbulent fraction (TF). Curves show turbulent fractions measured at different downstream locations.

Body force perturbed flows

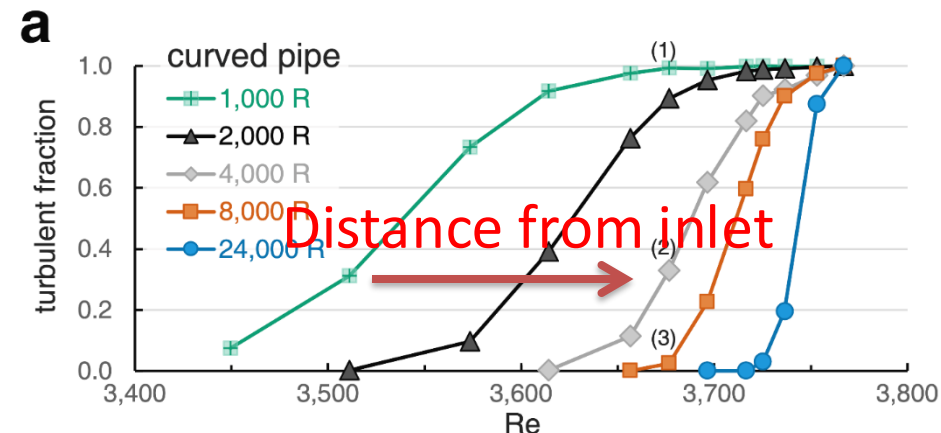
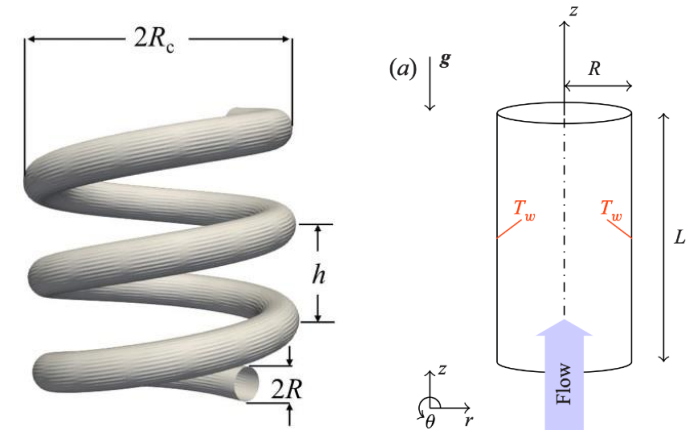
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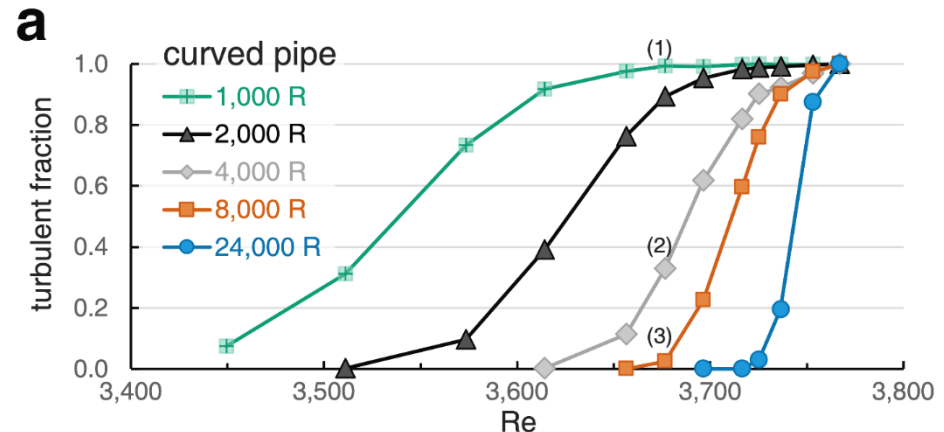
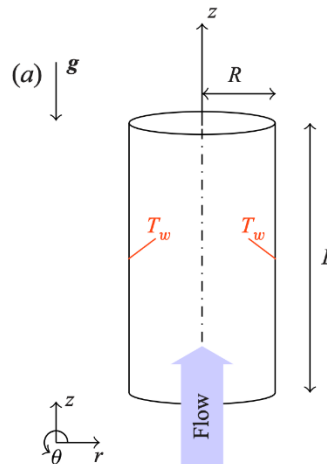
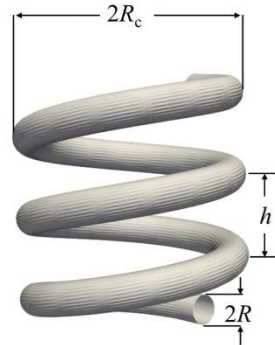


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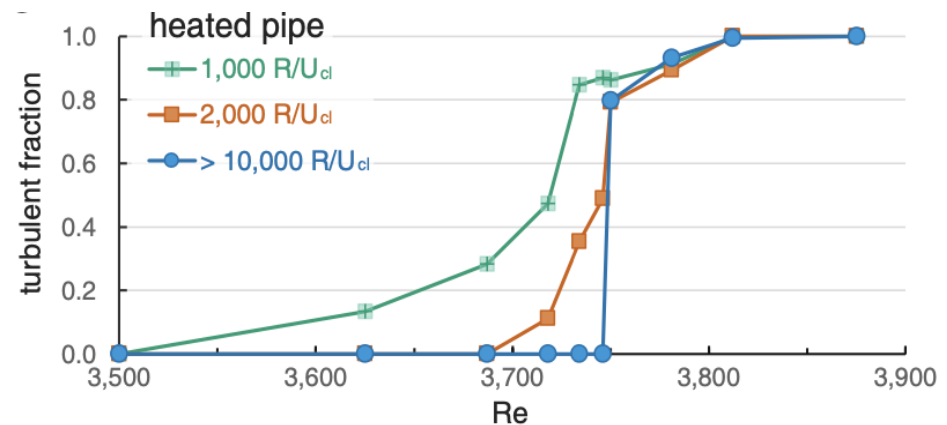
- Do body forces change the transition qualitatively?

Body forces can turn transition discontinuous

- How do body forces delay and sharpen the DP transition?
- How does the crossover from the continuous transition occur?
- Still a phase transition or just subcriticality?



Re dependence of the turbulent fraction (TF). Curves show turbulent fractions measured at different downstream locations.

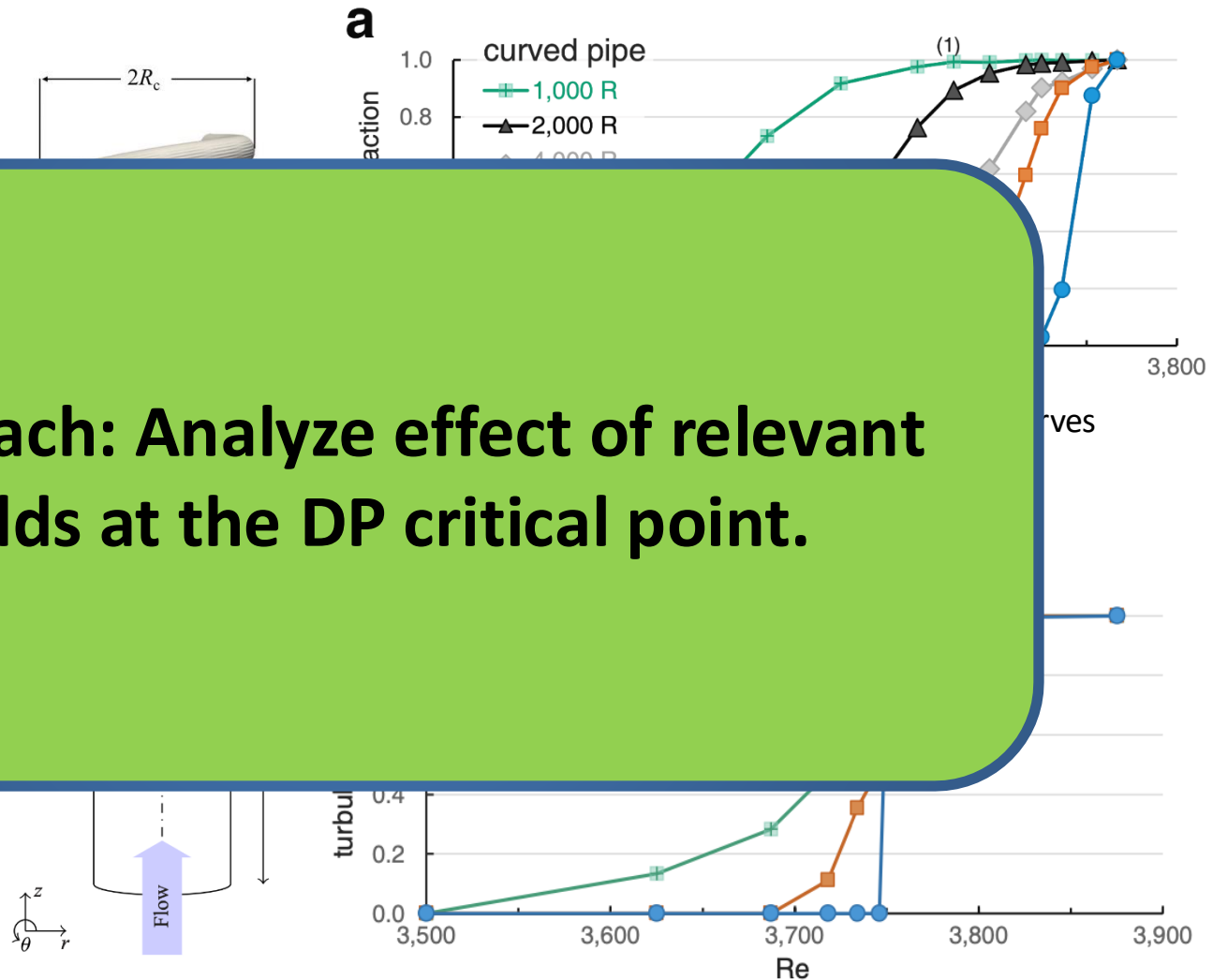


DNS data on heated flows. Discontinuity occurs for plot of turbulent fraction vs Re at long times

Body forces can turn transition discontinuous

- How do body forces delay and sharpen transition?
- How do cross-sectional fields at the DP critical point affect transition?
- Still a transition or just subcriticality?

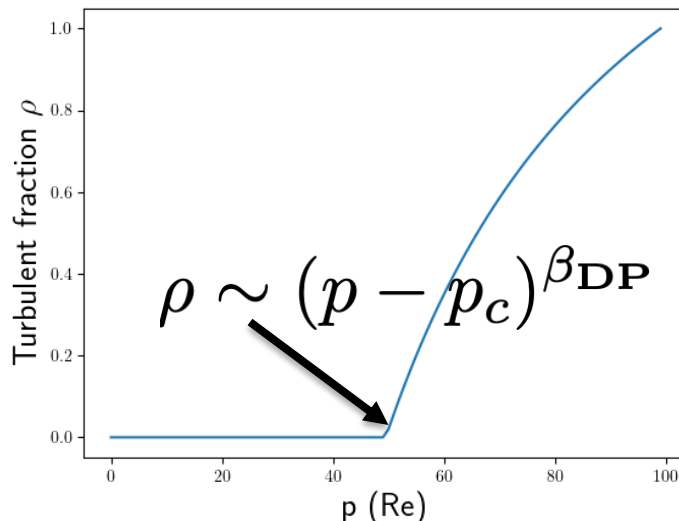
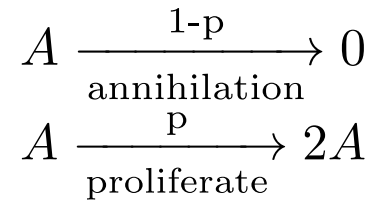
Approach: Analyze effect of relevant fields at the DP critical point.



DP mean field theory

- Body forces would likely cause flow instabilities, so we look at DP mean field theory and add higher order terms

$$\partial_t \rho = (2p - 1)\rho - p \cdot \rho^2$$



$A \Rightarrow$ Turbulence

$0 \Rightarrow$ Laminar

$\rho \Rightarrow$ Turbulent fraction in fluid

$p \Rightarrow$ Re

Body forces captured via higher order terms

- Body forces would likely cause flow instabilities, so we look at DP mean field theory and add higher order terms

$$\partial_t \rho = (2p - 1)\rho - p \cdot \rho^2 + \boxed{a(p)\rho^2 + b(p)\rho^3}$$

$A \xrightarrow{1-p} 0$
 annihilation
 $A \xrightarrow{p} 2A$
 proliferate
 $2A \rightarrow 3A$

- Physical consistency ($\rho = 1$ when $p=1$ i.e. turbulent state at large Re)

$$(2p - 1)\rho^2 - p\rho^2 + a\rho^2 + b\rho^3 \Big|_{p=1, \rho=1} = 0$$

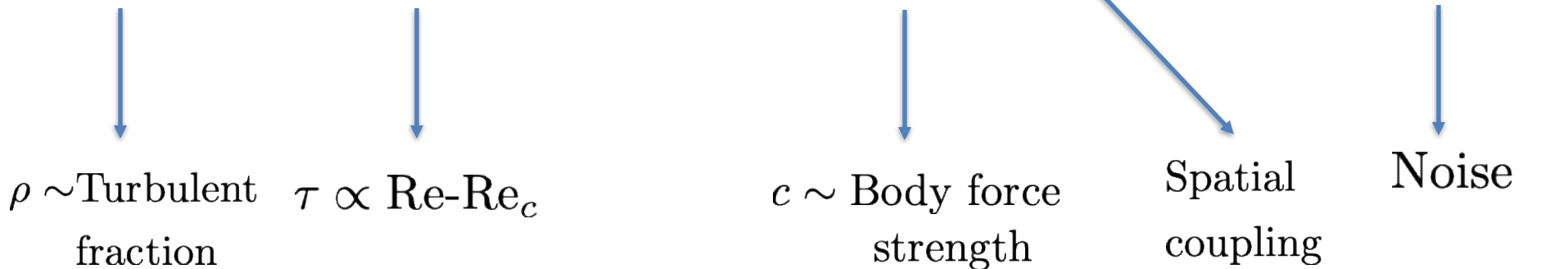
$$\implies \boxed{a + b = 0}$$

One effective Parameter!

Scaling reveals higher order terms are important

- Complete stochastic field theory description

$$\partial_t \rho(\vec{x}, t) = \tau \rho(\vec{x}, t) - g \rho(\vec{x}, t)^2 - \boxed{c \rho^3(\vec{x}, t)} + \Gamma \nabla^2 \rho(\vec{x}, t) + \eta(\mathbf{x}, t)$$

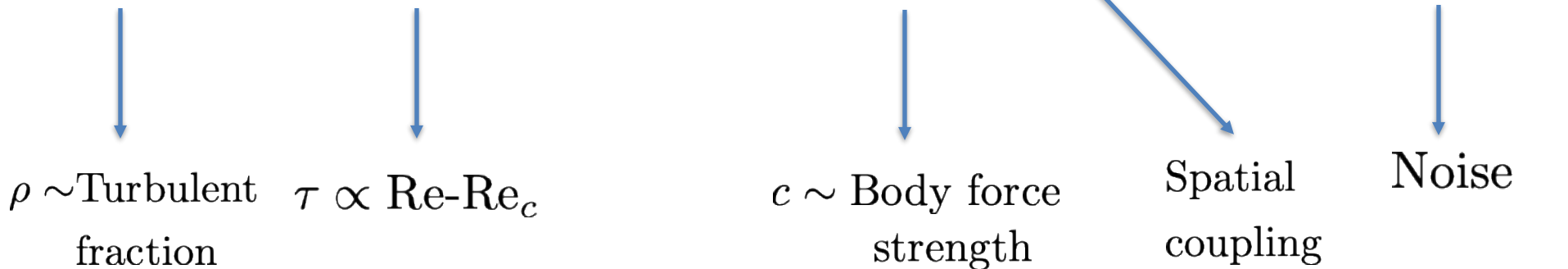


$$\langle \eta_a \eta_b \rangle = k \rho(a) \delta_{a,b}$$

Scaling reveals higher order terms are important

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Scaling Analysis

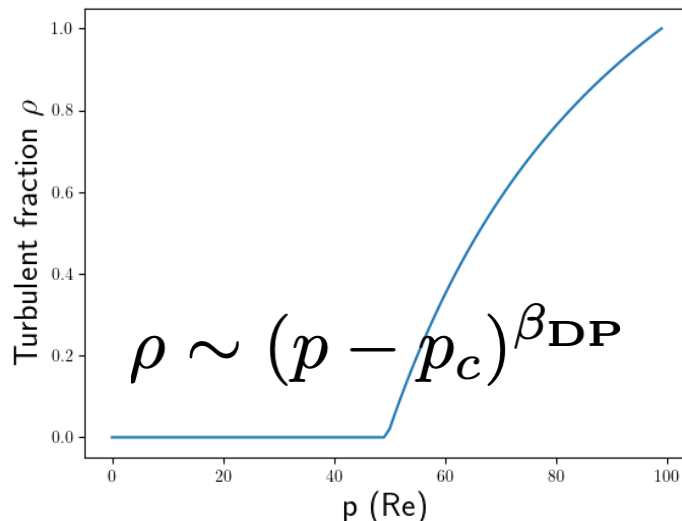
Tricritical Directed Percolation

Higher order terms allow for discontinuity

$$\frac{\partial \rho}{\partial t} = \delta p \rho - p \rho^2 + a \rho^2 (1 - \rho) = -\frac{\partial V}{\partial \rho}$$

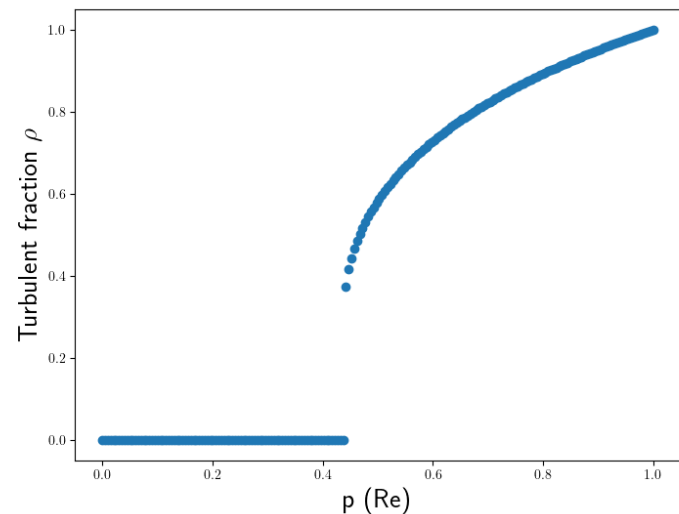
$$a < 1$$

V has only one minima; transition continuously from $\rho = 0$ to $\rho \neq 0$.



$$a > 1$$

V develops another minima; transition discontinuously from $\rho = 0$ to $\rho \neq 0$.

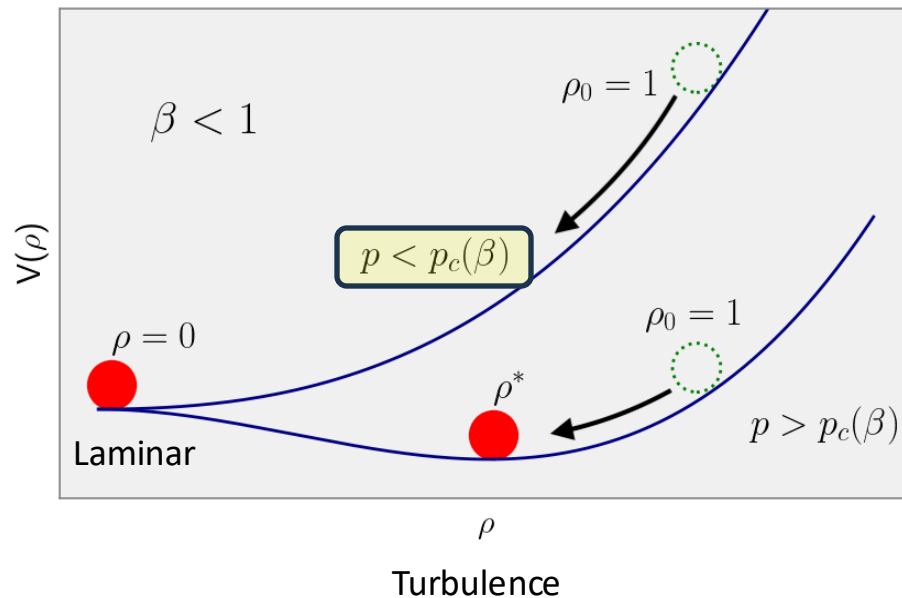


Nature of transitions

- Can be seen within Lyapunov potential

$$\partial_t \rho = \delta p \rho - p \rho^2 + \beta \rho^2 (1 - \rho) \longrightarrow$$

$$\frac{\partial \rho}{\partial t} = - \frac{\partial V}{\partial \rho}$$

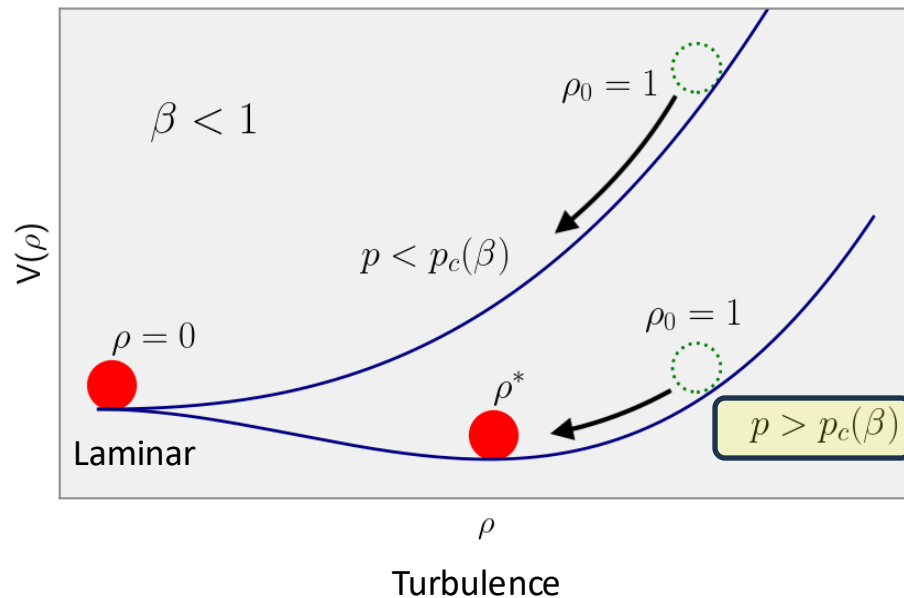


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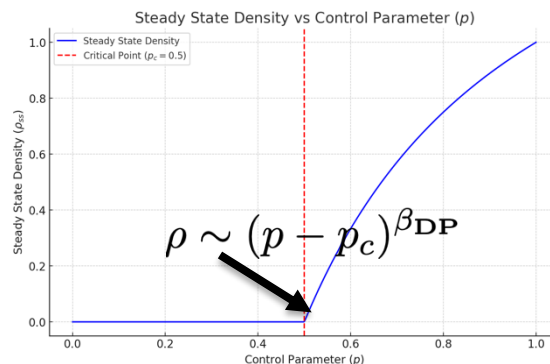
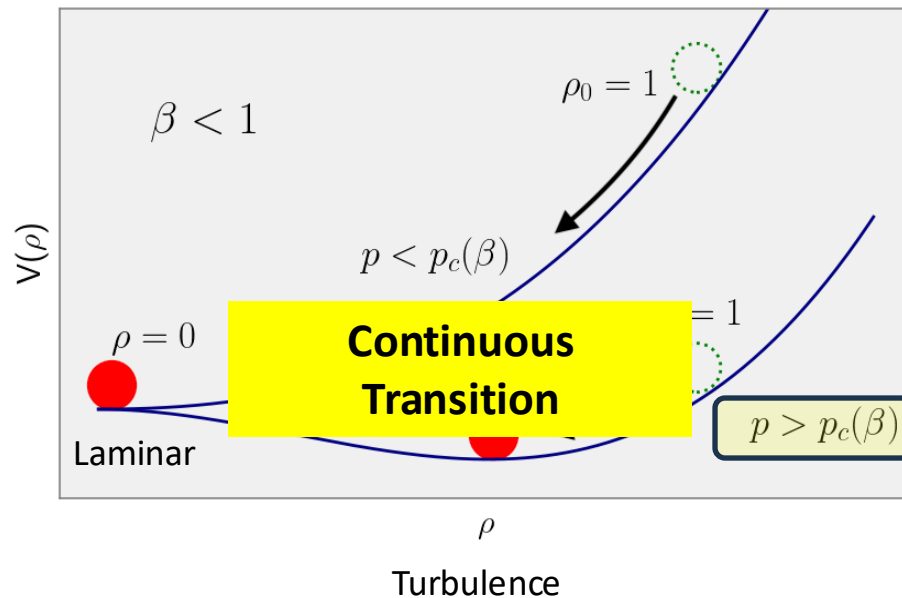


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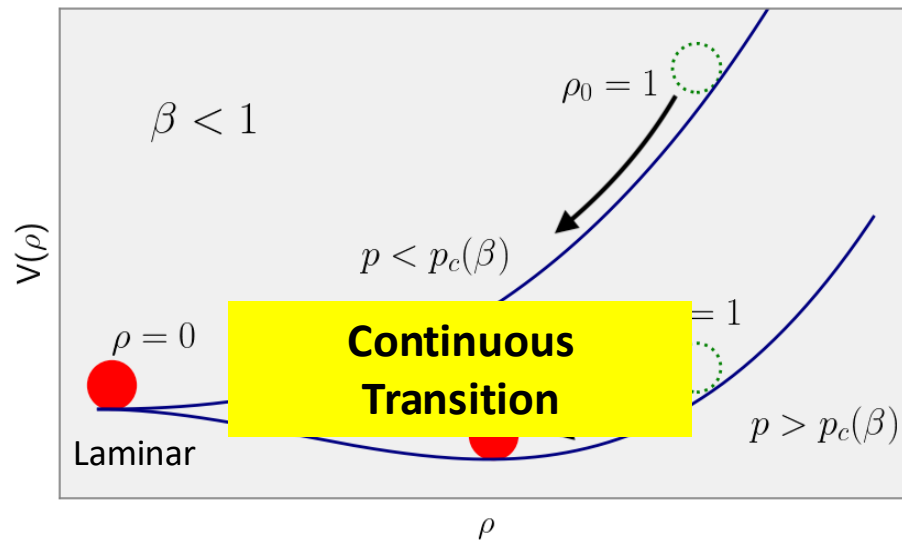


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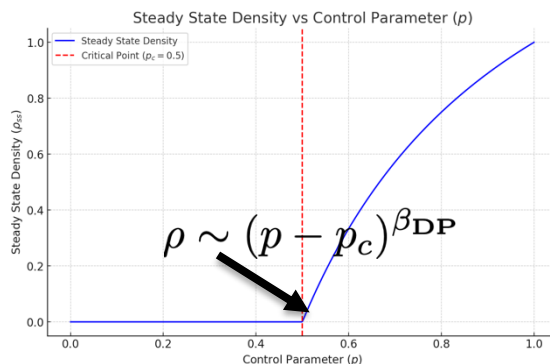
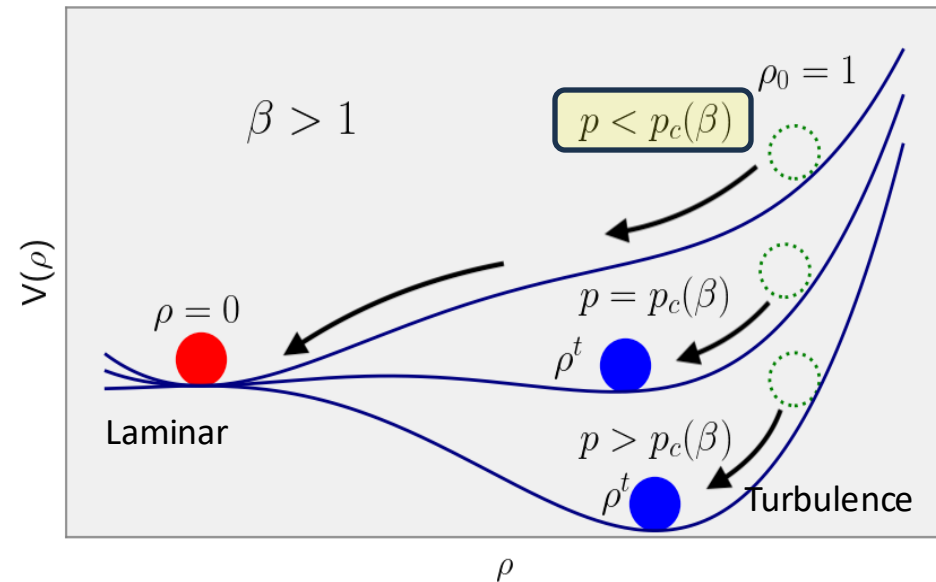
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Turbulence

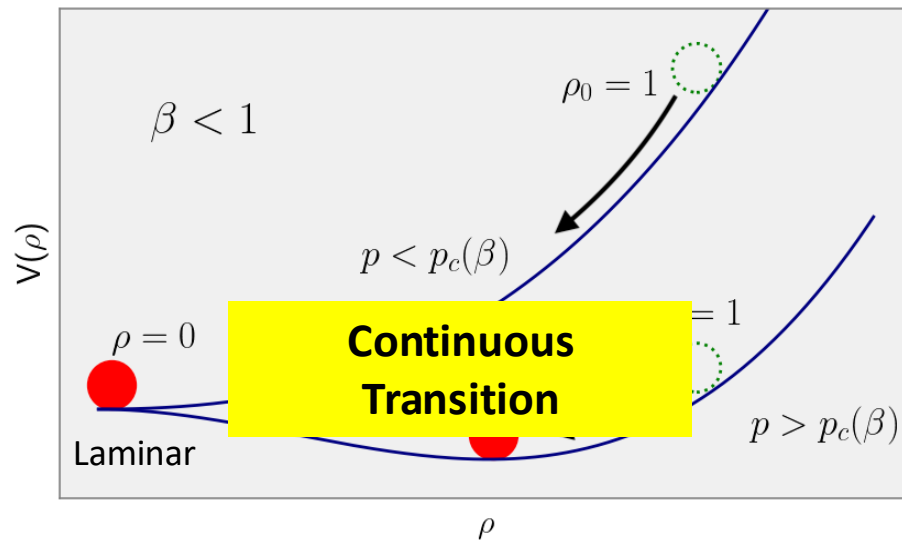


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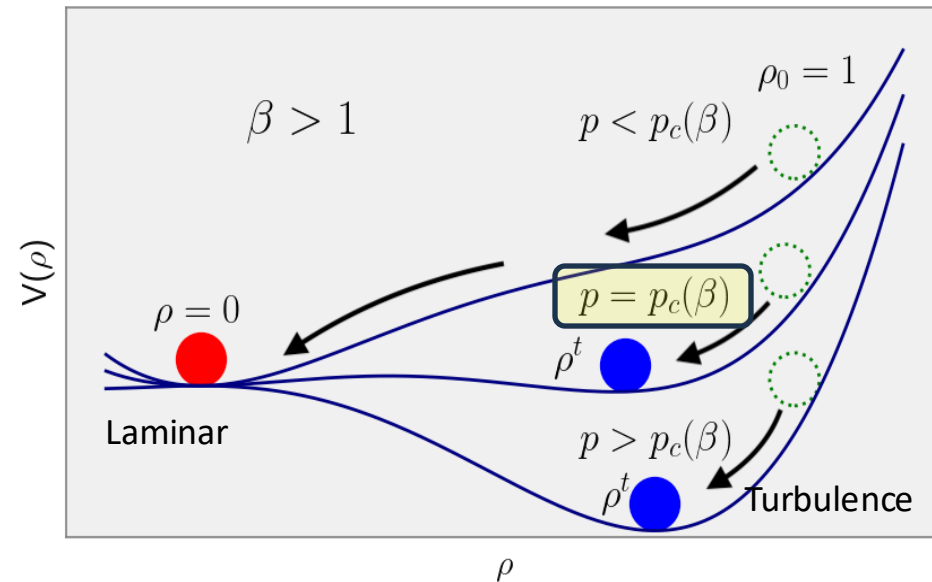
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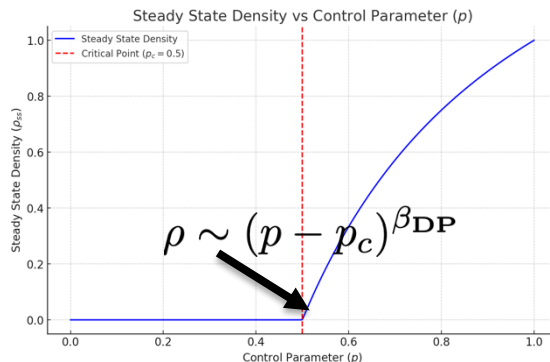
$$\frac{\partial \rho}{\partial t} = - \frac{\partial V}{\partial \rho}$$



Turbulence



Turbulence is Metastable

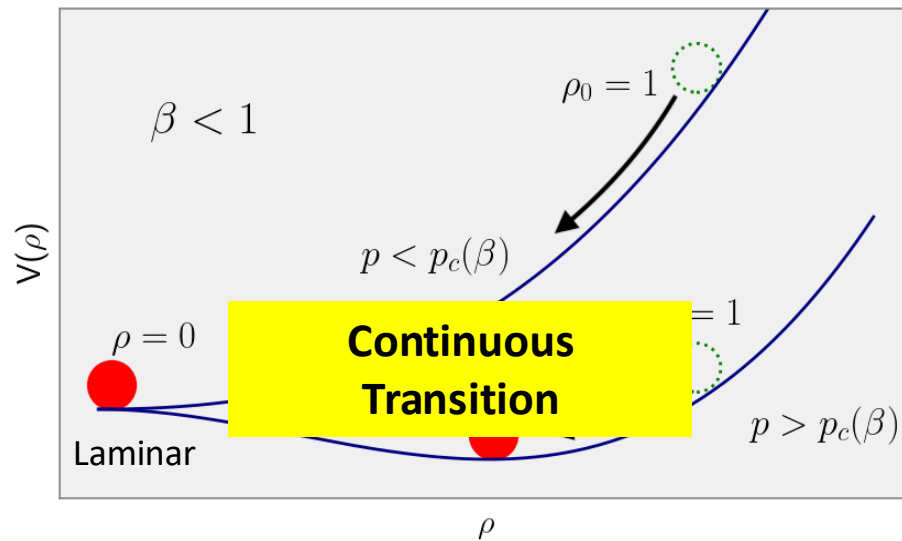


Nature of transitions

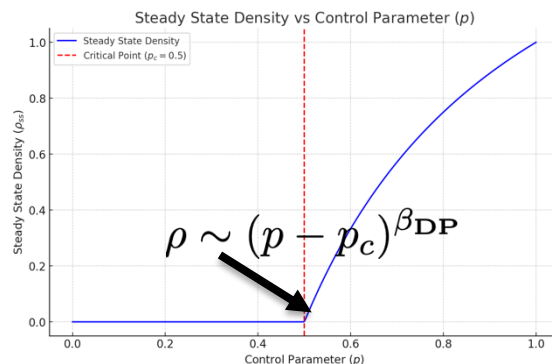
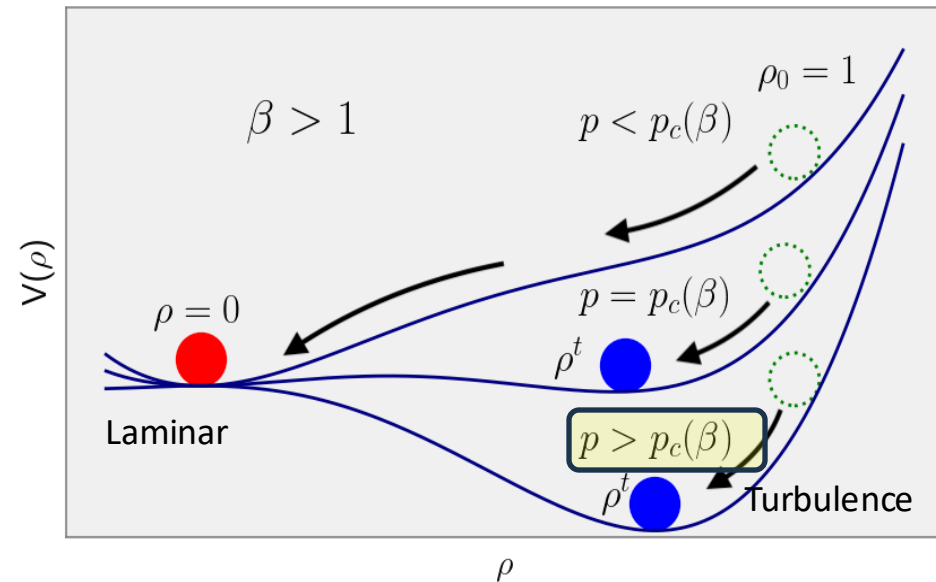
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Turbulence

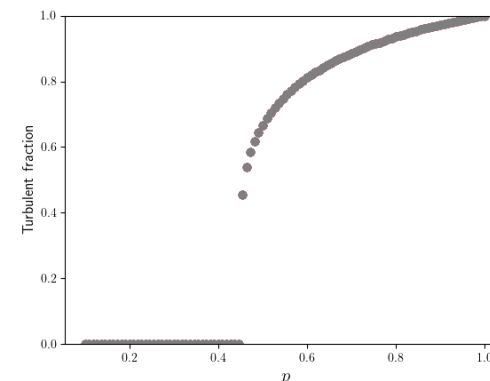
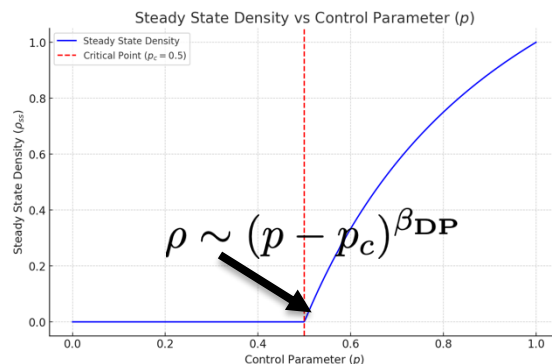
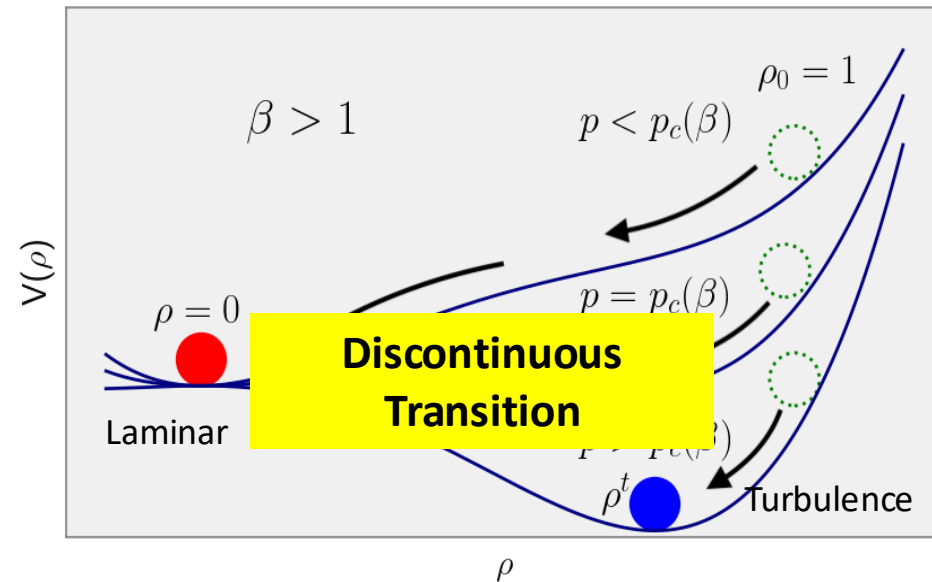
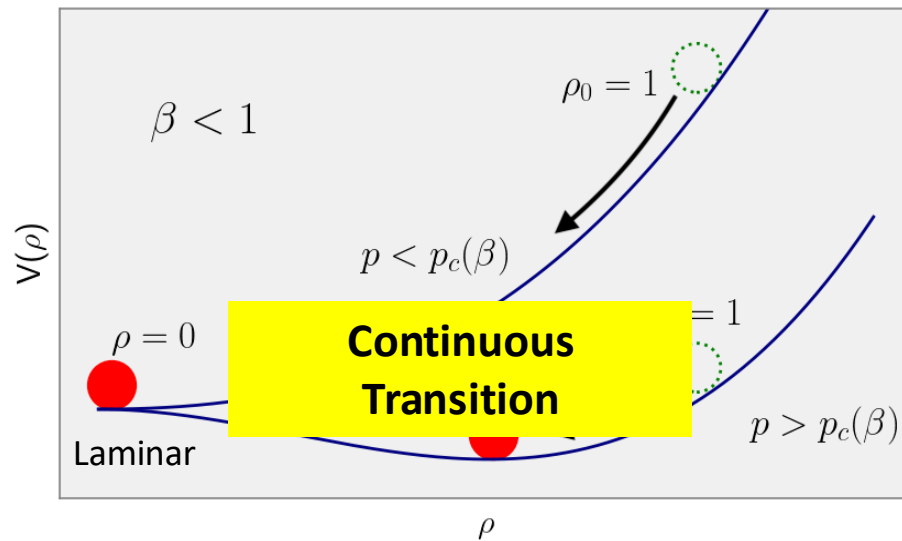


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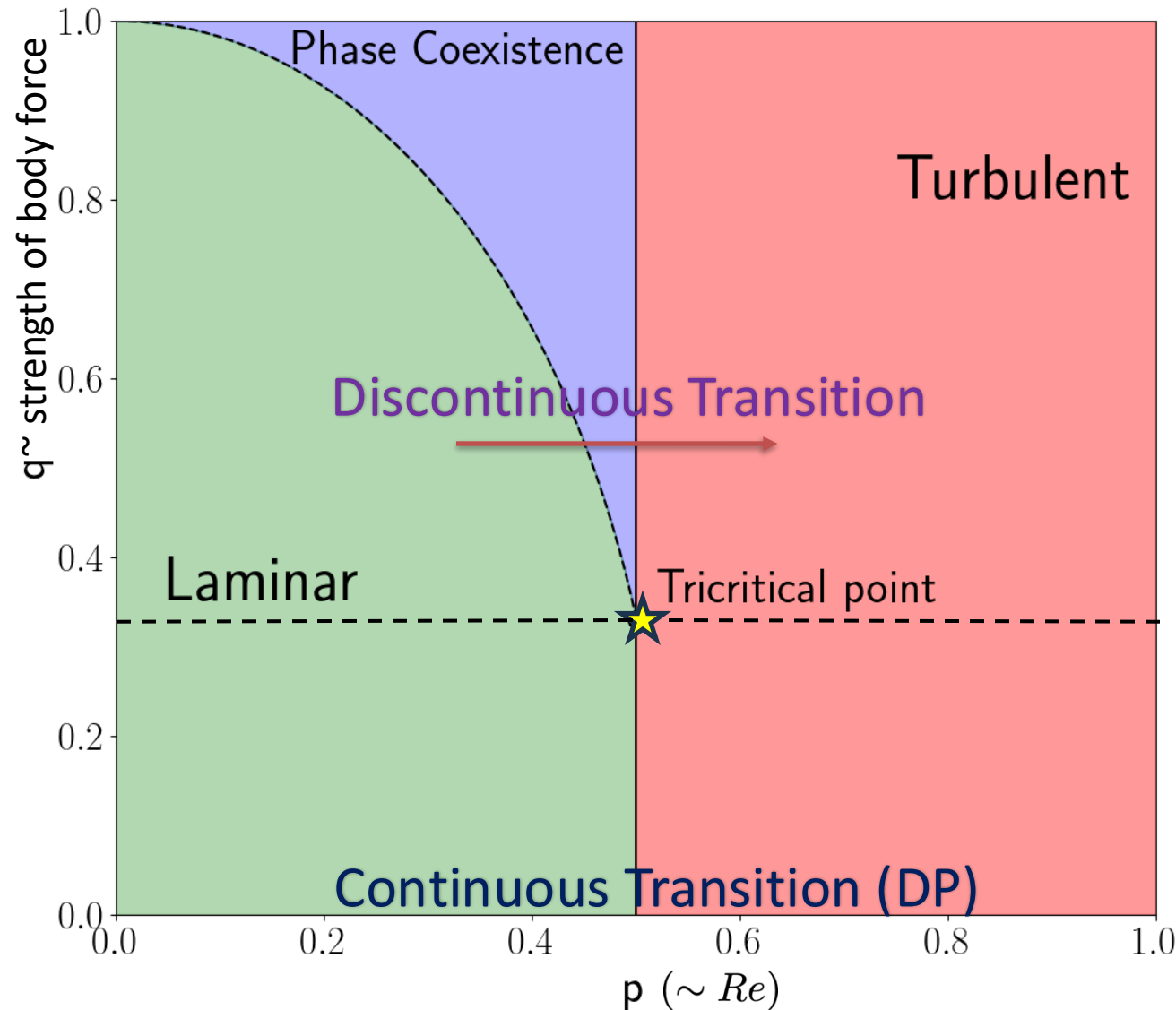
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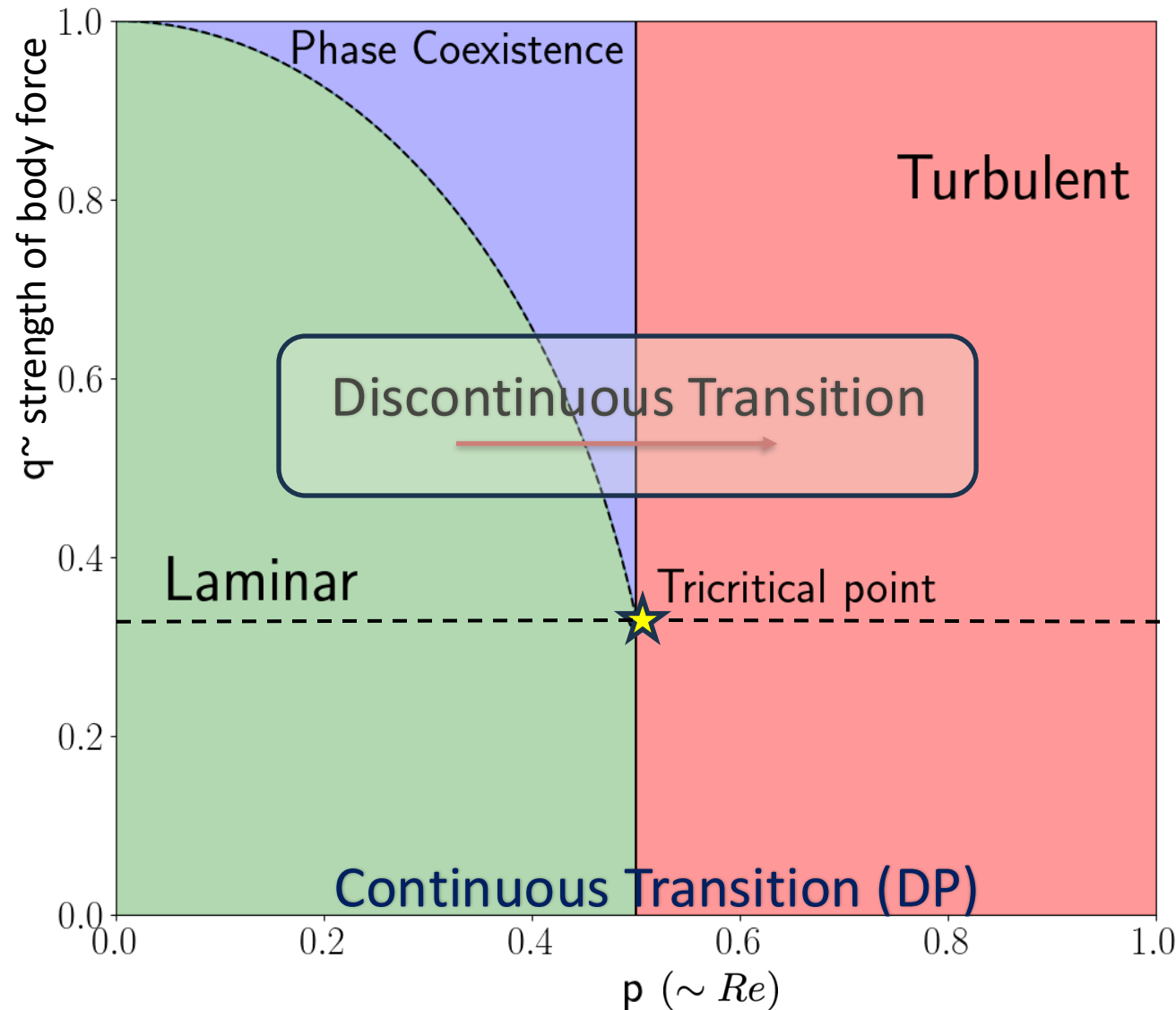
Prediction: Phase diagram

- $q < q_c$: Line of continuous DP transitions
- $q > q_c$: Line of discontinuous transitions
- Continuous and discontinuous lines join at Tricritical point.



Prediction: Phase diagram

- $q < q_c$: Line of continuous DP transitions
- $q > q_c$: Line of discontinuous transitions
- Continuous and discontinuous lines join at Tricritical point.



Prediction: Modified exponents

- Near tricritical point

- For near the transition point

$$\rho \sim (\tau/g)^{\beta_{DP}} \Big|_{\tau \ll \frac{g^2}{4c}}$$

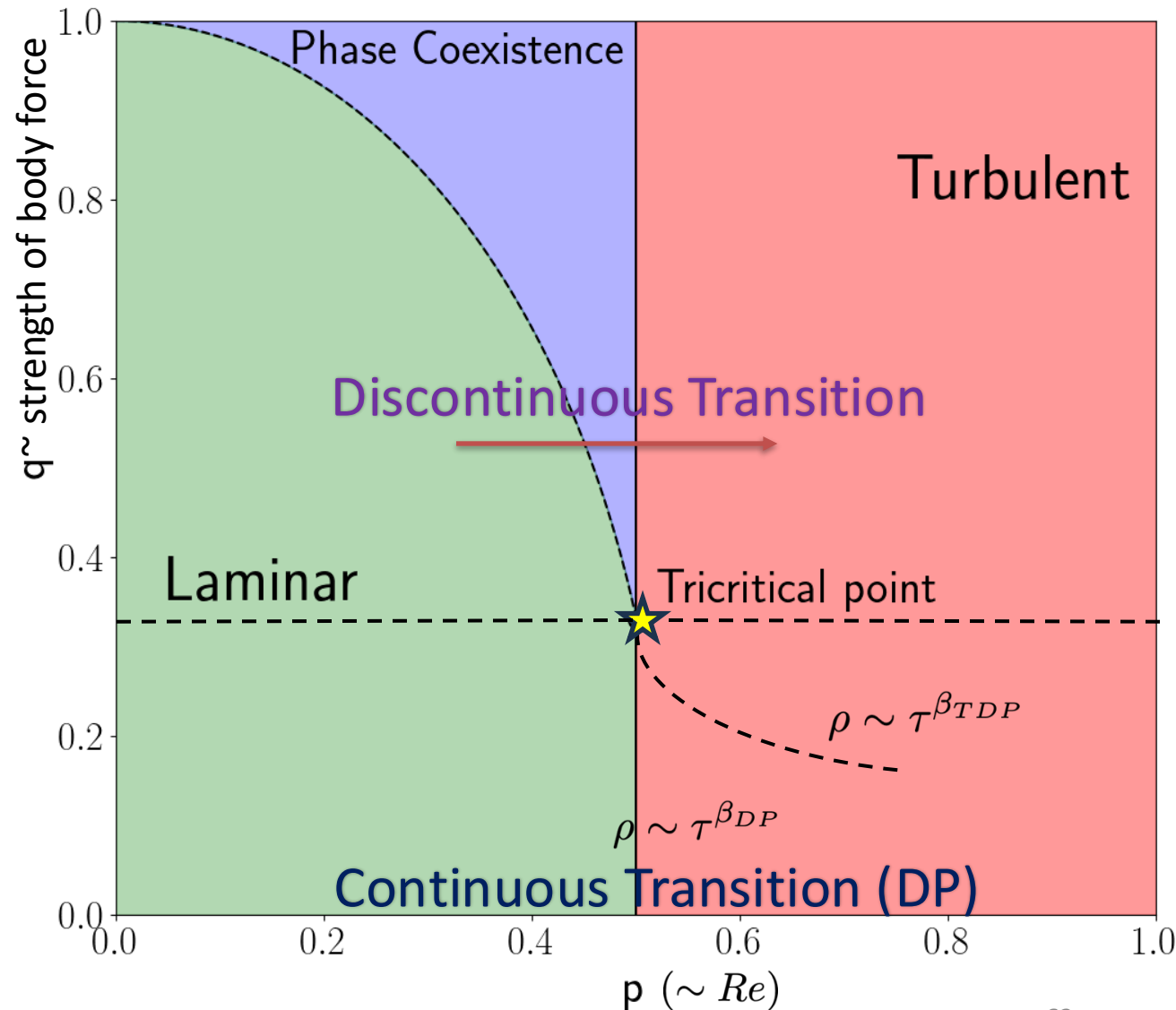
- But crossover at large values

$$\rho \sim \left(\frac{\tau}{c}\right)^{\beta_{TDP}} \Big|_{\tau \gg \frac{g^2}{4c}}$$

$$\tau \sim (Re - Re_c)/Re_c$$

- Mean field values

$$\beta_{DP}^{MF} = 1 \quad \beta_{TDP}^{MF} = 1/2$$



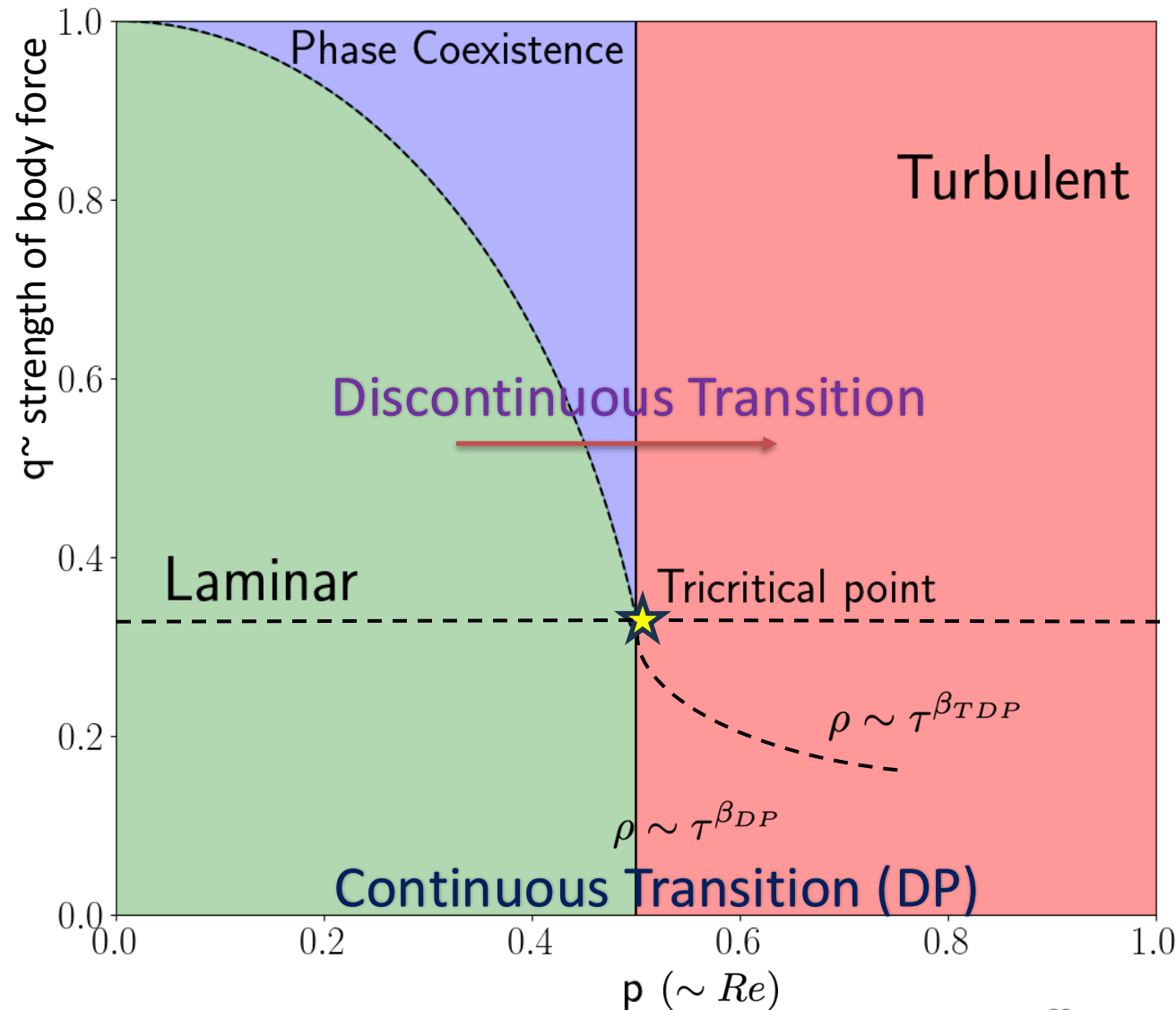
Prediction: Two parameter scaling relation

- Near tricritical point the new scaling relation is given by

$$\rho \sim g^{\beta_{TDP}/\phi} f(\tau g^{-1/\phi})$$

$$\tau \sim (Re - Re_c)/Re_c$$

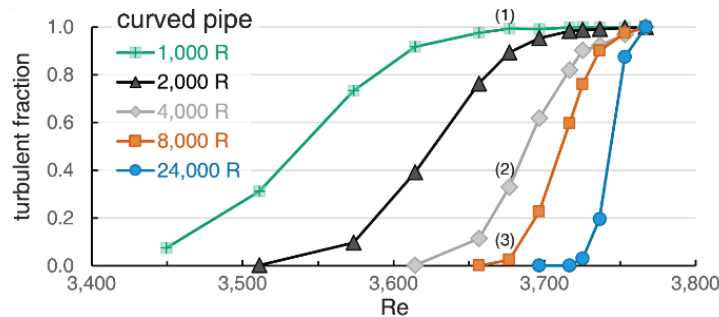
where g has linear dependence on both Re and body force strength near tricritical point.



Tricritical Directed percolation recapitulates the transition profiles

Spatial structure of turbulent fraction: curved pipes

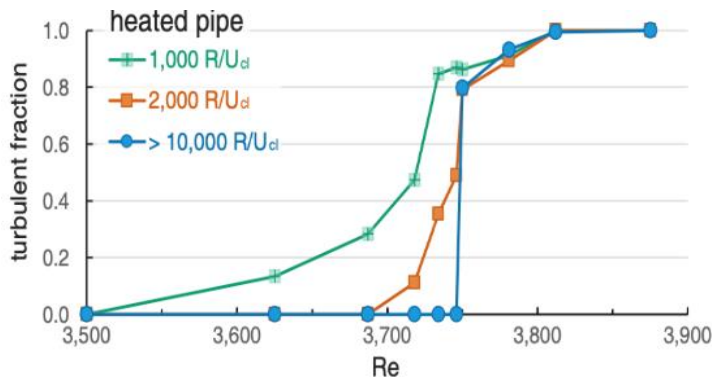
Experiment



Re dependence of the turbulent fraction measured at different downstream locations.

Turbulent fraction in a spatially-uniform pipe: heated pipe

DNS

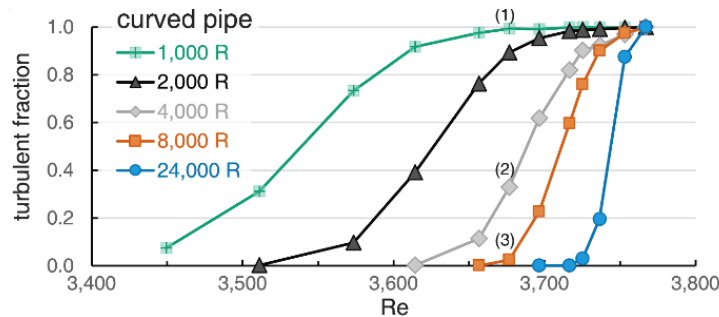


DNS data on heated flows.

Tricritical Directed percolation recapitulates the transition profiles

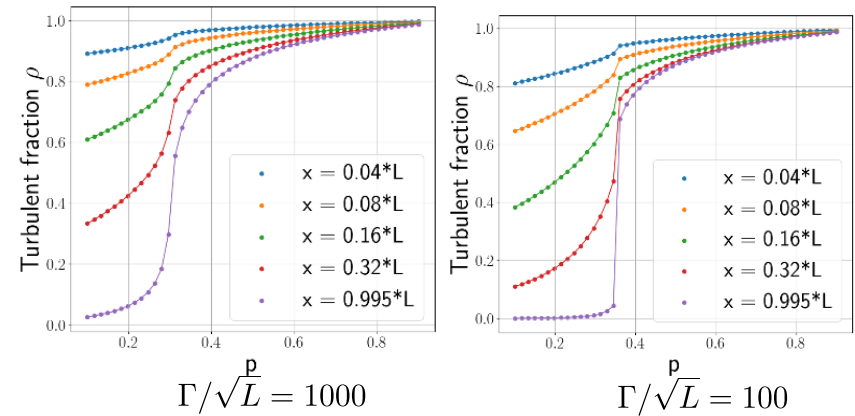
Spatial structure of turbulent fraction: curved pipes

Experiment



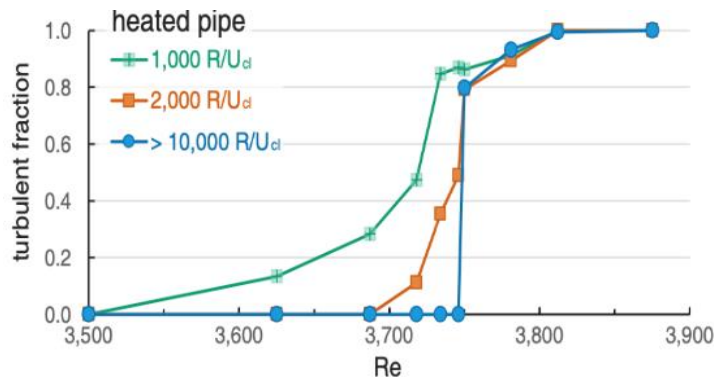
Re dependence of the turbulent fraction measured at different downstream locations.

Theory



Turbulent fraction in a spatially-uniform pipe: heated pipe

DNS



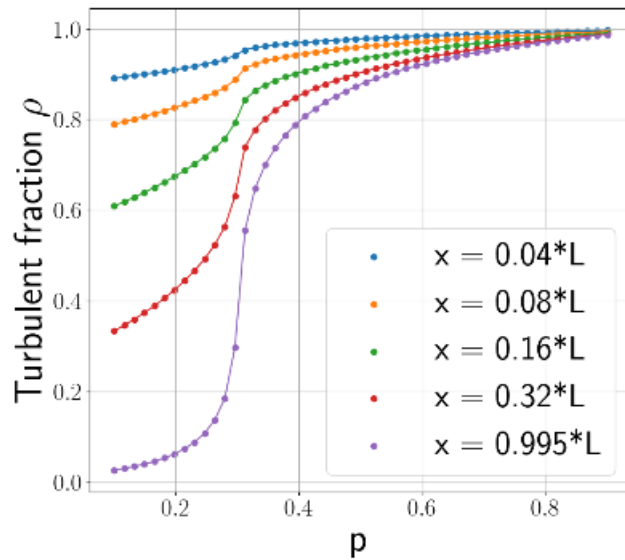
DNS data on heated flows.

Spatio-temporal turbulent fraction is correctly accounted

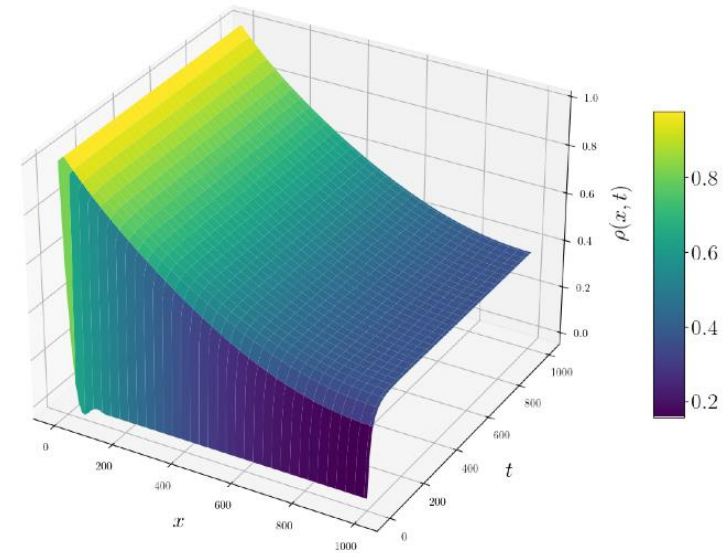
Spatial structure of turbulent fraction: curved pipes

$$\partial_t \rho(\mathbf{x}, t) = \tau_{p,q} \rho(\mathbf{x}, t) - g_{p,q} \rho(\mathbf{x}, t)^2 - c_{p,q} \rho(\mathbf{x}, t)^3 + \Gamma \nabla^2 \rho(\mathbf{x}, t)$$

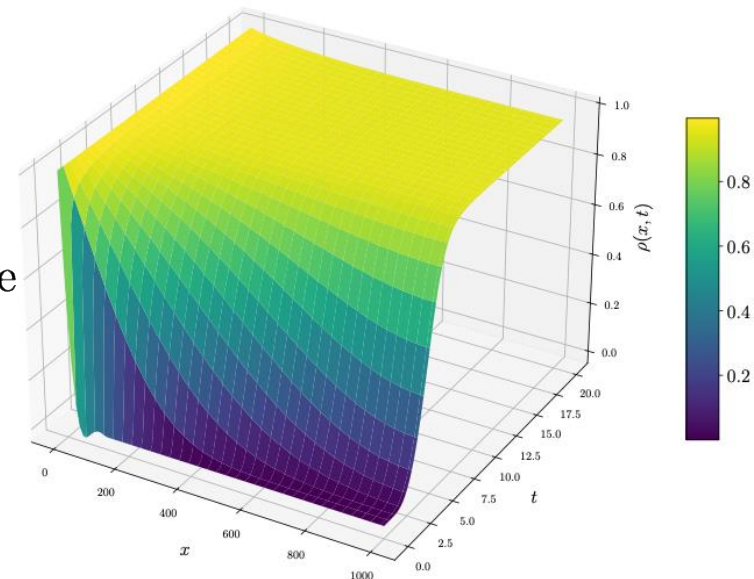
$p = 0.3 \quad \sim \text{low Re}$



$\Gamma / \sqrt{L} = 1000$



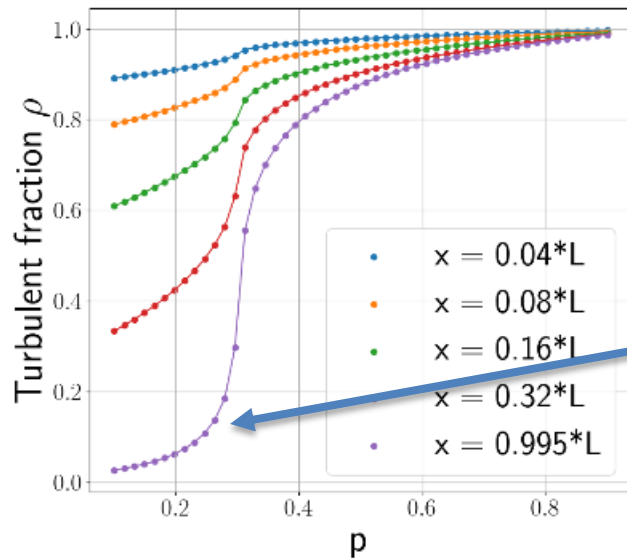
$p = 0.7 \quad \sim \text{high Re}$



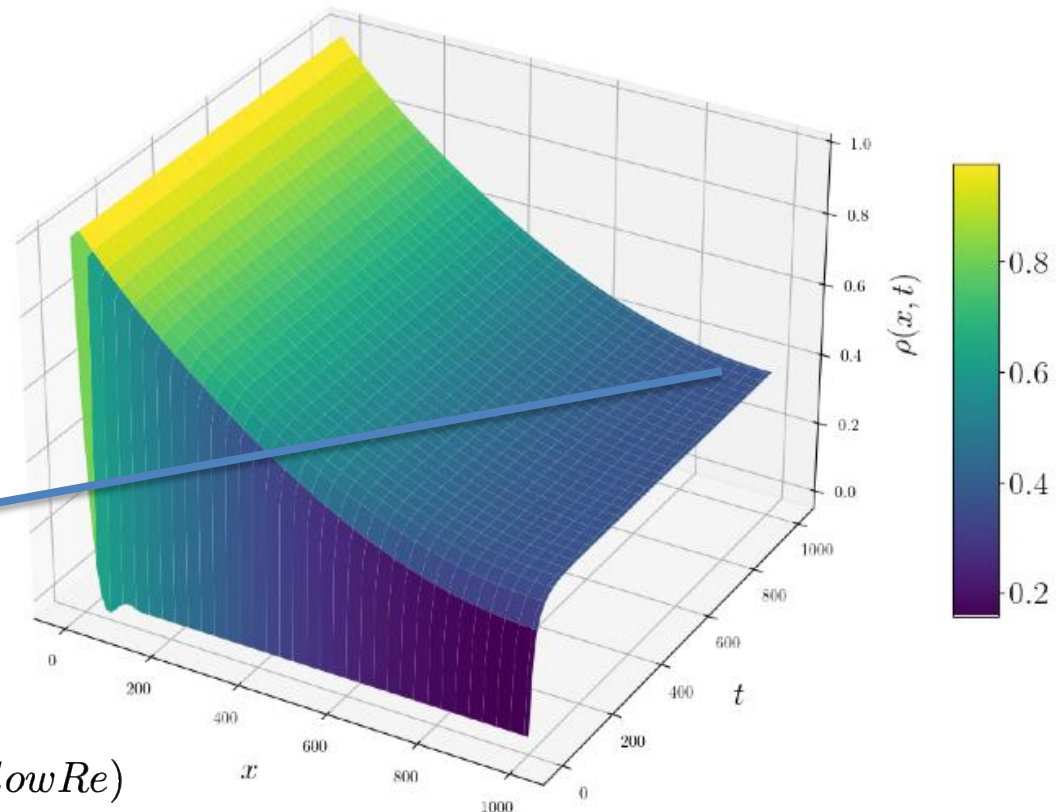
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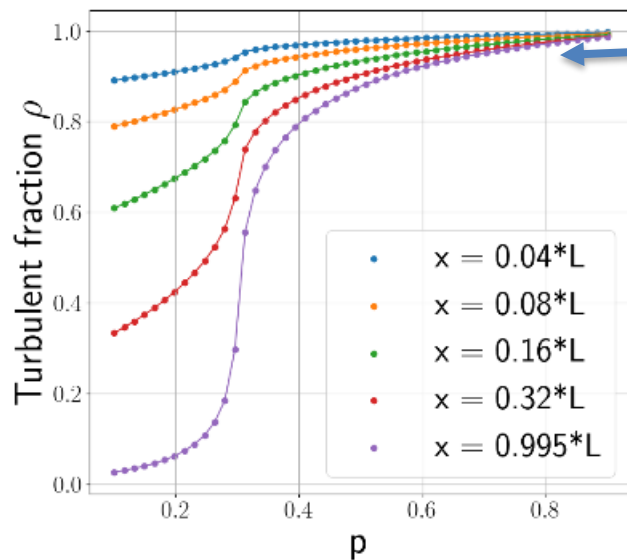
$\Gamma/\sqrt{L} = 1000$ $p = 0.3$ (lowRe)



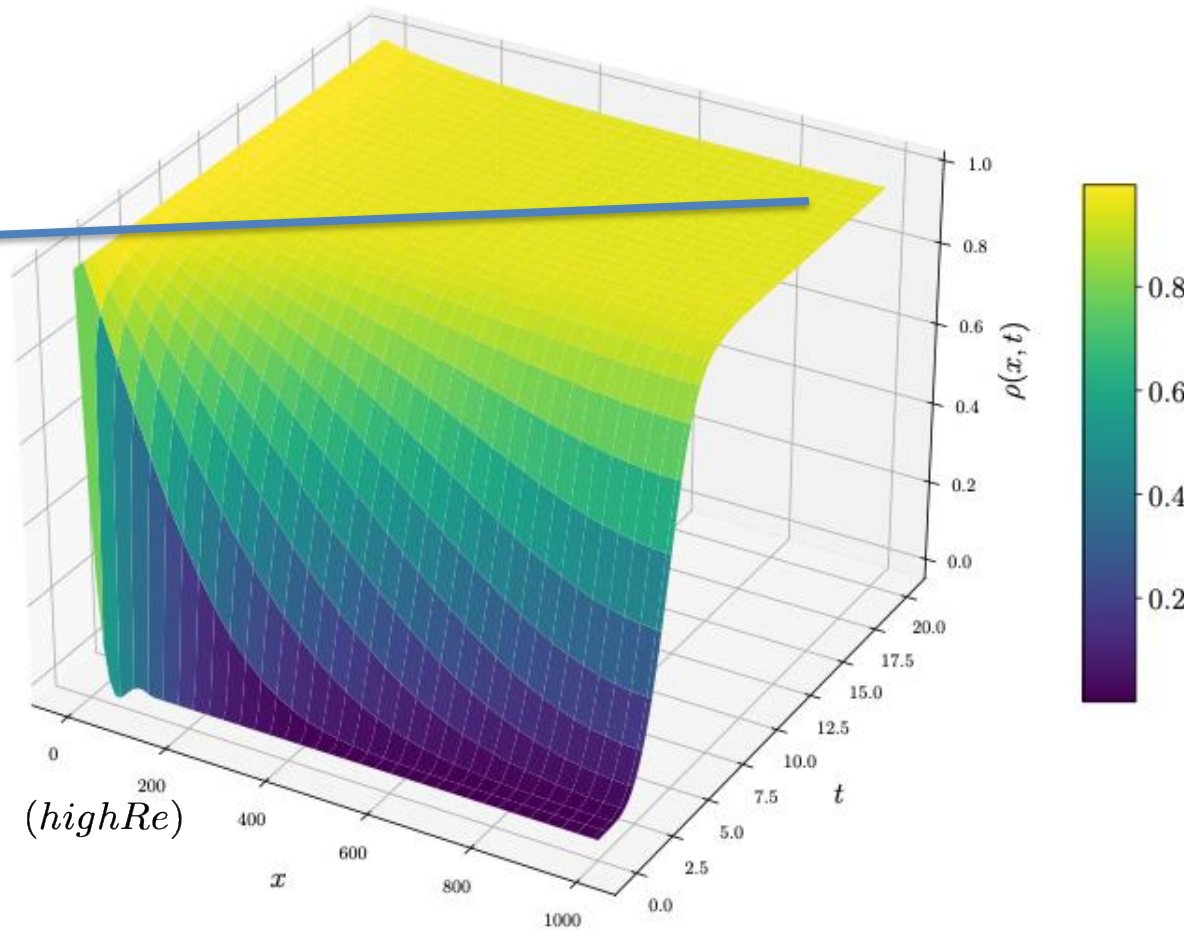
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Spatial structure of turbulent fraction: curved pipes

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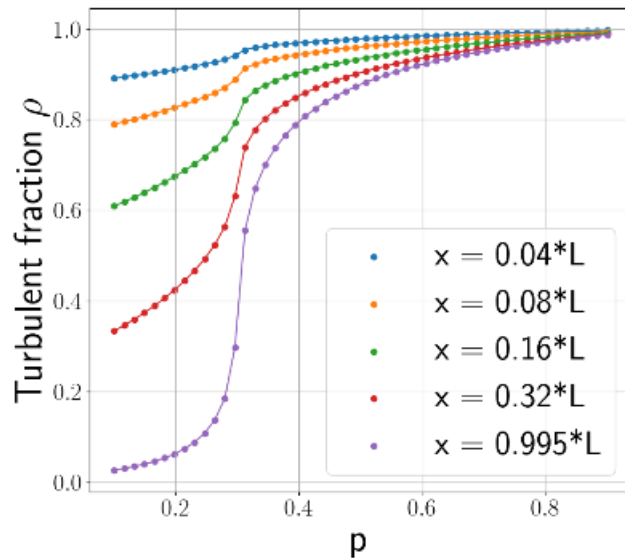
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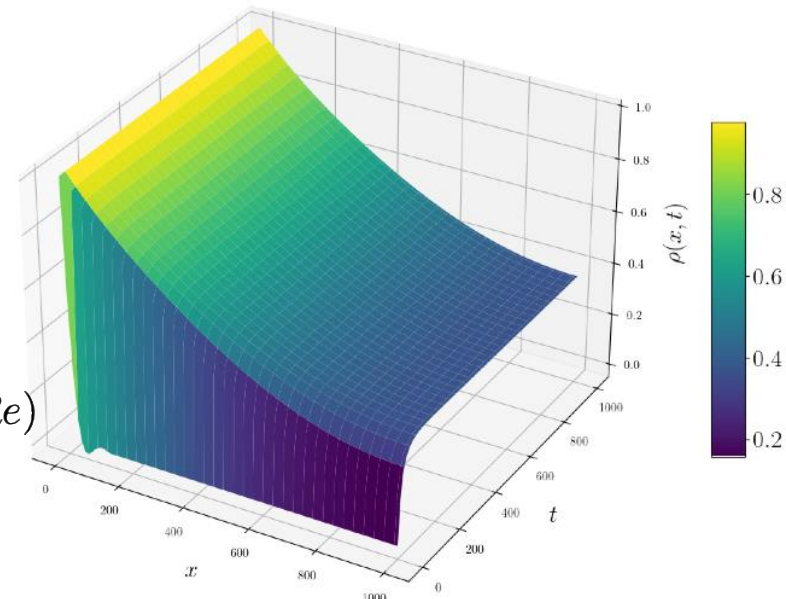
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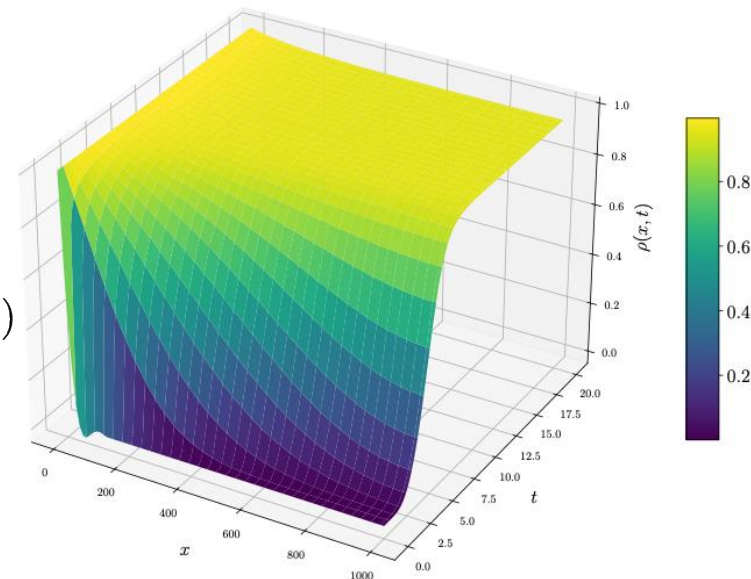


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$p = 0.3$ (*lowRe*)



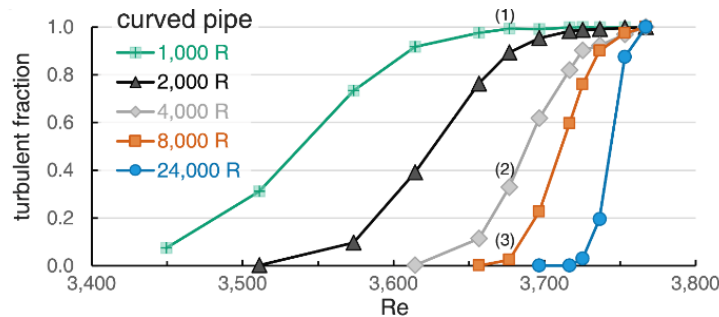
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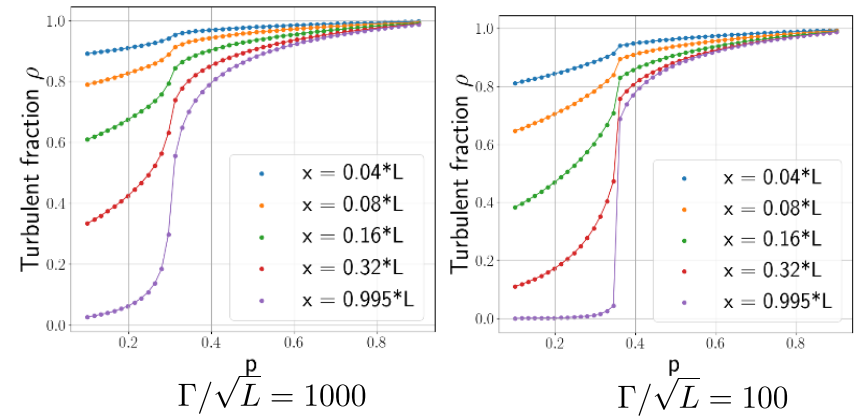
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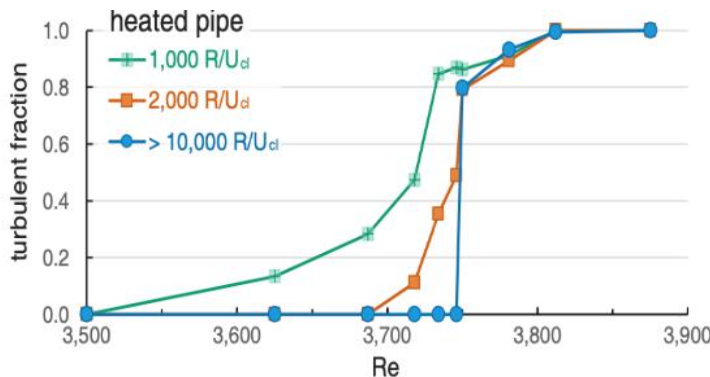
Re dependence of the turbulent fraction measured at different downstream locations.

Theory



Turbulent fraction in a spatially-uniform pipe: heated pipe

DNS

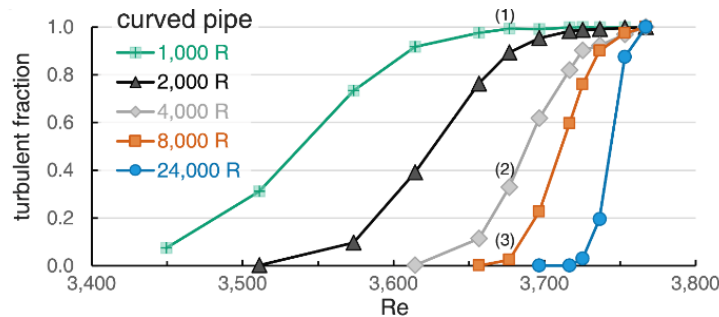


DNS data on heated flows.

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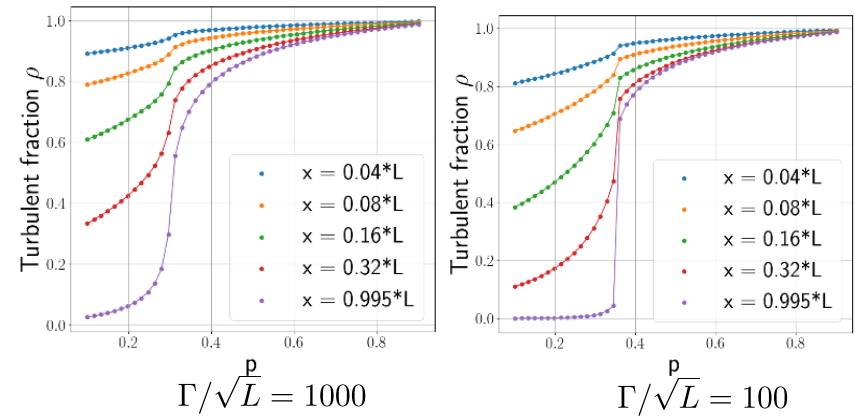
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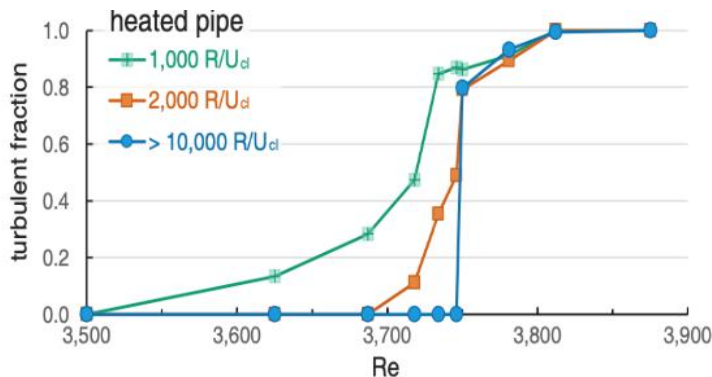
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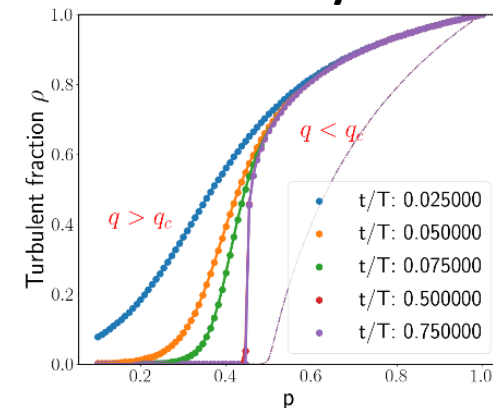
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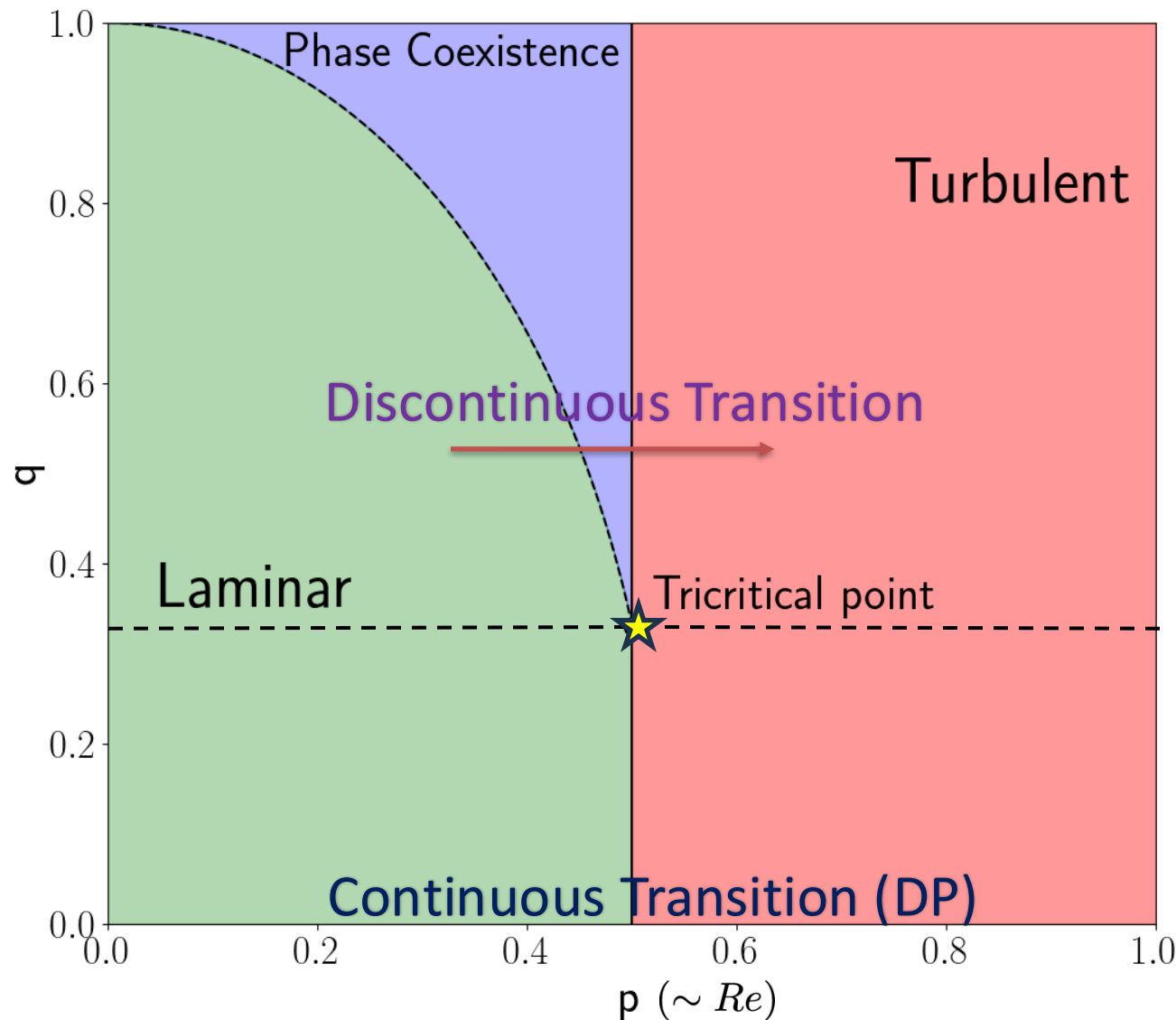
Theory



Plot of turbulent fraction vs time for a spatially uniform pipe

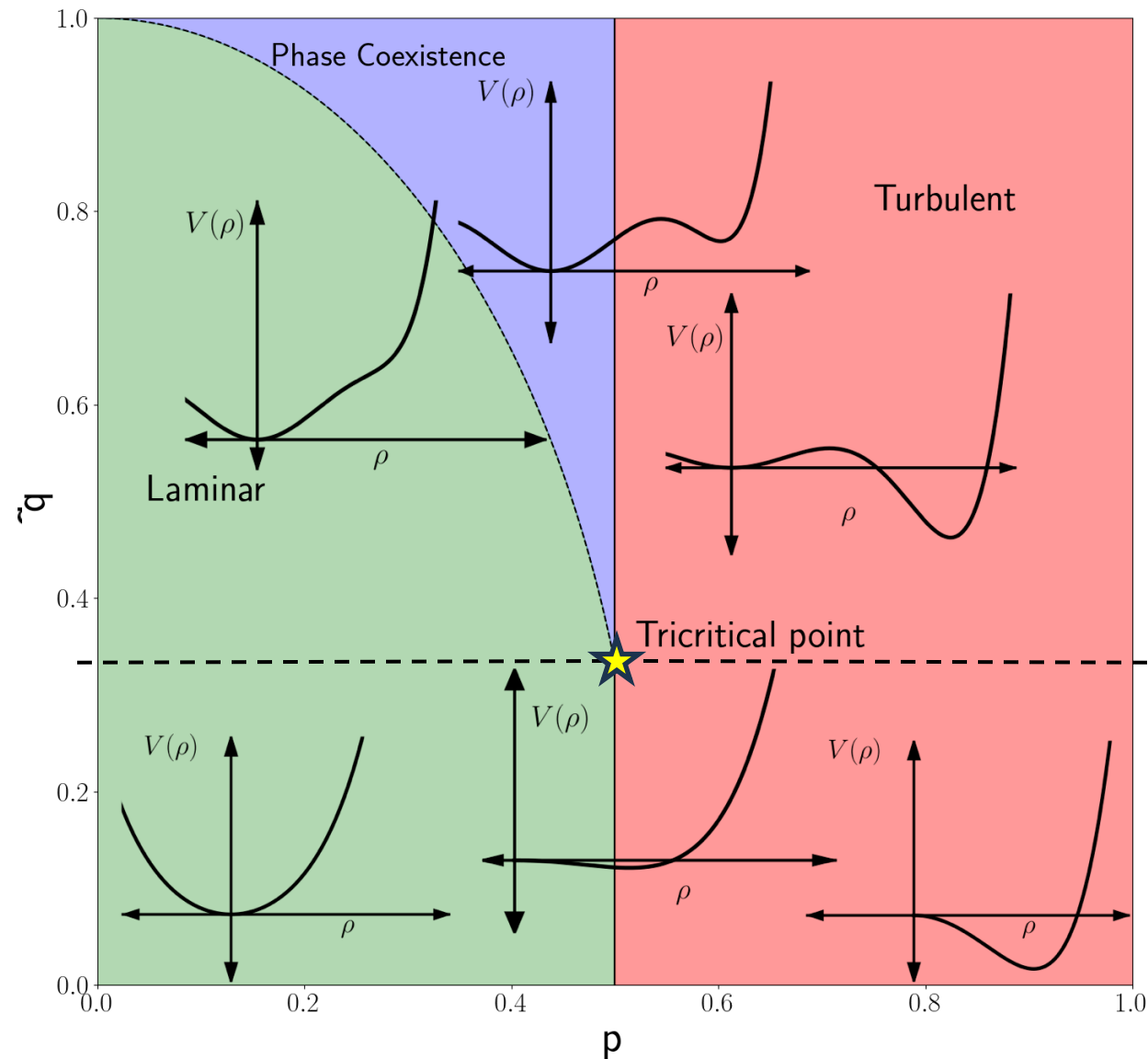
Kinetics of transition: Metastability

- $q < q_c$: Line of continuous DP transitions
- $q > q_c$: Line of discontinuous transitions
- Continuous and discontinuous lines join at Tricritical point.

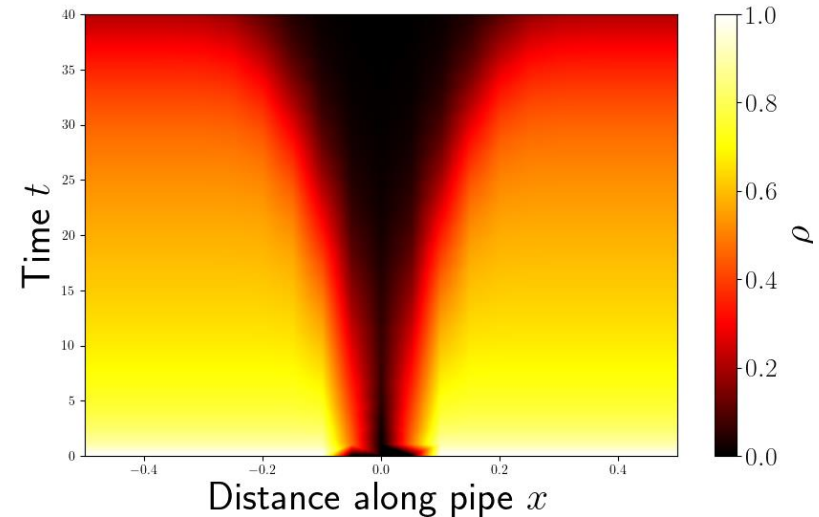
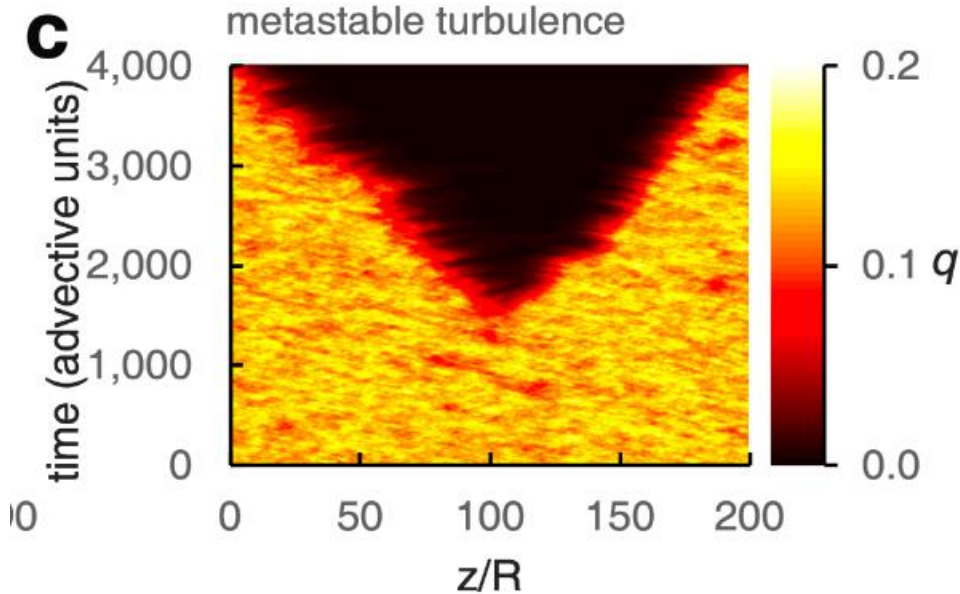


Kinetics of transition: Metastability

- In the discontinuous regime, phase coexistence occurs.
- Turbulence is metastable.



Relaminarization of metastable turbulence

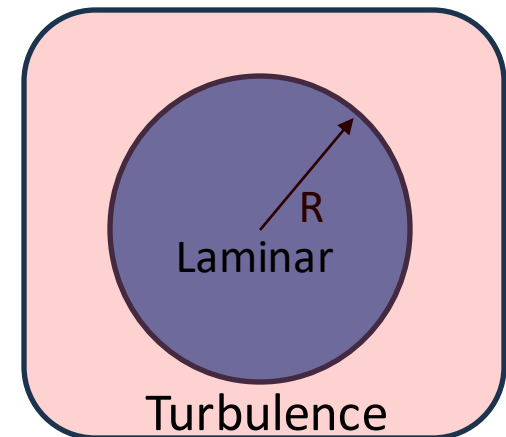


DNS

- Turbulence can be metastable, laminar bubble develops stochastically and expands.
- Threshold laminar bubble size R controlled by effective (non-equilibrium) surface tension

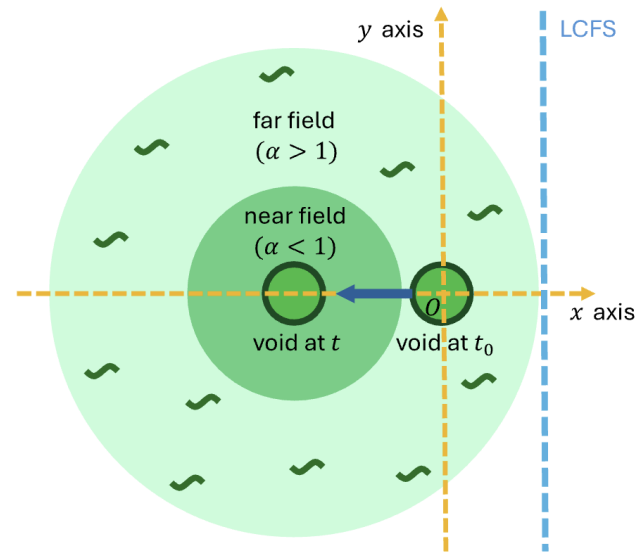
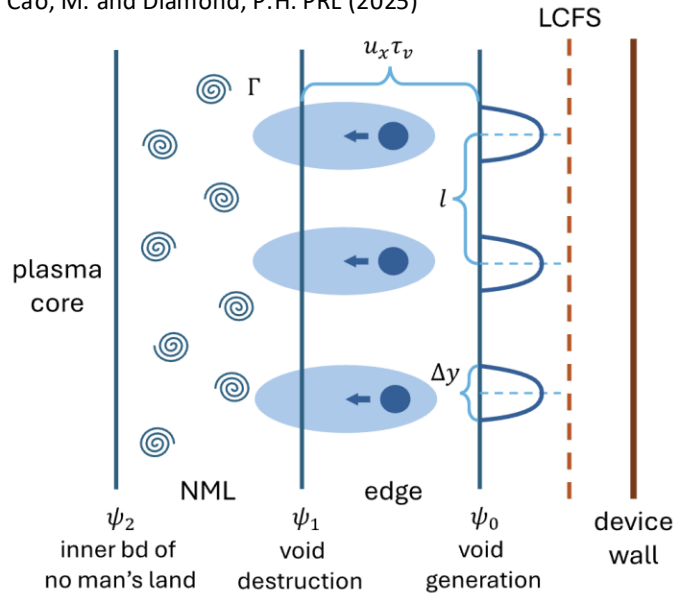
$$\sigma \sim (Re - Re_c)^{3/2}$$

Theory



Similarities to plasma turbulence

Cao, M. and Diamond, P.H. PRL (2025)



Cao, M. and Diamond, P.H. PRL (2025)

- Voids in plasma turbulence
 - Coherent *inward-propagating* objects, radiate drift waves
 - Energize turbulence and form a broad turbulent layer.
- In our theory
 - Laminar/turbulent blobs similarly drive the kinetics
 - Lead to relaminarization, boundary driven transitions etc.

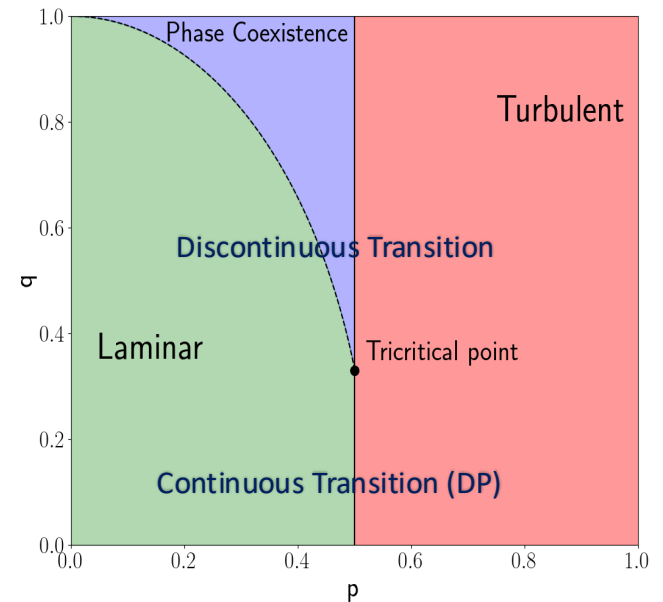
Summary

Critical point under perturbation?

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- We uncover a tricritical fixed point
- New exponents, mean field phase diagram and scaling functions are predicted
- Can experimentally checked to vindicate/falsify our analysis.

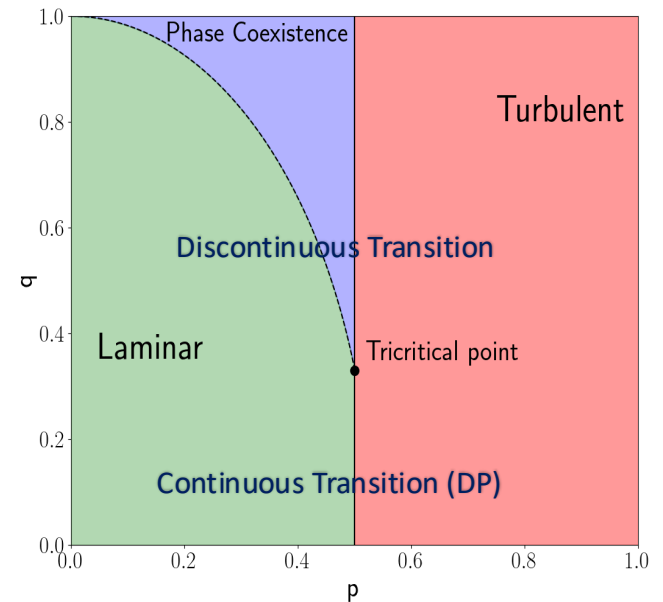


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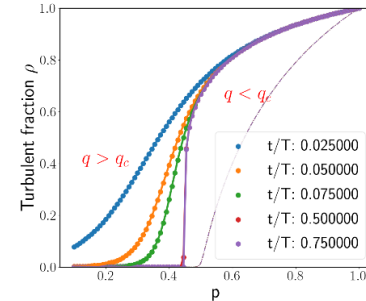
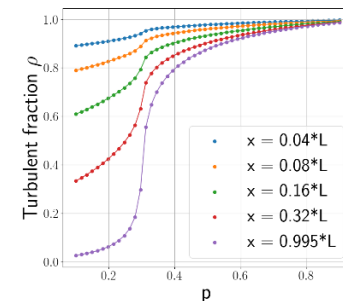
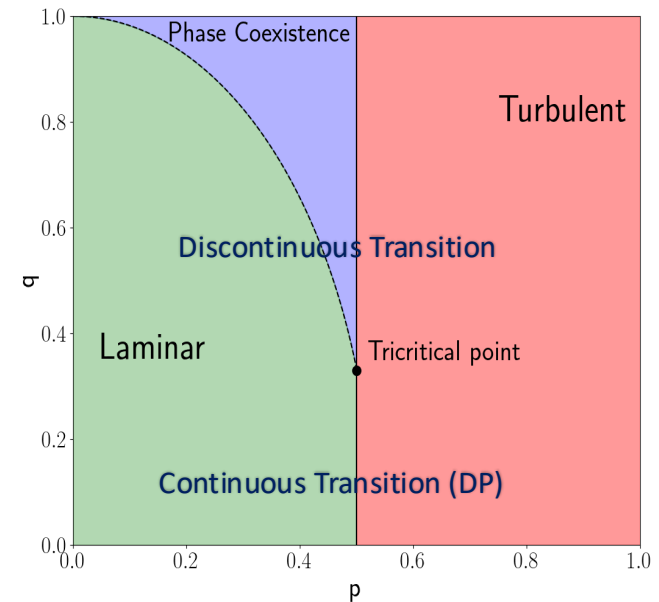
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Transition profiles?

- Tricritical DP explains sharpening and delay of transition
- Accounts for profiles of heated and curved pipes.



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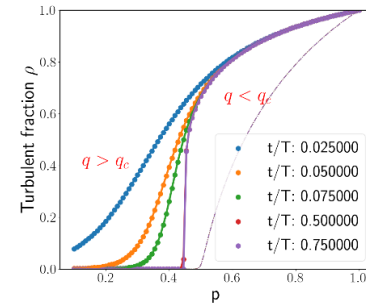
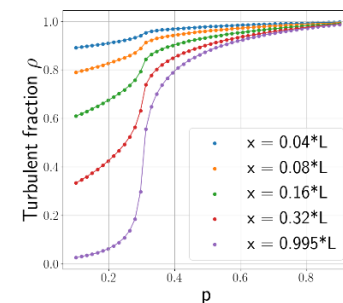
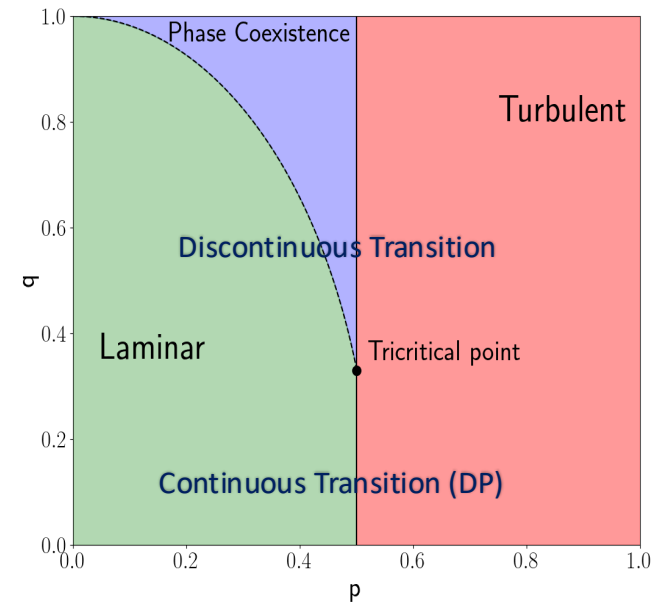
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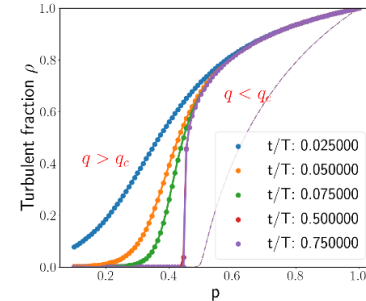
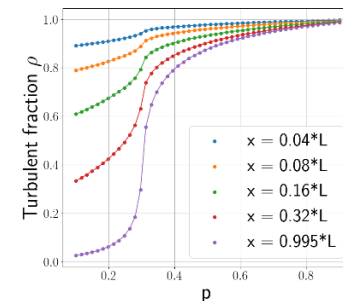
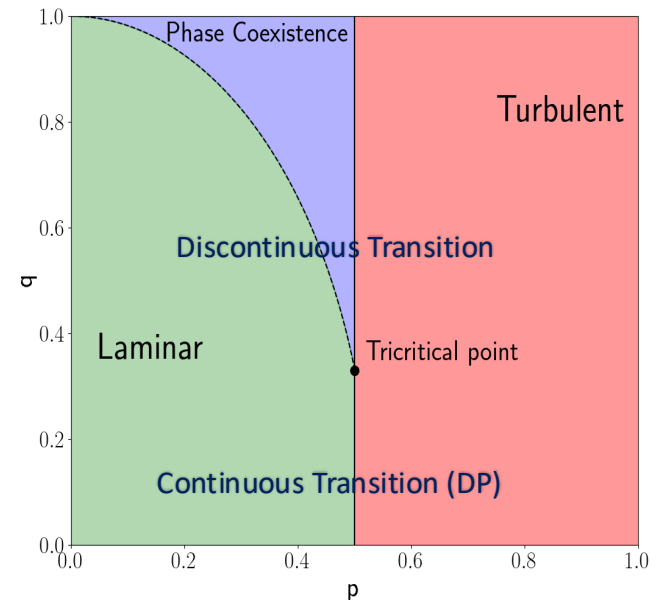
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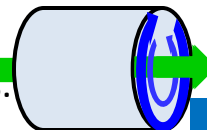
- Tricritical DP explains sharpening and delay of transition
- Accounts for profiles of heated and curved pipes.

Minimal models, Landau theory etc. work?

- Yes! Focus on emergent long wavelength modes
- Effective field theory is stochastic predator-prey dynamics.



Turbulence



Azimuthal mean flow

Laminar modes

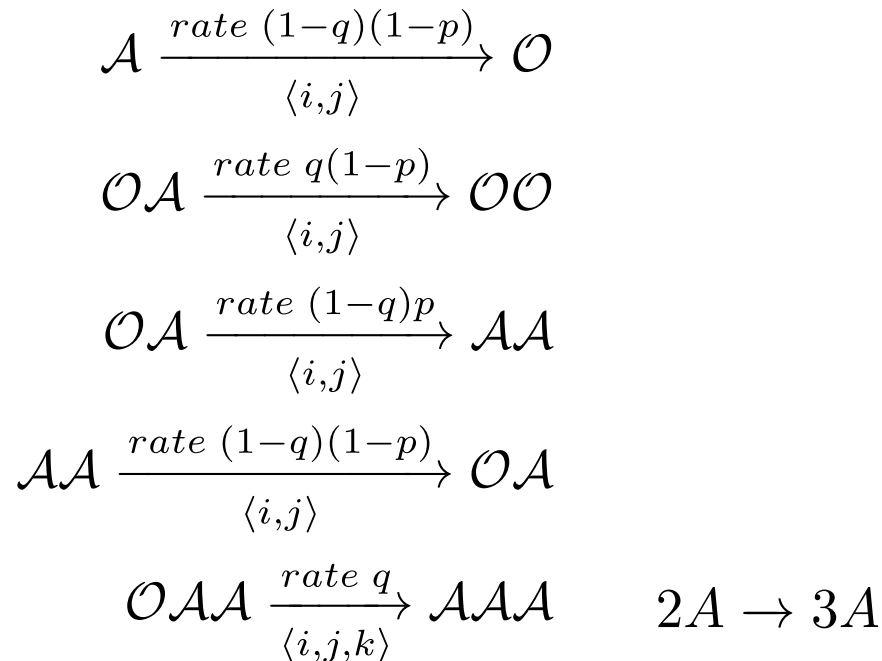
Turbulent modes

Azimuthal modes

Reserve Slides

Tricritical Directed Percolation

- Both discontinuous and continuous transition. A microscopic lattice model that shows same phenomenology is given by **Tricritical Contact Process (TCP)**



- A **Tricritical Point** in addition to usual DP critical point.
- Upper critical (spatial) dimension $d_c = 3$, only log corrections for MFT exponents/scaling functions

Tricritical Directed Percolation

- Both discontinuous and continuous transition. A microscopic lattice model that shows same phenomenology is given by **Tricritical Contact Process (TCP)**

$$\mathcal{A} \xrightarrow[\langle i,j \rangle]{\text{rate } (1-q)(1-p)} \mathcal{O}$$

$$\mathcal{O}\mathcal{A} \xrightarrow[\langle i,j \rangle]{\text{rate } q(1-p)} \mathcal{O}\mathcal{O}$$

$$\mathcal{O}\mathcal{A} \xrightarrow[\langle i,j \rangle]{\text{rate } (1-q)p} \mathcal{A}\mathcal{A}$$

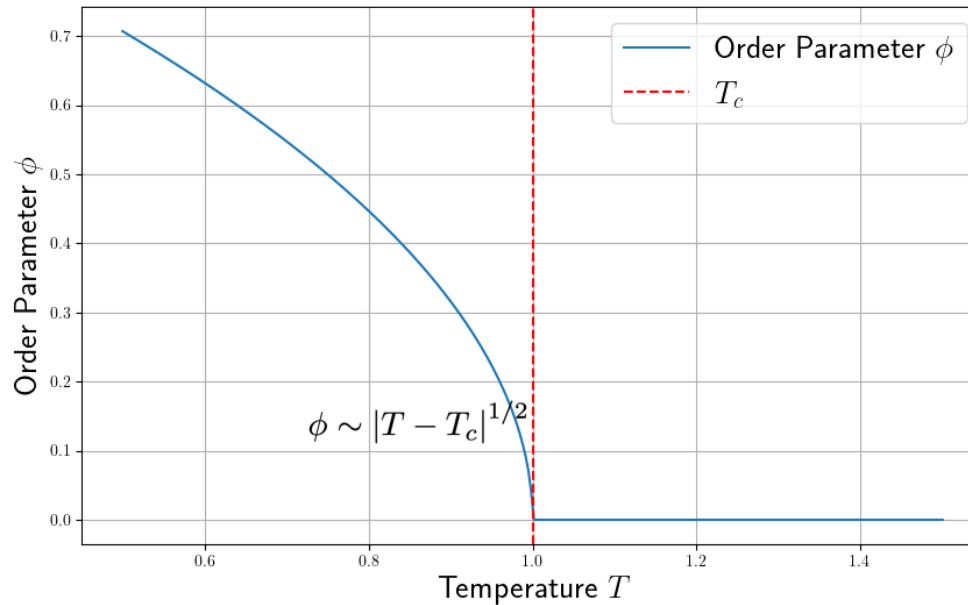
$$\mathcal{A}\mathcal{A} \xrightarrow[\langle i,j \rangle]{\text{rate } (1-q)(1-p)} \mathcal{O}\mathcal{A}$$

$$\mathcal{O}\mathcal{A}\mathcal{A} \xrightarrow[\langle i,j,k \rangle]{\text{rate } q} \mathcal{A}\mathcal{A}\mathcal{A}$$

$$2\mathcal{A} \rightarrow 3\mathcal{A}$$

- A **Tricritical Point** in addition to usual DP critical point.
- Upper critical (spatial) dimension $d_c = 3$, only log corrections for MFT exponents/scaling functions

Tricritical point



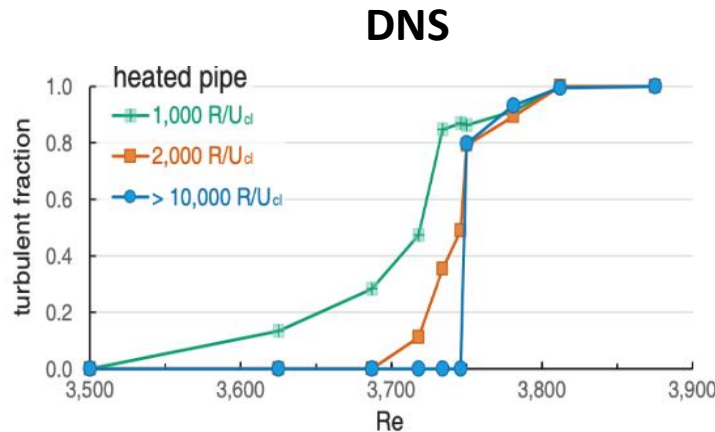
- Critical point:
 - Requires tuning one parameter (e.g. $T - T_c$)

$$\mathcal{F}_{GL} = a_2\phi^2 + a_4\phi^4 \quad \frac{\partial \mathcal{F}_{GL}}{\partial \phi} = 0$$

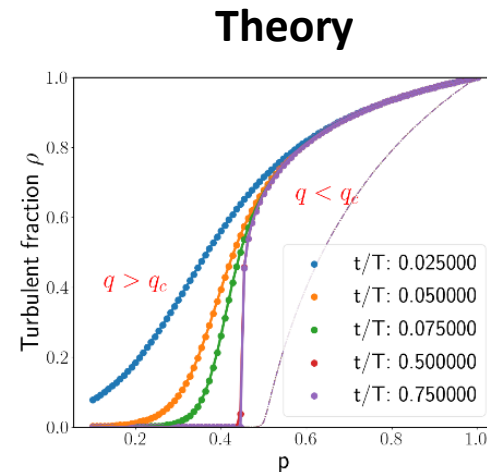
$$\phi = 0 \implies \text{Disordered} \quad \phi \neq 0 \implies \text{Ordered}$$

Tricritical Directed percolation recapitulates the transition profiles

Turbulent fraction in a spatially-uniform pipe: heated pipe

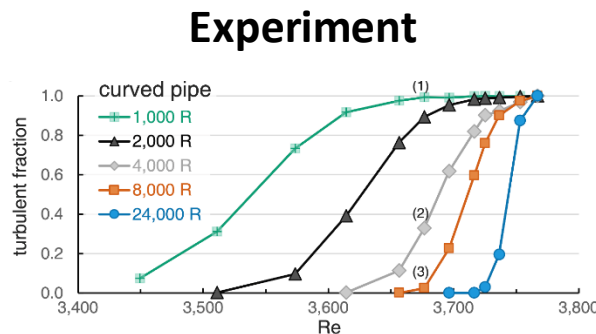


DNS data on heated flows.

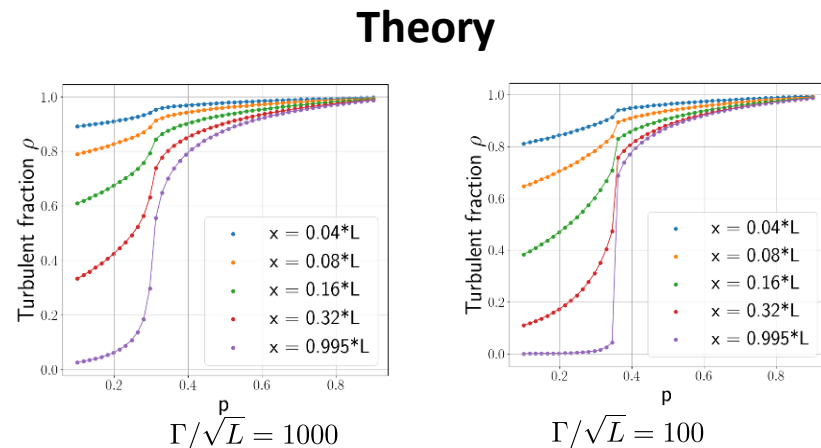


Plot of turbulent fraction vs time for a spatially uniform pipe

Spatial structure of turbulent fraction: curved pipes



Re dependence of the turbulent fraction measured at different downstream locations.



$$\Gamma/\sqrt{L} = 1000$$

$$\Gamma/\sqrt{L} = 100$$

Directed Percolation under body forces

- Since body forces would likely cause additional flow instabilities and disturbances, we look at usual DP MFT and add higher order terms

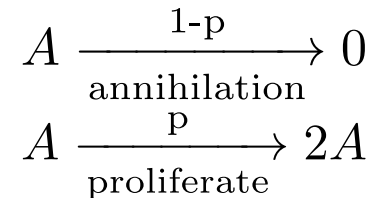
$$\partial_t \rho = (2p - 1)\rho - p \cdot \rho^2$$

$\rho \equiv$ turbulent fraction/ density of A

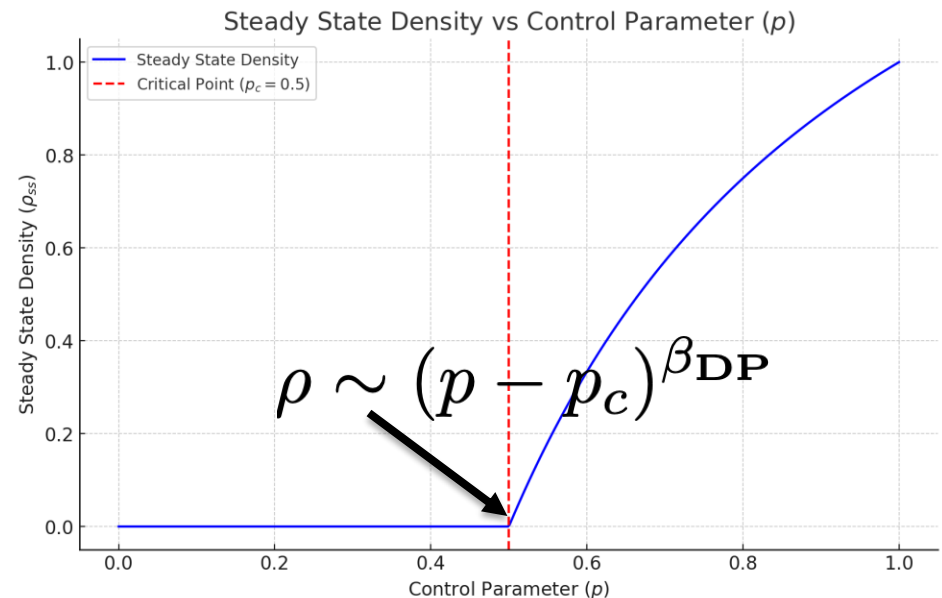
$p \equiv Re$

$0 \equiv laminar$

$A \equiv turbulent$



- Steady state: ρ vs p



Model: Directed Percolation under Body forces

- Since body forces would likely cause additional flow instabilities and disturbances, we look at usual DP MFT and add higher order terms

$$\partial_t \rho = (2p - 1)\rho - p \cdot \rho^2 + \boxed{\beta(p)\rho^2 + \gamma(p)\rho^3}$$

$$A \xrightarrow[annihilation]{1-p} 0$$

$$A \xrightarrow[proliferate]{p} 2A$$

- For physical consistency
 - $\rho = 0$ when $p=0$ (complete laminarity when no birth-processes occur i.e. absorbing phase exists) → Trivially satisfied by the above equation
 - $\rho = 1$ when $p=1$ (complete turbulence when absorbing phase is destroyed) → gives

$$(2p - 1)\rho^2 - p\rho^2 + \beta\rho^2 + \gamma\rho^3|_{p=1, \rho=1} = 0$$

$$\implies \beta + \gamma = 0 \implies \boxed{\gamma = -\beta} \quad \text{One effective Parameter!}$$

Scaling reveals higher order terms are important

- Relevant at transition?

$$\delta p \rightarrow \lambda \delta p = \lambda^{\beta_{DP}} \delta p, \quad t \rightarrow \lambda^{-1} t = \lambda^{-\nu_{\parallel}} t$$

$$\frac{\partial \rho}{\partial t} = \delta p \rho - p \rho^2 \quad \longrightarrow \quad \lambda^2 \frac{\partial \rho}{\partial t} = \lambda^2 \delta p \rho - \lambda^2 p \rho^2$$

$$\frac{\partial \rho}{\partial t} = \delta p \rho - p \rho^2 + a \rho^2 (1 - \rho) \quad \longrightarrow \quad \lambda^2 \frac{\partial \rho}{\partial t} = \lambda^2 (\delta p \rho - p \rho^2) + \underbrace{\lambda^2 a \rho^2}_{\text{term is RG relevant DP}} \cdot 1$$

term is RG relevant
DP

- Complete stochastic mean field description

$$\partial_t \rho(\vec{x}, t) = \tau \rho(\vec{x}, t) - g \rho(\vec{x}, t)^2 - \boxed{c \rho^3(\vec{x}, t)} + \Gamma \nabla^2 \rho(\vec{x}, t) + \eta(\mathbf{x}, t)$$



Tricritical Directed Percolation

Modified mean field theory

- New mean field theory

$$\partial_t \rho = \delta p \rho - p \rho^2 + \beta \rho^2 (1 - \rho)$$

- Relevant at transition? Scaling by DP critical exponents must leave MFT invariant

$$\delta p \rightarrow \lambda \delta p = \lambda^{\beta_{DP}} \delta p, \quad t \rightarrow \lambda^{-1} t = \lambda^{-\nu_{\parallel}} t$$

$$\frac{\partial \rho}{\partial t} = \delta p \rho - p \rho^2 \quad \Rightarrow \quad \lambda^2 \frac{\partial \rho}{\partial t} = \lambda^2 \delta p - \lambda^2 p \rho^2$$

$$\partial_t \rho = \delta p \rho - p \rho^2 + \beta \rho^2 (1 - \rho) \quad \Rightarrow \quad \lambda^2 \frac{\partial \rho}{\partial t} = \lambda^2 (\delta p - p \rho^2) + \lambda^2 \beta \rho^2 \cdot 1 \Big|_{\rho \approx 0; 1-\rho \sim 1}$$

\beta is RG marginal

DP

- As a reaction process:
$$\begin{array}{l} A \xrightarrow{1-p} 0 \\ A \xrightarrow{p} 2A \\ 2A \xrightarrow{\beta} 3A \end{array}$$

@ Hof group comment:
0 state remains absorbing since no process like $0 \rightarrow A$ present.

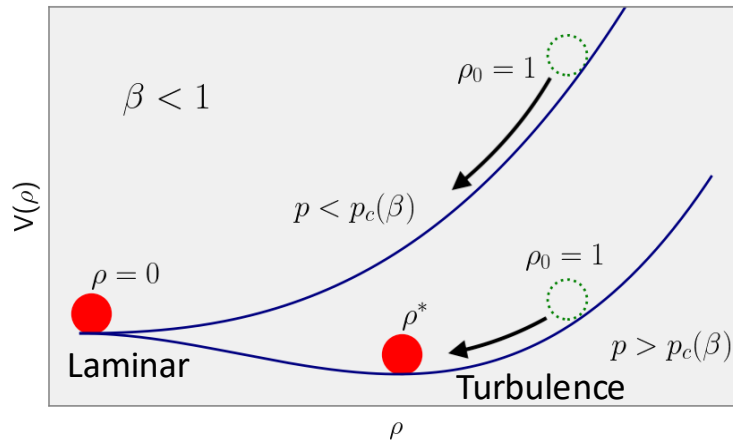
Valid absorbing state transition

Nature of transitions

- Can be seen within Lyapunov potential

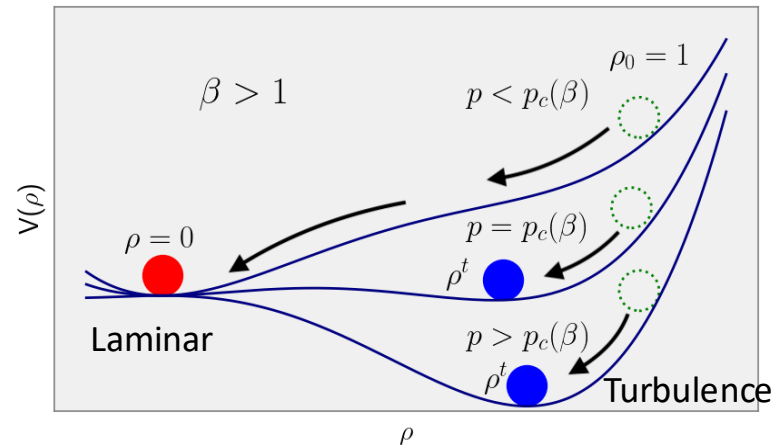
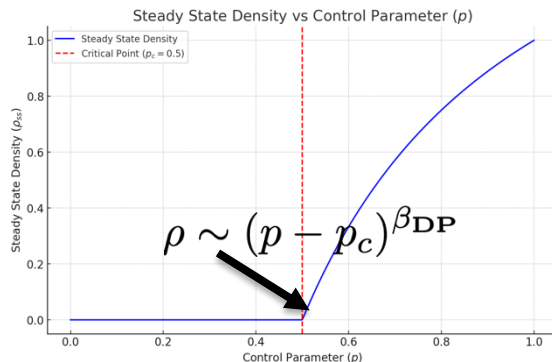
$$\partial_t \rho = \delta p \rho - p \rho^2 + \beta \rho^2 (1 - \rho) \longrightarrow$$

$$\frac{\partial \rho}{\partial t} = - \frac{\partial V}{\partial \rho}$$



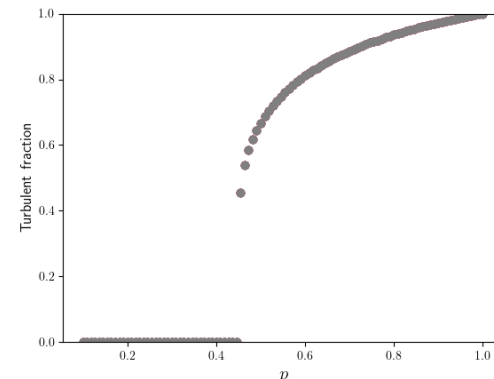
$$\beta < 1$$

V has only one minima; transition **continuously** from $\rho = 0$ to $\rho \neq 0$.



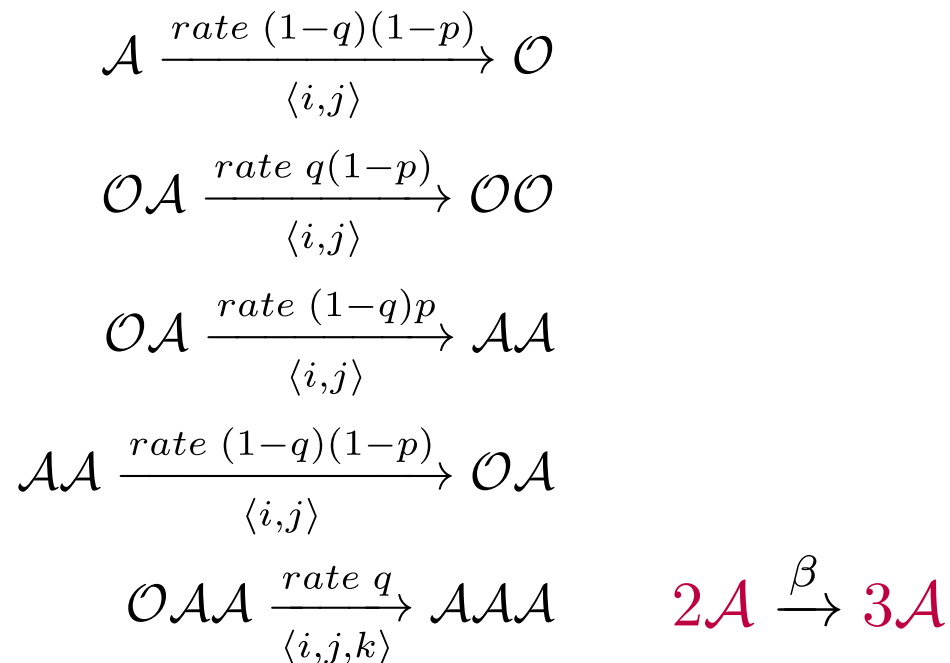
$$\beta > 1$$

V develops another minima; transition **discontinuously** from $\rho = 0$ to $\rho \neq 0$.



Tricritical Directed Percolation

- New MFT can show both discontinuous and continuous transition. A microscopic lattice model that shows same phenomenology is given by **Tricritical Contact Process (TCP)**



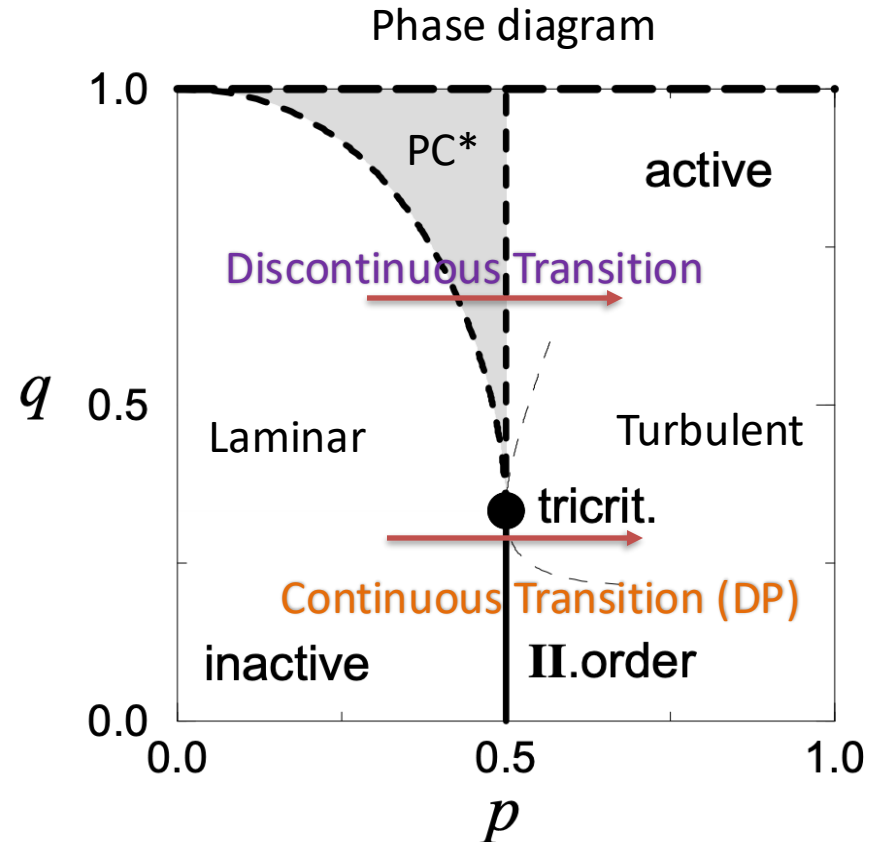
- New model has same MFT, can show cont/discont transition depending on q , and has a **Tricritical Point** in addition to usual DP critical point.

Tricritical Directed Percolation (TDP)

- $q < q_c$: Line of continuous DP transitions
- $q > q_c$: Line of discontinuous transitions
- Continuous and discontinuous lines join at Tricritical point.
- Away from Tricritical point:
 - 2nd order transition is in DP class
 - 1st order transition -> no universality
- At Tricritical point:
 - Continuous transition, $\beta_{TDP} = \frac{1}{2}$
 - Near tricritical point

$$\rho \sim (p - p_{tc})^{\beta_{TDP}} \cdot f\left(\frac{q - q_c}{(p - p_{tc})^\phi}\right)$$

$\phi = 1/2$ is the crossover exponent,
characterizing shift from DP to TDP
universality class

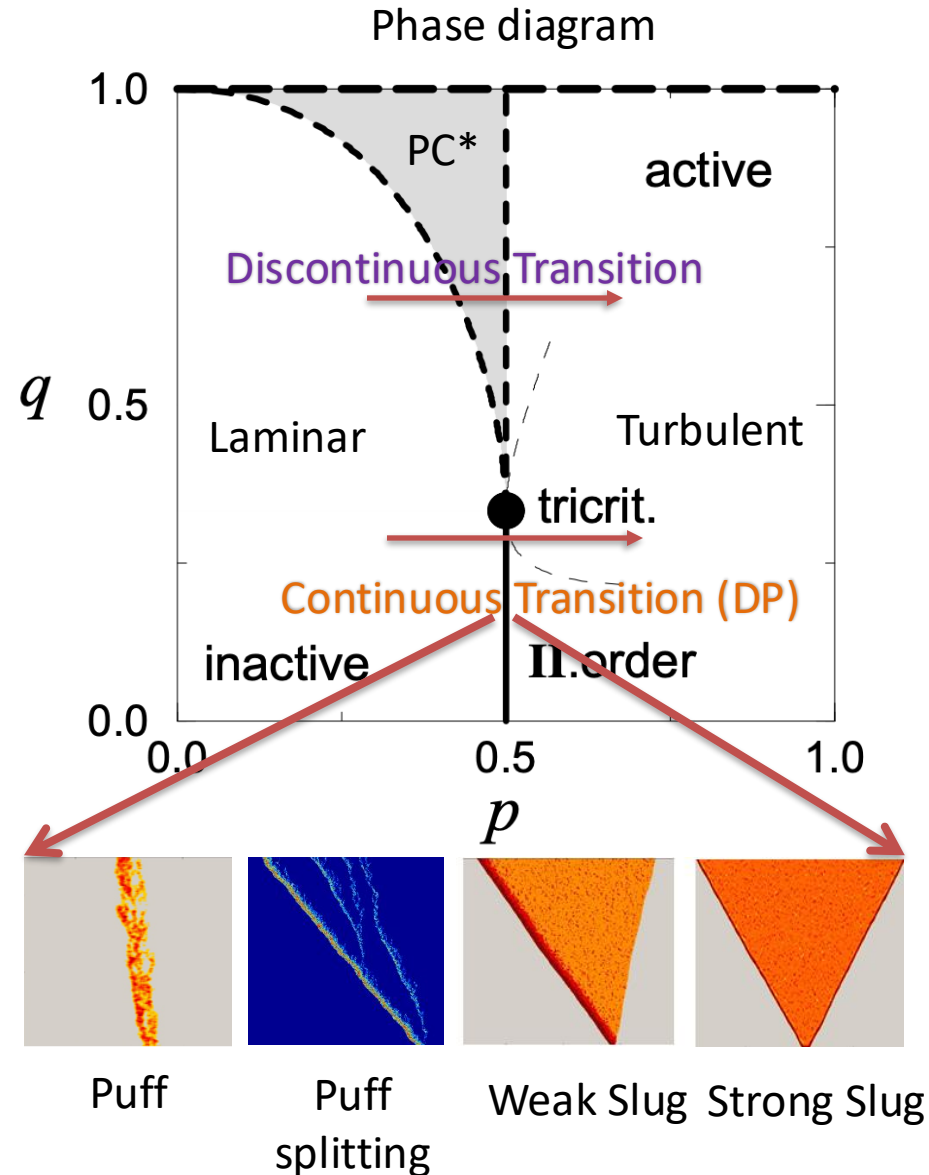


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- Continuous and discontinuous lines join at Tricritical point.
- Away from Tricritical point:
 - 2nd order transition is in DP class
 - 1st order transition -> no universality
- At Tricritical point:
 - Continuous transition, $\beta_{TDP} = \frac{1}{2}$
 - Near tricritical point

$$\rho \sim (p - p_{tc})^{\beta_{TDP}} \cdot f\left(\frac{q - q_c}{(p - p_{tc})^\phi}\right)$$

$\phi = 1/2$ is the crossover exponent, characterizing shift from DP to TDP universality class

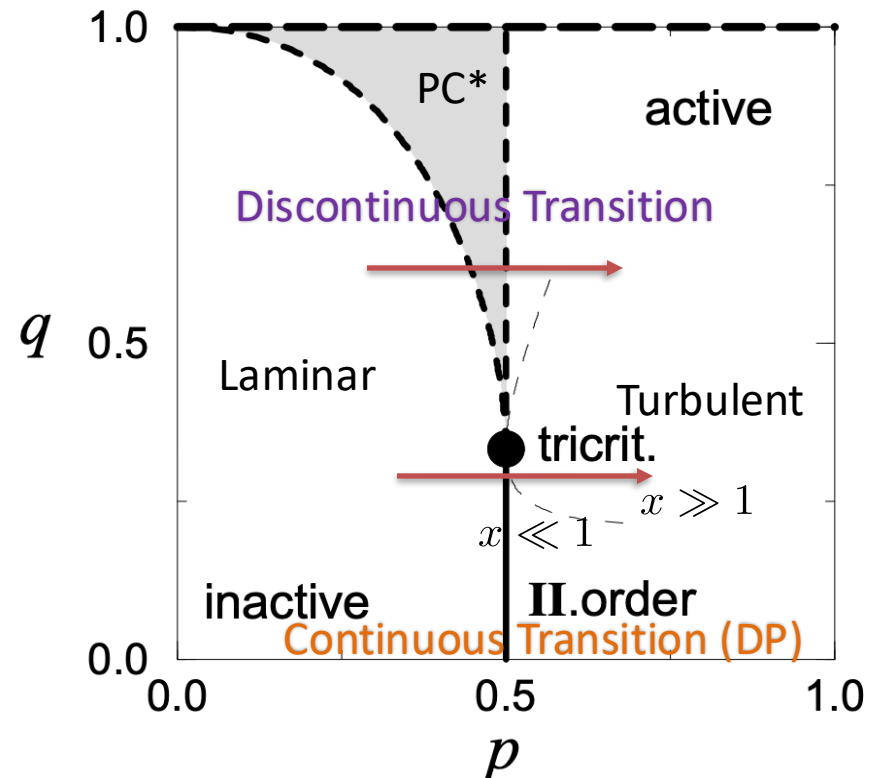


Tricritical Directed Percolation (TDP)

- At Tricritical point:
 - Continuous transition, exponent $\beta_{TDP} = \frac{1}{2}$
 - Near tricritical point

$$\rho \sim (q - q_{tc})^{\beta_{TDP}/\phi} \cdot \tilde{f}\left(\frac{p - p_c}{(q - q_{tc})^\phi}\right)$$

$$\tilde{f}_{tDP}(x, 1, 0) \sim \begin{cases} x^{\beta_{DP}} & \text{if } x \ll 1 \\ x^{\beta_t} & \text{if } x \gg 1 \end{cases}$$



- Critical Dimension: 3 (= 4 for usual DP)
 - MFT can be used for D=3 flows, with possibly logarithmic corrections to exponents/scaling functions

TDP in pipe flows

- To make contact with expt data:
 - Turbulent fraction inserted at start of pipe :
 - Modify the MFT to include space

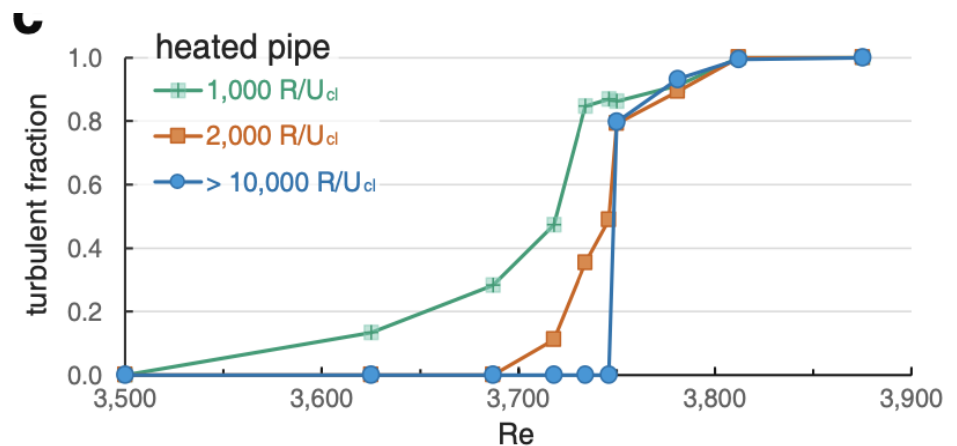
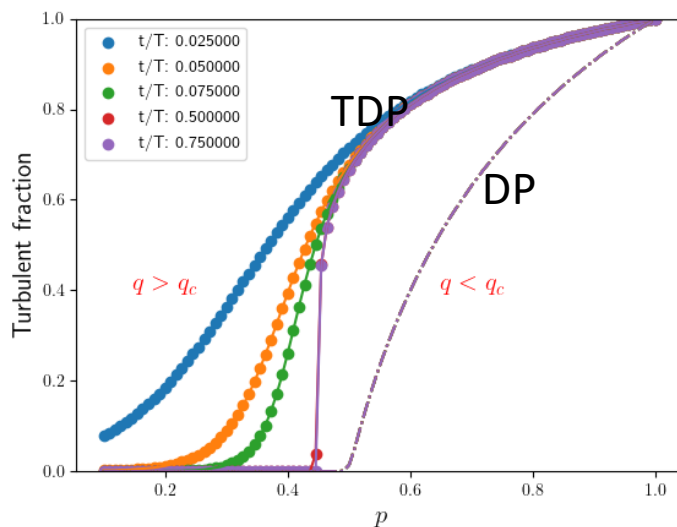
$$\partial_t \rho = \tau_{p,q} \rho - g_{p,q} \rho^2 - \boxed{c_{p,q} \rho^3} + \boxed{\Gamma \nabla^2 \rho}$$

MFT corresponding to TDP

Spatial inhomogeneity

$$\begin{aligned}\tau_{p,q} &= 2p - 1 \\ g_{p,q} &= p - (2 - p)q \\ c_{p,q} &= q\end{aligned}$$

- Spatially uniform pipe simulations:



Expt data on heated pipe

TDP in pipe flows

- To make contact with expt data:
 - Turbulent fraction inserted at start of pipe :
 - Modify the MFT to include space

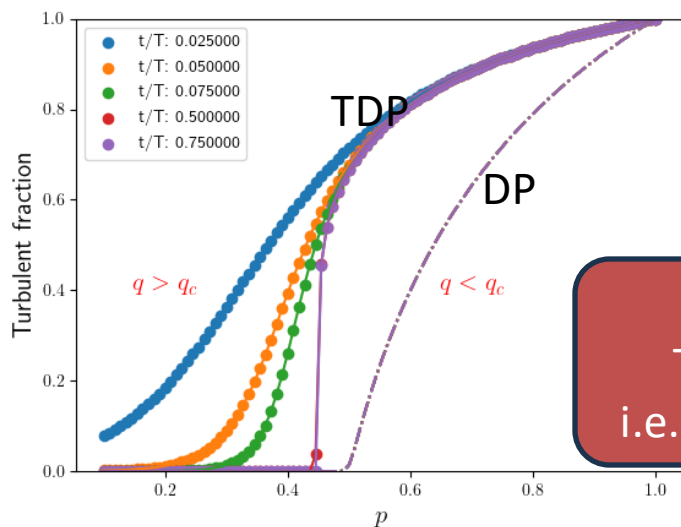
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MFT corresponding to TDP

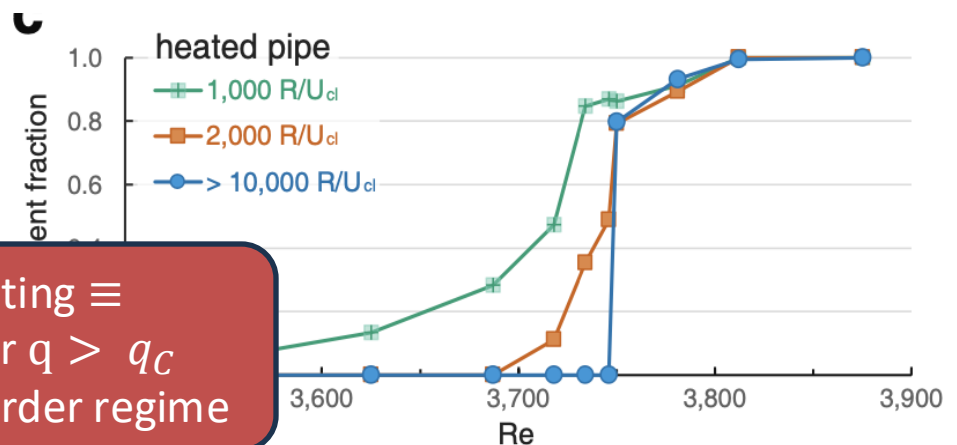
Spatial inhomogeneity

$$\begin{aligned}\tau_{p,q} &= 2p - 1 \\ g_{p,q} &= p - (2 - p)q \\ c_{p,q} &= q\end{aligned}$$

- Spatially uniform pipe simulations:



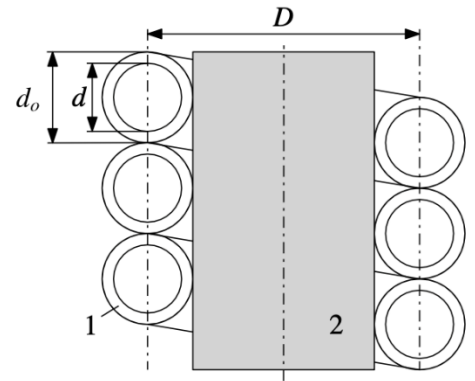
Heating $\equiv$
TDP for $q > q_c$
i.e. first order regime



Expt data on heated pipe

TDP in pipe flows

- Now we consider the case for curved pipes.
 - Include proper body forces:
 - Curvature driven flow instability, input it via higher order terms in the Landau theory
 - Include spatial dependence
 - Boundary conditions: $\rho = 1$ at boundary inlet
 - Allow for Spatially inhomogeneity



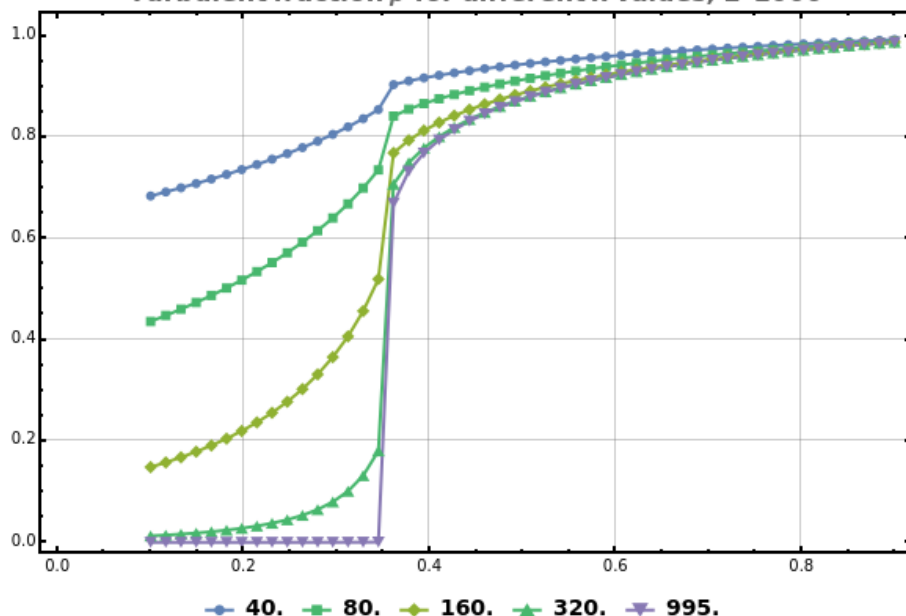
MFT for curved pipes

- Since hydrostatic force is expected to not strongly deform the flow profile, it should likely result in a modification at a lower order than curvature (cubic)

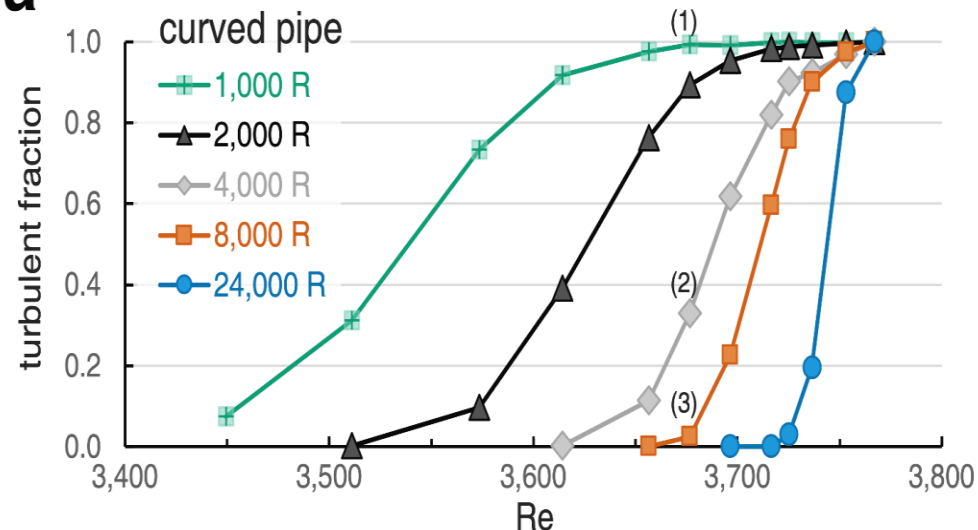
$$\partial_t \rho = \tau_{p,q} \rho - g_{p,q} \rho^2 - c_{p,q} \rho^3 + \Gamma \nabla^2 \rho$$

MFT corresponding accounting curvature as a body force

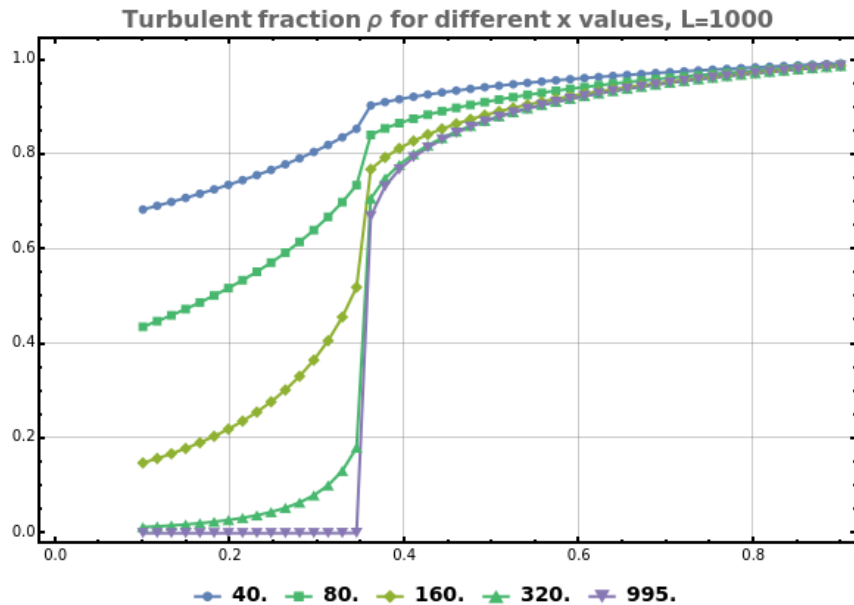
Turbulent fraction ρ for different x values, $L=1000$



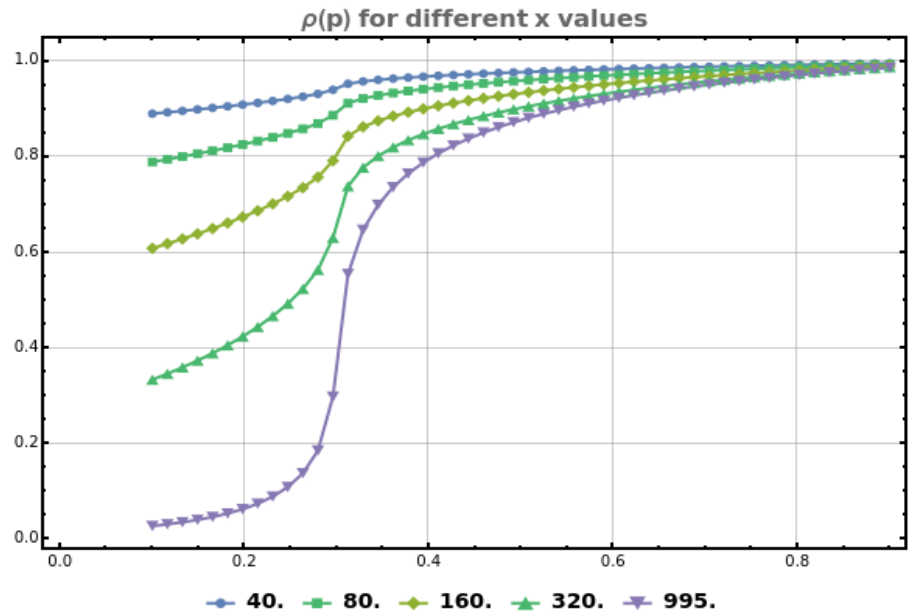
a



- Depending on spatial coupling strength, the profiles can be smoothened/discontinuity avoided.



$$\Gamma/\sqrt{L} = 100$$

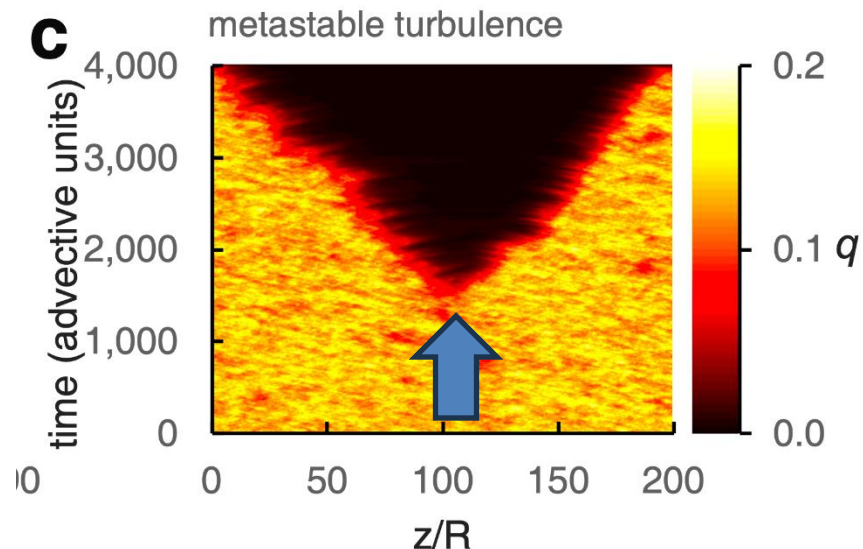
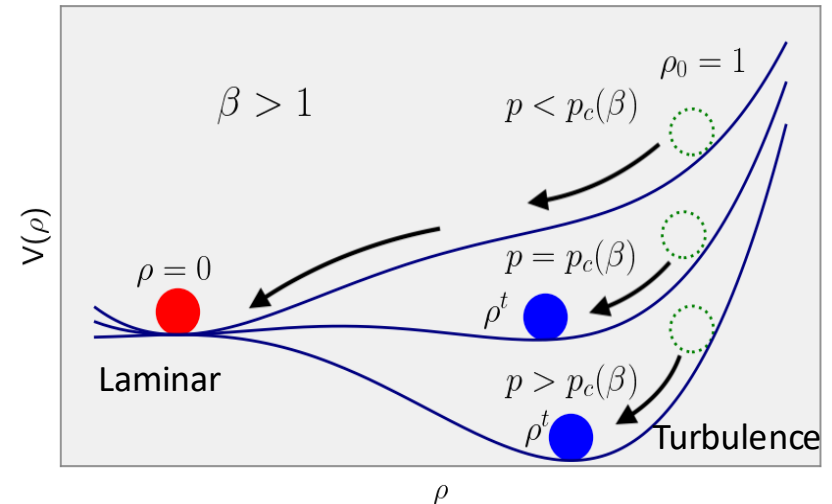


$$\Gamma/\sqrt{L} = 1000$$

- We cannot, apriori, bound Γ from the expt values since it might not be in steady state for curved pipe.
- However, qualitatively, the smoothing can be explained by the Laplacian term in MFT.

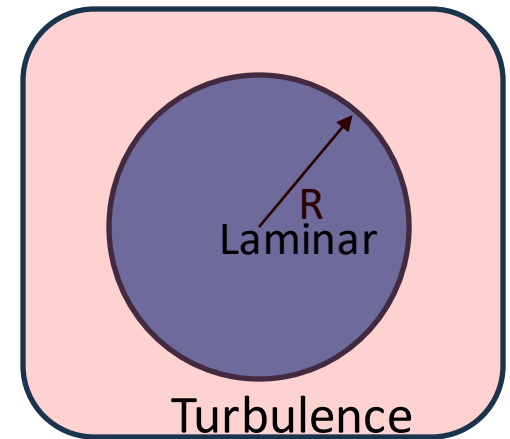
Nucleation of critical droplets

- First order transition
 - Possible that turbulence is metastable.
- Initializing a fully turbulent flow will recover laminarity through a nucleation process.



Dynamical Nucleation

- A laminar blob of radius R
 - Has lower free energy contribution from volume but energy required to create the laminar-turbulence interface i.e. surface tension σ



$$\Delta E = -\frac{4\pi R^3}{3} \cdot (V_{turbulent} - V_{laminar}) + 4\pi \cdot R^2 \cdot \sigma$$

$$\sigma = 2 \cdot \int_{\rho=0}^{\rho=\rho^*} \left[\frac{\Gamma \cdot \Delta V_{lyapunov}}{2} \right]^{1/2}$$

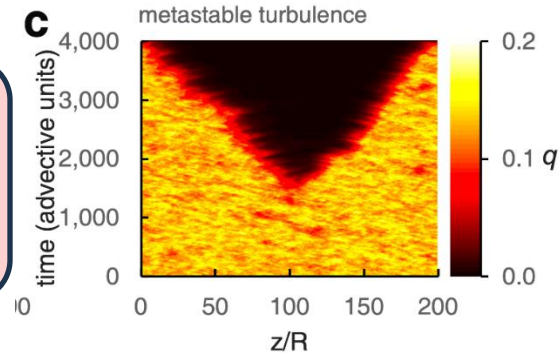
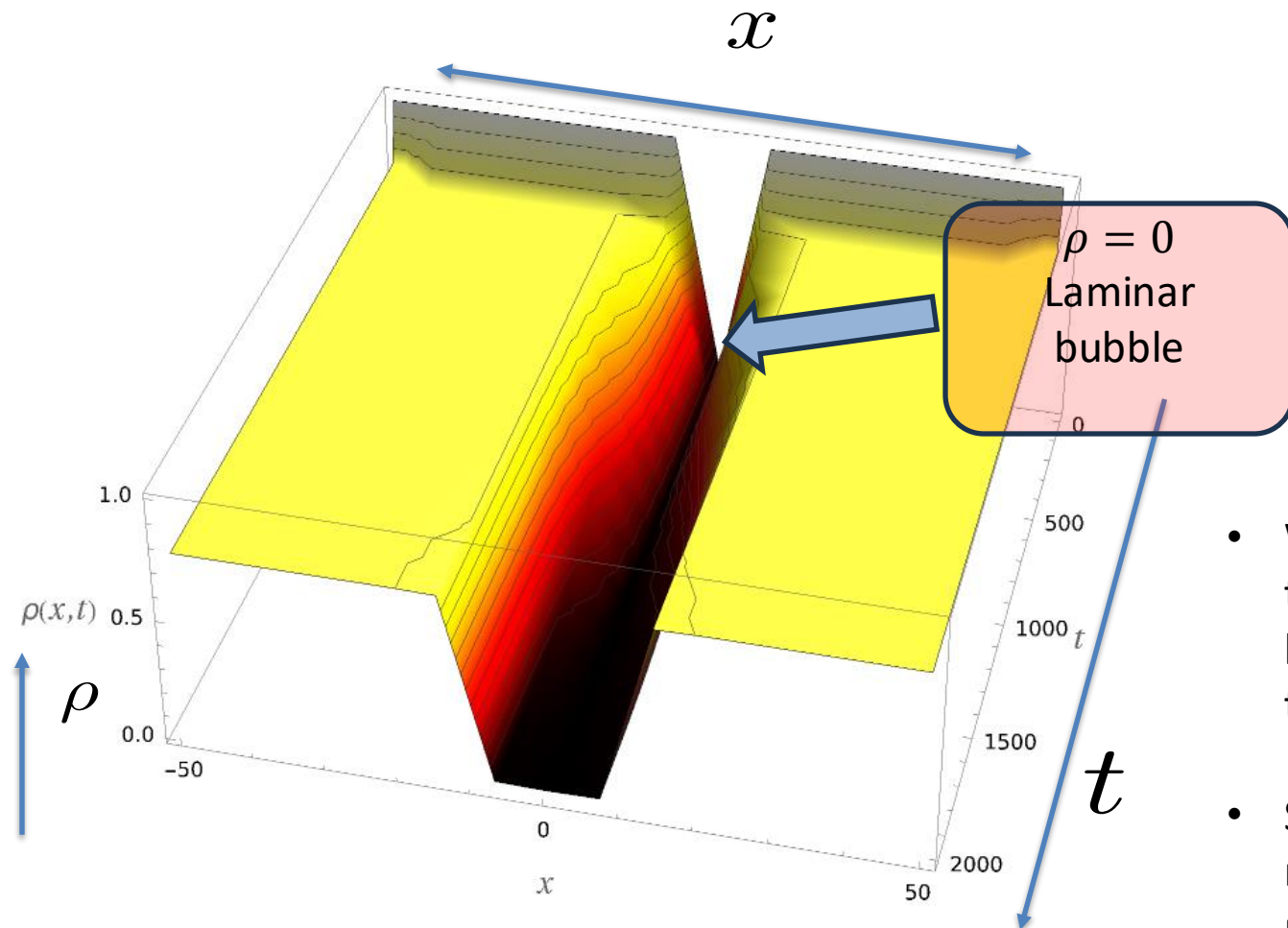
- For $R > \text{critical droplet size } R_c$, laminar blob expands unto and covers metastable turbulence.

$$R_c = \frac{2 \cdot \sigma}{V_{turbulent}} \propto \sqrt{\Gamma}$$

- Assuming Γ as constant near the transition and neglecting any higher order dependence of q , measurement of R_c can provide a value for Γ
- An additional check can be from surface tension measurements, which are expected to scale near the tricritical point as

$$\sigma \sim |p - p_c|^{3/2}$$

Relaminarization of flow



- We initialize a fully turbulent flow $\rho = 1$ with a laminar bubble $\rho = 0$ in the centre (see fig).
- Since turbulence is metastable (for current parameters), the laminar section expands, ultimately relaminarizing the flow.

Summary

- Body forces are expected to be a relevant variable at the DP transition
- Adding RG relevant terms, we uncover a tricritical fixed point, still in the larger class of absorbing phase transitions.
- The results qualitatively explain the delay and sharpening of the laminar-turbulence transition, in both heated and curved pipes.
- The spatial variation of the turbulent fraction is in good semi-quantitative agreement with experiment.
- A possible next step would be see if any possible scaling could be worked out between from the data of turbulent fraction at different pipe locations. Currently we're seeing how to extract this info.

Reserve Slides

MFT with spatial inhomogeneity

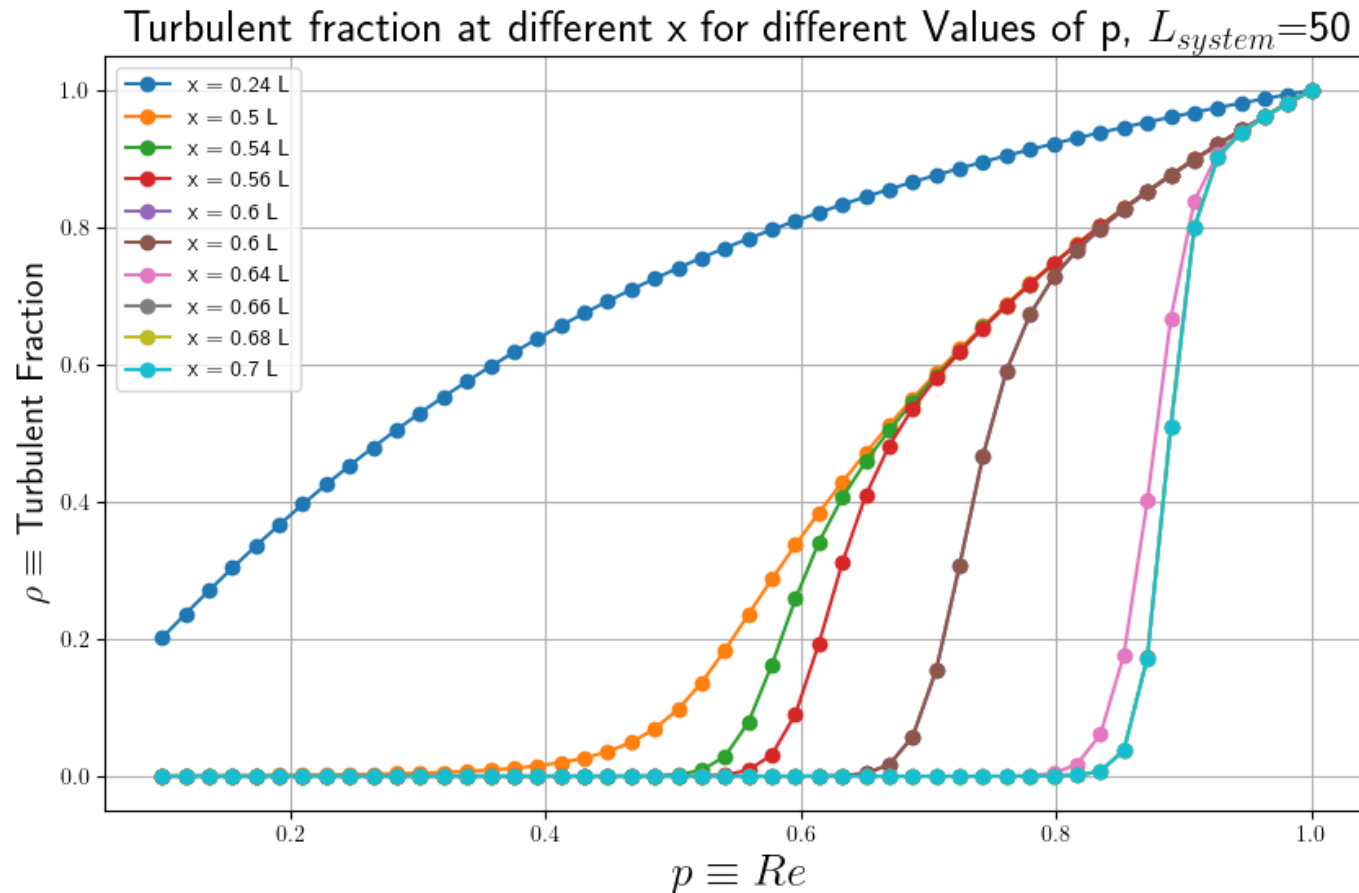
- To make contact with curved pipe data:
 - Turbulent fraction inserted at start of pipe :
 - Modify the MFT to include space
 - Set initial turbulent fraction =1 near $x=0$

$$\frac{\partial \rho}{\partial t} = (\delta p - p h) \cdot \rho - p \rho^2 + p h + \boxed{\Gamma \nabla^2 \rho}$$

Lowest order
RG relevant
term

- Include proper body forces:
 - Hydrostatic force: $F = \rho g d \rightarrow h(x) = c x$
 - Centrifugal force: $F = \text{const} \rightarrow h = \lambda$
 - $h \rightarrow h(x) = c x + \lambda$ i.e. linearly decreasing

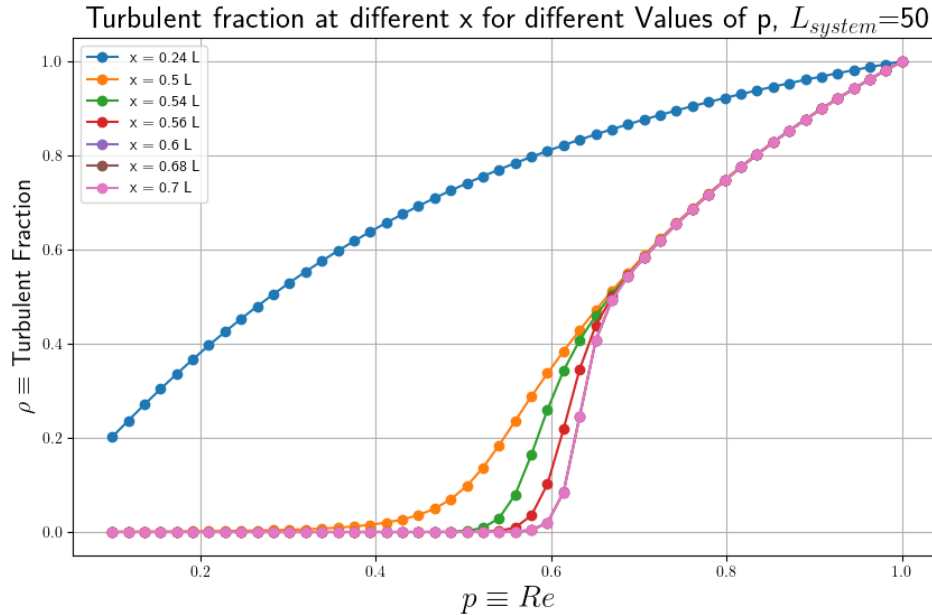
Results



- Qualitative Agreement: DP transition delayed to higher Re , Sharpening down the pipe

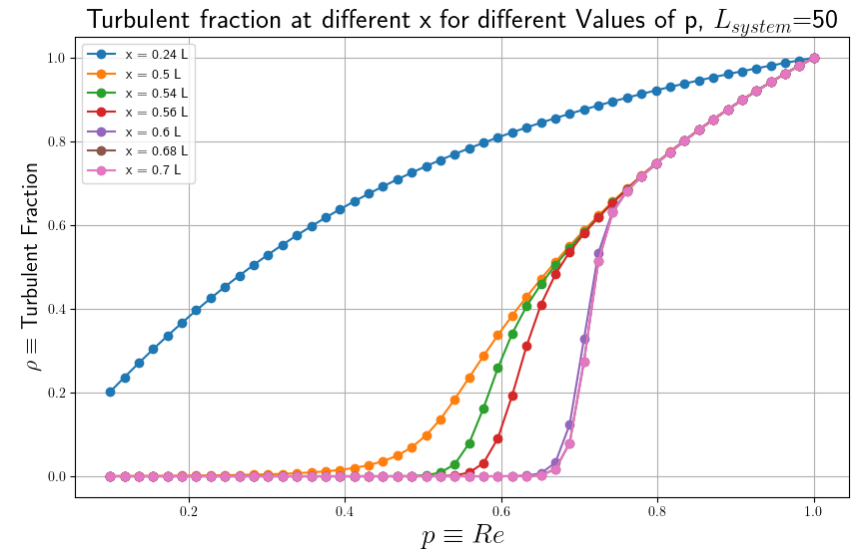
Sensitivity to centrifugal force term

- Effect of changing λ

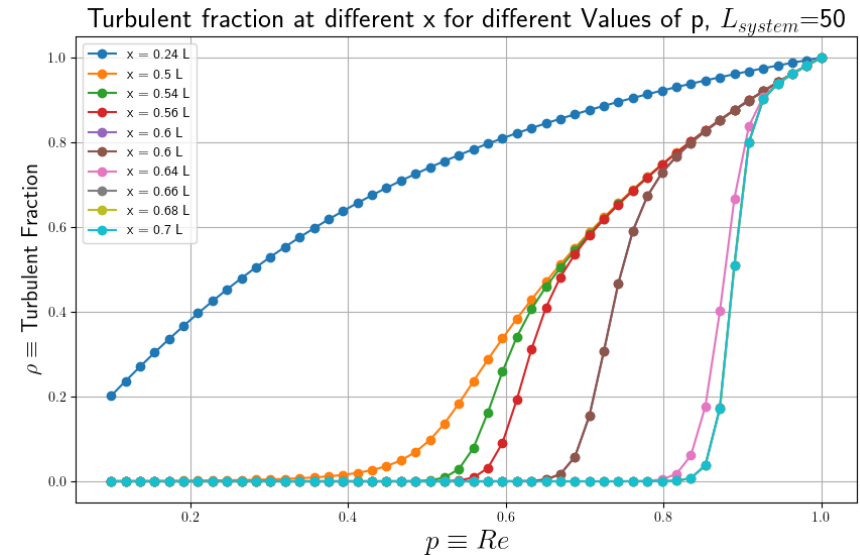
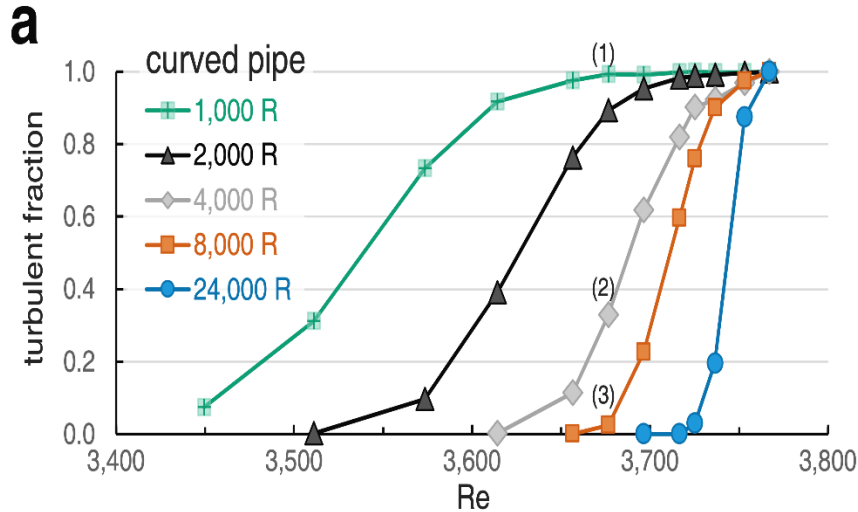


$$\lambda = 5 * 10^{-7}$$

- Lower λ delays the transition further down the pipe
- This is scaled λ , but here we did not carefully track non-dimensionalization



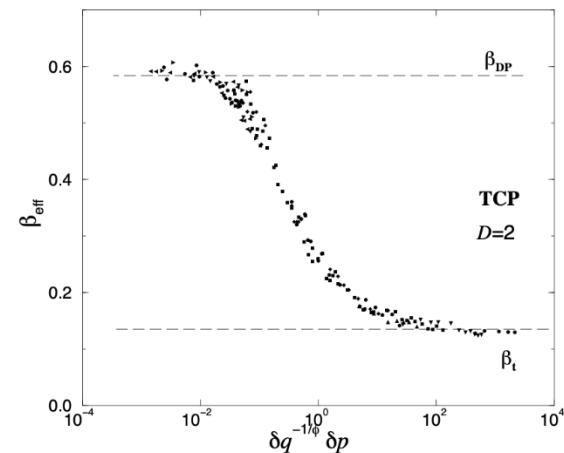
$$\lambda = 5 * 10^{-10}$$



- Consequences:
 - Qualitative agreement gives support to MFT, moreover recent work¹ shows that the critical region where MFT fails is roughly $1Re$ for straight pipes
 - Since the transition curves depend on λ , curves could in principle be data collapsed, but here it would depend on 2 parameters i.e. one might expect a one parameter family of data collapsed curves indexed by λ

Tricritical points

- Need to tune two parameters (Expt)
- Plot effective exponent and see crossover

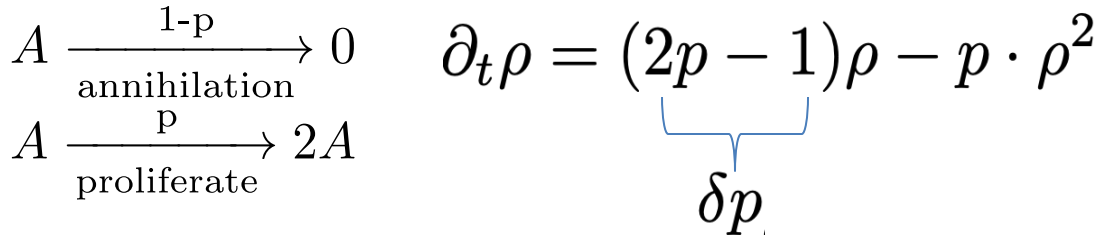


	$\text{TDP}_{D=2}$	$\text{TDP}_{D>3}$	$\text{DP}_{D=2}$ [46, 47, 35, 43]	$D = 3$ [48, 35, 43]	$\text{DP}_{D>4}$
β	0.14 ± 0.02	1/2	0.5834 ± 0.0030	0.813 ± 0.009	1
$\nu_{\perp}$	0.59 ± 0.08	1/2	0.7333 ± 0.0075	0.584 ± 0.005	1/2
$\nu_{\parallel}$		1	1.2950 ± 0.0060	1.110 ± 0.010	1
σ	1.12 ± 0.05	3/2	2.1782 ± 0.0171	2.049 ± 0.026	2
γ'	0.93 ± 0.06	1/2	0.2998 ± 0.0162	0.126 ± 0.023	0
γ	1.00 ± 0.06	1	1.5948 ± 0.0184	1.237 ± 0.023	1
$\eta_{\perp}$	0.42 ± 0.24	1	1.5912 ± 0.0148	1.783 ± 0.016	2
D_{f}	1.76 ± 0.05	2	1.2044 ± 0.0091	1.608 ± 0.019	2
β'		1	$\beta = \beta'$	$\beta = \beta'$	1
δ		1	0.4505 ± 0.0010	0.732 ± 0.004	1
α		1/2	$\alpha = \delta$	$\alpha = \delta$	1
θ		0	0.2295 ± 0.0010	0.114 ± 0.004	0
z		2	1.7660 ± 0.0016	1.901 ± 0.005	2
$\frac{\tilde{X}(+1,0)}{\tilde{X}(-1,0)}$	0.35 ± 0.05	1/2	0.25 ± 0.01	0.65 ± 0.03	1
U	0.84 ± 0.04		0.704 ± 0.013	0.61 ± 0.02	1/2

Puff absence for reported curved pipe data

Modified mean field theory describes discontinuity, uncovers a tricritical point

Regular directed percolation

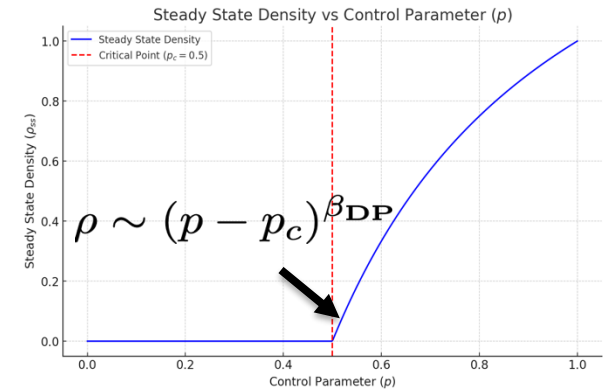


$$\rho \equiv A$$

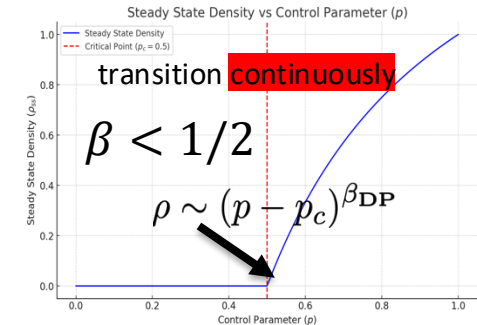
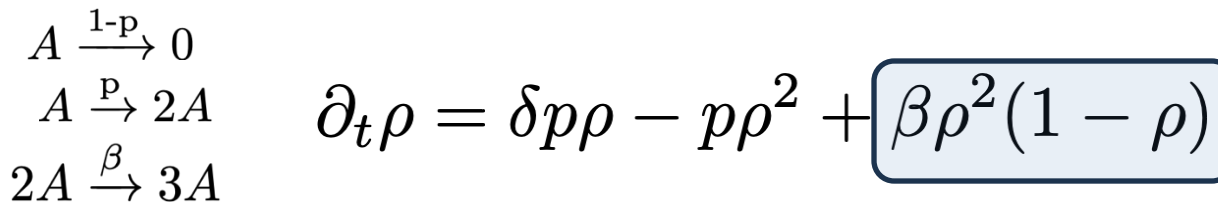
$$p \equiv Re$$

$$0 \equiv \text{laminar}$$

$$A \equiv \text{turbulent}$$

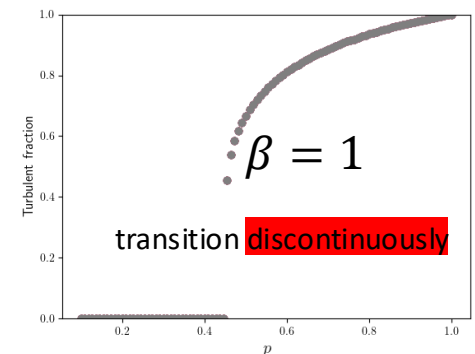


Directed percolation with higher order marginal term



Near $\beta = 1/2$, cubic term can become relevant, leading to tricritical point

Depending on $\beta < 1/2$ or $\beta > 1/2$, new mean field theory predicts a continuous DP or a discontinuous transition.



Roughness-induced criticality

- Answered the question of what would be the signatures of turbulence as a dynamic critical phenomenon?
 - Critical: theory predicts new scaling laws in Re , roughness (r)
 - Experimental Agreement in 2D Soap Films
- **Roughness-induced**: laminar pipe flow is linearly stable, but boundary roughness is a relevant variable, coupling to turbulent state.

Roughness as Body force

- Using conformal mapping , it was shown by Nigel that roughness can be transformed into **two-body force** terms in NSE.
- This provided a link between body forces and roughness, where these can be thought of as conjugate fields that couple to turbulence (i.e. order parameter).

Friction factor of two-dimensional rough-boundary turbulent soap film flows

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(Received 11 August 2008; revised manuscript received 29 May 2009; published 25 June 2009)

We use momentum-transfer arguments to predict the friction factor f in two-dimensional turbulent soap film flows with rough boundaries (an analog of three-dimensional pipe flow) as a function of Reynolds number Re and roughness r , considering separately the inverse energy cascade and the forward enstrophy cascade. At intermediate Re , we predict a Blasius-like friction factor scaling of $f \propto Re^{-1/2}$ in flows dominated by the enstrophy cascade, distinct from the energy cascade scaling of $Re^{-1/4}$. For large Re , $f \sim r$ in the enstrophy-dominated case. We use conformal map techniques to perform direct numerical simulations that are in satisfactory agreement with theory and exhibit data collapse scaling of roughness-induced criticality, previously shown to arise in the three-dimensional pipe data of Nikuradse.

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PACS number(s): 47.27.ek, 47.27.nf

Turbulent flows are marked by rich structure over a range of scales—they host fluctuations, vortices, tangles, and other coherent structures that continue to defy a detailed analytical understanding [1,2]. When parametrized in terms of the typical flow speed U , characteristic length scale L and kinematic viscosity of the fluid ν three-dimensional turbulence exhibits universal phenomena as the Reynolds number $Re = UL/\nu \rightarrow \infty$. Most famously, in a theory referred to as K41 [3,4], the dependence of the fluctuation energy spectrum $E(k)$ on the wave number and mean energy-transfer rate $\bar{\epsilon}$ occurs in a way that is independent of ν : $E(k) = \bar{\epsilon}^{2/3} k^{-5/3}$ for values of the wave number in the so-called inertial range, intermediate between the scales of forcing and the scales where molecular viscosity becomes significant. In this inertial range, turbulent eddies break up into smaller eddies through a mechanism, which is to a first approximation Hamiltonian, and results in a cascade of energy to smaller length scales [5].

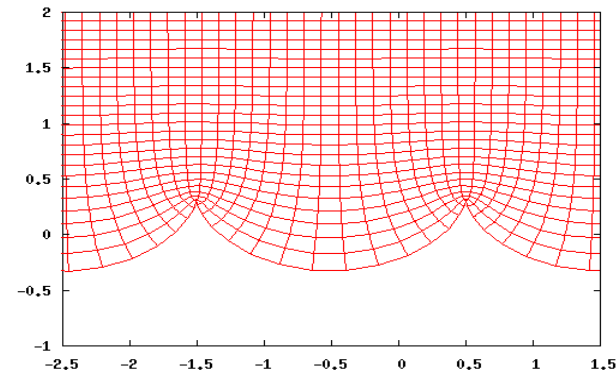
During the 1930s, Nikuradse undertook a systematic series of measurements of the pressure drop across a turbulent pipe flow as a function of Re [6] and also as a function of

tions. Second, Goldenfeld [13] pointed out that the power-law behavior of Nikuradse's friction factor data in the regimes $Re \rightarrow \infty$ and $r/D \rightarrow 0$ was analogous to critical phenomena, where the inverse Reynolds number and roughness play similar roles to, for example, the coupling constant and external magnetic field in an Ising model. Consequently, the dependence of Nikuradse's data on Re and r can be collapsed onto a universal function with sufficient precision for intermittency corrections to be extracted [14]. These results show that the friction factor reflects the nature of the turbulent state through its dependence on the energy spectrum and that the turbulent state is itself a manifestation of a nonequilibrium critical point at $Re = \infty$ and $r/D \rightarrow 0$.

The purpose of this Rapid Communication is to test the claims of Refs. [12,13] in a context where detailed calculations are in principle possible: the case of two-dimensional (2D) soap film turbulence [15,16]. Here, a soap film is supported between two vertical wires, and the draining flow provides a versatile laboratory for exploring two-dimensional turbulence [16]. It is well understood that the nature of turbulence in 2D is different from three-dimensional: there is no vortex stretching, for example. Nevertheless, turbulent phe-

Representation of Roughness

- Want to generate walls with a particular lengthscale or set of lengthscales using the conformal map.
- At lower boundary ($v=0$), try:
$$z = w + re^{ikw}$$
- Coordinate singularities when



$$|g'|^2 = (1 - rk \sin(ku)e^{-kv})^2 + (rk \cos(ku)e^{-kv})^2 = 0$$
$$ku = \frac{\pi}{2} + n\pi, e^{-kv} = rk$$

- $rk < 1$ in order to prevent singularities inside the computational domain.
- Parameters r and k .

Conformal Mapping

- Navier-Stokes equation before mapping:

$$\frac{\partial \mathbf{V}}{\partial t} + \mathbf{V} \cdot \nabla \mathbf{V} = \nu \nabla^2 \mathbf{V} - \nabla \mathbf{P}$$

- Map coordinates: $x + iy = f(u + iv)$

$$\bar{\mathbf{V}} = (y_v V_x - x_v V_y) \hat{\mathbf{u}} + (x_u V_y - y_u V_x) \hat{\mathbf{v}}$$

$$|g'|^2 \equiv x_u^2 + x_v^2 \quad \mathbf{A} \equiv \begin{bmatrix} x_u y_{uv} + x_v x_{uv} \\ x_u x_{uv} - x_v y_{uv} \end{bmatrix}$$

- Navier-Stokes equation after mapping:

$$|g'|^2 \frac{\partial \bar{\mathbf{V}}}{\partial t} + (\bar{\mathbf{V}} \cdot \nabla) \bar{\mathbf{V}} = \nu \nabla^2 \bar{\mathbf{V}} + \frac{|\bar{\mathbf{V}}|^2}{|g'|^2} \mathbf{A} + \frac{2\nu}{|g'|^2} \mathbf{A}^\perp (\nabla \times \bar{\mathbf{V}})_z - \nabla \mathbf{P}$$

Meaning of New Terms

$$|g'|^2 \frac{\partial \bar{\mathbf{V}}}{\partial t} + (\bar{\mathbf{V}} \cdot \nabla) \bar{\mathbf{V}} = \nu \nabla^2 \bar{\mathbf{V}} + \frac{|\bar{\mathbf{V}}|^2}{|g'|^2} \mathbf{A} + \frac{2\nu}{|g'|^2} \mathbf{A}^\perp (\nabla \times \bar{\mathbf{V}})_z - \nabla \mathbf{P}$$

- Two new **body forces** as a result of the mapping:

$$\frac{|\bar{\mathbf{V}}|^2}{|g'|^2} \mathbf{A}$$

Body force due to acceleration around contours of the boundary

$$\frac{2\nu}{|g'|^2} \mathbf{A}^\perp (\nabla \times \bar{\mathbf{V}})_z$$

Body force due to curvature in the map corresponding to vorticity of the real-space velocity field.

Roughness-induced criticality

- Answered the question of what would be the signatures of turbulence as a dynamic critical phenomenon?
- **Critical**: theory predicts new scaling laws in Re , roughness (r)
- **Experimental Agreement** in 2D Soap Films
- **Roughness-induced**: laminar pipe flow is linearly stable, but boundary roughness is a relevant variable, coupling to turbulent state.



- Roughness/ Body forces
 - Can couple to turbulent fraction (order) and promote it
 - Likely a relevant scaling field at transition

Intermittency corrections

Mehrafarin & Pourtolami (2008)

1. Definition of intermittency exponent

$$\langle \delta v_l^2 \rangle \sim (\langle \varepsilon_l \rangle l)^{2/3} \sim l^{2/3+\eta}$$

2. Momentum transfer $f \sim \frac{v_r}{V}$ $f \sim r^\alpha$ $\alpha = \frac{1}{3} + \frac{\eta}{2}$

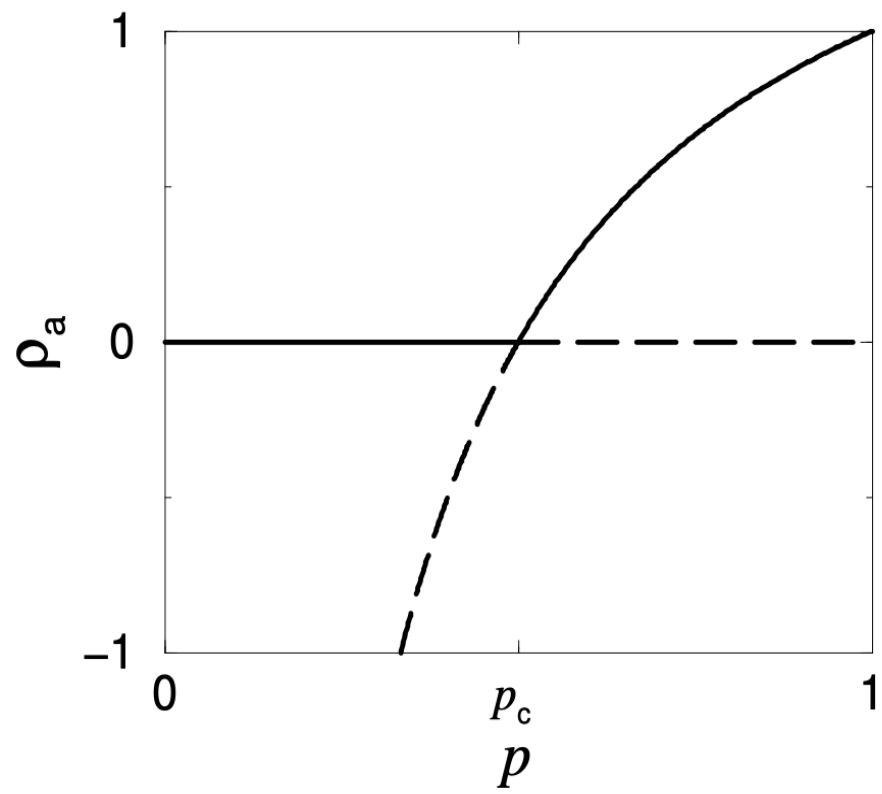
3. Flow transformation under rescaling

- **Boundary layer viscosity** $\sim r v_r$ (length $\times$ velocity), i.e., as $\sim r^{\alpha+1}$.

$$f\left(l \frac{r}{R}, l^{-\alpha-1} \text{Re}\right) = l^\alpha f\left(\frac{r}{R}, \text{Re}\right)$$

4. Putting it all together

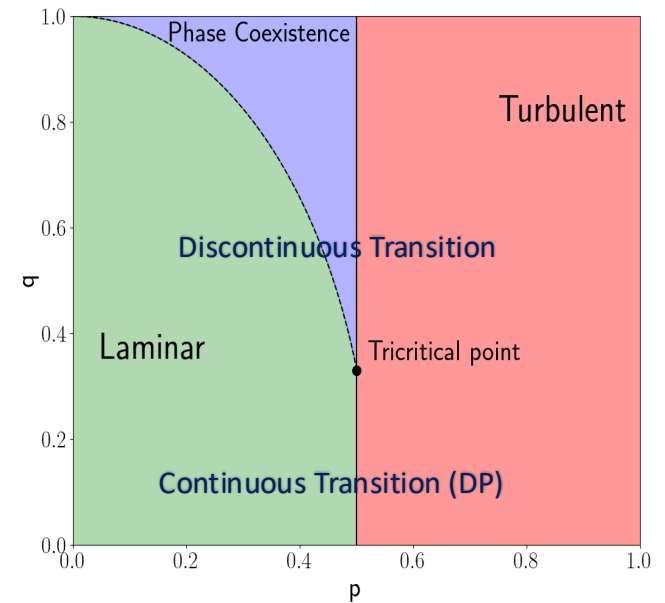
$$f = \text{Re}^{-(2+3\eta)/(8+3\eta)} g\left(\frac{r}{R} \text{Re}^{6/(8+3\eta)}\right)$$



Summary

Body forces are expected to be a relevant variable at the DP transition

- We uncover a tricritical fixed point
- New exponents, mean field phase diagram and scaling functions are predicted
- Can experimentally checked to vindicate/falsify our analysis.

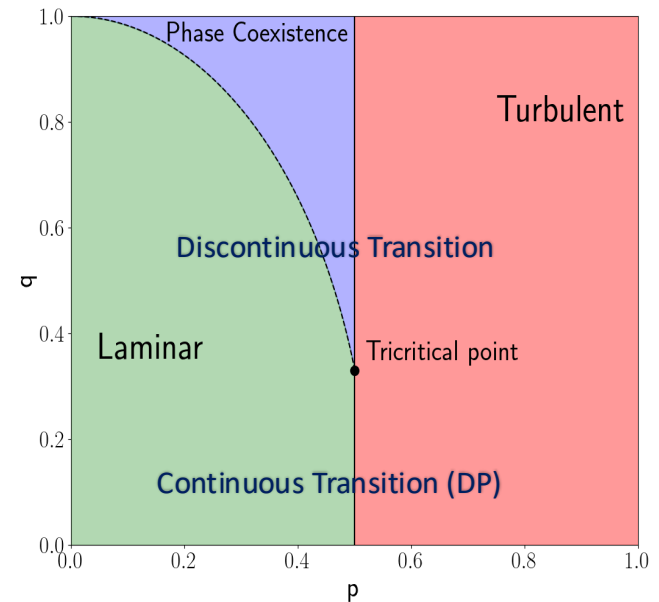


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How do body forces sharpen/delay the transition?

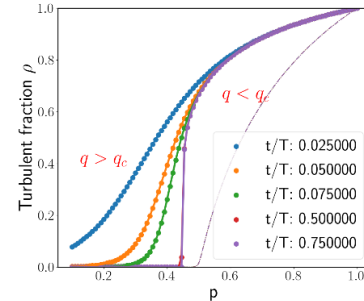
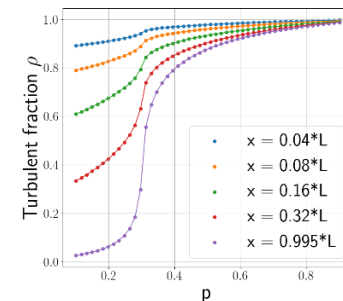
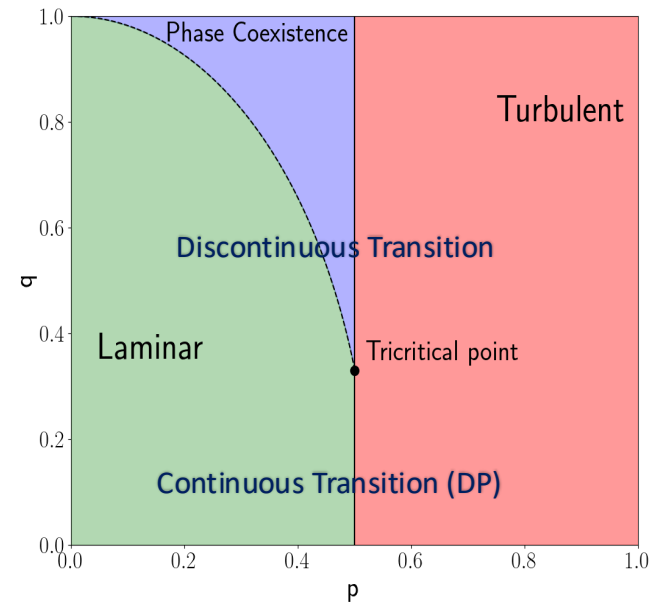


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Tricritical DP qualitatively explains the turbulent fraction profiles



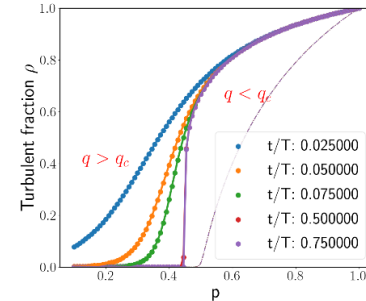
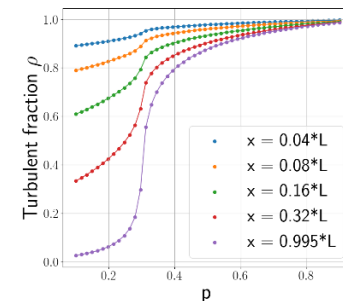
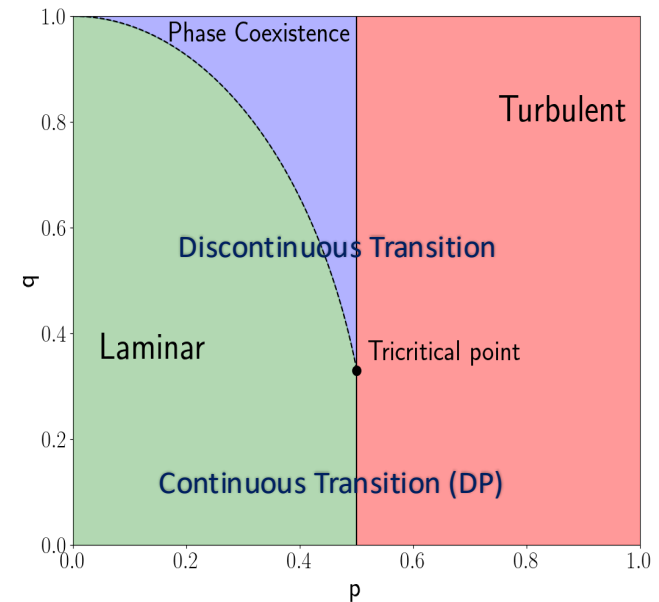
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- New exponents, mean field phase diagram and scaling functions are predicted
- Can experimentally checked to vindicate/falsify our analysis.

Tricritical DP qualitatively explains the turbulent fraction profiles

- Describes the delay of critical point and sharpening of the laminar-turbulence transition
- Accounts for profiles of heated and curved pipes.



Summary

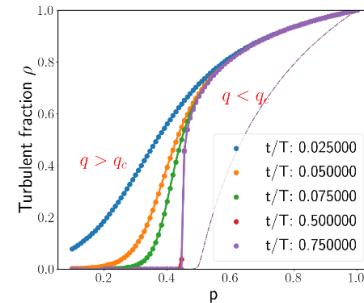
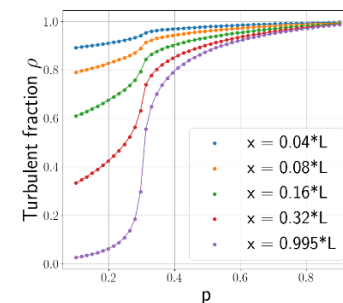
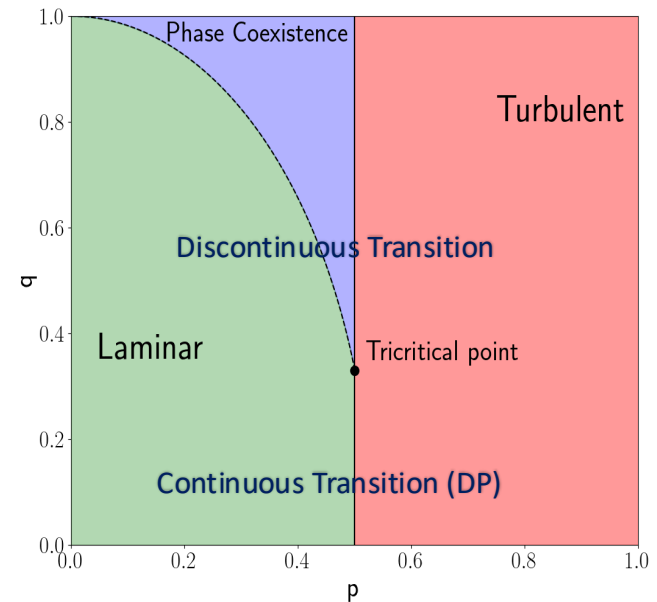
Body forces are expected to be a relevant variable at the DP transition

- We uncover a tricritical fixed point
- New exponents, mean field phase diagram and scaling functions are predicted
- Can experimentally checked to vindicate/falsify our analysis.

Tricritical DP qualitatively explains the turbulent fraction profiles

- Describes the delay of critical point and sharpening of the laminar-turbulence transition
- Accounts for profiles of heated and curved pipes.

Is coarse graining possible?



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Critical scaling can be extracted without explicit use of Navier Stokes equation

